Bonds, Business Cycles and Financial Crises*

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Abstract

Do corporate bond markets stabilize corporate financing when banks are impaired? Theory delivers an ambiguous aggregate prediction: firms with bond-market access may substitute away from banks, while constrained firms may lose financing and bank loans and bonds may be complements. I study which force dominates using a new long-run dataset of primary bond issuance by issuer sector for 19 countries from the nineteenth century to the present. Non-financial corporate bond issuance falls sharply and persistently after systemic banking crises: three years after crisis onset, issuance is roughly one-half below its pre-crisis level. By contrast, government issuance rises strongly, while financial-sector issuance increases only temporarily. The NFC contraction is robust to alternative normalizations and historical periods and is substantially deeper in crises with severe risky-debt-market repricing. Countries with deeper pre-crisis bond markets are associated with stronger subsequent investment growth, but this relationship is not distinctly stronger during banking crises. Thus, while theory permits substitution by firms with market access, the aggregate response is a contraction rather than a corporate spare-tire expansion.

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1 Introduction

The idea that capital markets can serve as a "spare tire" when banks retrench has long shaped debates about financial-system resilience. Reflecting on the Asian financial crises, Greenspan (1999) put the argument starkly: "The lack of a spare tire is of no concern if you do not get a flat. East Asia had no spare tires." In this view, firms facing impaired bank lending can turn to bond markets, allowing market finance to cushion disruptions in bank credit. Yet theory does not imply that aggregate corporate bond issuance must rise during banking crises. Firms with bond-market access may substitute away from banks, while marginal or constrained firms may lose access altogether. Moreover, bank loans and bonds can be complements for some firms, so tighter bank intermediation can reduce rather than increase bond borrowing (Crouzet, 2018). Crisis-period flight to safety may further redirect investor demand toward safer public claims (Caballero and Krishnamurthy, 2008). Because these forces operate in opposite directions, theory gives no unambiguous prediction for aggregate non-financial corporate bond issuance during banking crises.

Existing empirical evidence does not resolve this ambiguity. During the Global Financial Crisis, U.S. firms with bond-market access substituted toward market debt as bank lending contracted (Adrian et al., 2013). Yet this response was selective and dynamic: the corporate bond market itself froze after Lehman before reopening faster than syndicated lending, with the recovery concentrated among larger and higher-quality issuers (Becker and Benmelech, 2025). Internationally, issuance declined across many markets during the GFC, whereas COVID-19 produced an unusually broad expansion under extensive policy support (Bruno et al., 2024). Evidence that some firms substitute toward bonds therefore does not establish that this margin is large enough to stabilize corporate financing in the aggregate. Therefore, this paper asks: Do corporate bond markets function as an aggregate spare tire during systemic banking crises? This question is distinct from whether credit booms predict subsequent financial crises or recessions: I condition on realized systemic banking-crisis onsets and trace how primary-market financing evolves thereafter.

The paper makes three contributions. First, I assemble a new long-run, cross-country dataset on primary bond issuance by issuer sector, drawing on statistical yearbooks, central-bank publications, securities-market statistics, and archival capital-issue records. The harmonized data cover 19 countries from the nineteenth century to 2024 and 54 distinct banking-crisis onsets. The core NFC series contains 1,251 country-year observations for 17 countries and covers 46 crises. The data distinguish, where possible, government, financial-corporate, non-financial-corporate, and total-corporate issuance. This decomposition is central to the paper: the spare-tire hypothesis concerns whether non-financial firms replace impaired bank credit with market debt, whereas financial-corporate and government issuance capture different crisis-period financing margins. Gross primary issuance measures securities brought to market in a given year rather than changes in out-

standing stocks, allowing me to observe primary-market activity directly during financial stress.

Second, I estimate how sectoral primary bond issuance responds to systemic banking crises. NFC issuance contracts sharply and persistently. In the baseline estimates, gross issuance is about 38 log points below its pre-crisis level after one year and roughly one-half lower after three years. Relative to ordinary issuance dynamics, cumulative NFC issuance growth at the three-year horizon is 112 log poin below comparable non-crisis country-years. By contrast, government issuance rises sharply and persistently, while financial-corporate issuance increases only temporarily before declining at medium horizons. The NFC contraction appears in both pre- and post-World War II crises and is robust to scaling issuance by GDP, investment, and lagged bank credit.

The aggregate contraction is compatible with substitution by individual firms that retain market access. Security-level evidence from Australia illustrates one way this can occur: around the 1931 banking panic, some established firms continued to issue, but the number of active private issuers fell sharply and long-maturity issuance nearly disappeared. Across countries, NFC issuance also contracts substantially more in crises accompanied by large increases in corporate bond spreads. Together, the evidence shows that continued access for selected firms can coexist with falling aggregate primary issuance and substantial debt-market repricing.

Third, I ask whether deeper bond markets nevertheless improve real resilience after banking crises. This question relates closely to Grjebine et al. (2018), who show that economies with a larger bond-financing share and stronger bond–loan substitution recover faster from recessions. I find that countries with deeper pre-crisis NFC bond markets likewise experience stronger subsequent investment recovery. However, the relationship is not distinctly stronger during banking crises than in normal periods. The evidence therefore associates bond-market depth with broader investment resilience, but does not reveal a distinct crisis-specific spare-tire effect.

Taken together, the results distinguish between firms having access to bond markets and bond markets stabilizing financing in the aggregate. Selected firms may substitute toward market debt, and deeper bond markets are associated with stronger investment outcomes. But across systemic banking crises, these margins are not large enough to generate an aggregate NFC issuance expansion. Instead, NFC issuance contracts while the composition of primary bond-market activity shifts toward governments and, temporarily, financial corporations.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the construction of the dataset and documents the long-run evolution of corporate and government bond issuance. Section 4 studies issuance dynamics around banking crises and the relationship between bond-market depth and real recovery. Section 5 concludes.

2 Literature Review

This paper relates to four strands of literature: (i) credit booms and macroeconomic fluctuations, (ii) corporate bond markets and crises, (iii) theories of financing choices and financial frictions, and (iv) the economic history of bond markets.

Credit Cycles, Banking Crises, and Macroeconomic Fluctuations A robust finding in macro-finance is that credit booms predict financial crises and severe recessions (Schularick and Taylor, 2012; Jordà et al., 2013). The composition of credit matters: household debt booms are especially dangerous, while the sectoral allocation of corporate bank loans during credit booms also has striking implications for subsequent downturns (Mian et al., 2017; Müller and Verner, 2023). Yet in this line of work, corporate bonds are often omitted or treated inconsistently across countries and time. For example, Müller and Verner (2023) focus on bank loans and note that adding bonds does not materially change their aggregate results, while Ivashina et al. (2024) study corporate debt booms largely through the lens of bank lending. This paper differs from this literature in both its object and timing. The credit-cycle literature asks whether leverage expansions predict subsequent financial crises and the severity of recessions. I instead condition on realized systemic banking-crisis onsets and study the subsequent response of primary bond issuance across issuer sectors. The aim is therefore not to test whether bonds predict crises, but to ask whether market finance stabilizes aggregate corporate financing once the banking system becomes impaired.

Corporate Bond Markets and Crises A second strand studies directly how firms tap bond markets in times of stress. Evidence from recent crises shows that substitution into bond finance is selective and dynamic rather than frictionless. Adrian et al. (2013) document that U.S. firms with bond-market access increased bond financing as bank lending contracted during the GFC. Becker and Benmelech (2025) refine this picture using high-frequency issuance data. Both the corporate bond and syndicated-loan markets froze following Lehman, but bond issuance recovered substantially faster: by early 2009 the primary bond market had reopened, while syndicated lending remained depressed. This resilience was concentrated among higher-quality issuers, consistent with evidence that access to bond markets becomes increasingly selective during downturns (Erel et al., 2012; Becker and Ivashina, 2014).

Market functioning itself can also deteriorate during banking crises. Foley-Fisher et al. (2019) show that the collapse of AIG's securities-lending activity reduced secondary-market liquidity in exposed corporate bonds, while Boyarchenko et al. (2025) show that broader corporate-bond-market distress predicts subsequent real activity.

Cross-country evidence further emphasizes that substitution requires access to an alternative functioning market. Using issuance-level data across 50 economies, Cortina et al. (2021) show that firms move away from debt markets directly hit by crises and toward less-

affected bond, loan, domestic, or international markets. These switches can partly offset contractions in affected markets, but are concentrated among large firms. Consistent with this state dependence, Bruno et al. (2024) find that the GFC was associated with lower corporate bond issuance across most regions, whereas the COVID-19 shock produced an unusually broad increase in bond issuance, particularly in local-currency markets. In the United States, direct Federal Reserve interventions further strengthened this margin: Boyarchenko et al. (2022) show that primary issuance accelerated following the announcement of the corporate credit facilities, even before secondary-market spreads had fully normalized. Taken together, this literature suggests that a bond-market "spare tire" is conditional on issuer quality, market depth, intermediary functioning, access to alternative markets, and policy support. This paper asks whether these selective substitution margins are sufficiently large to stabilize aggregate sectoral bond issuance across a much broader set of countries and historical banking crises. At the macroeconomic level, Grjebine et al. (2018) find that economies with greater bond–loan substitution recover faster from recessions. My contribution is to ask whether these firm- and episode-specific substitution margins are large enough to appear in aggregate NFC primary issuance around systemic banking crises.

In contrast to work centered on individual firms, modern episodes, or recession recoveries, the long-run sectoral panel directly measures the aggregate NFC issuance response across historical banking crises.

Theories of Financing Choices and Financial Frictions Models of financing choice emphasize that access to market debt is heterogeneous. Firms with stronger reputations or greater pledgeable income can access bond markets on terms unavailable to more constrained firms (Diamond, 1991; Holmstrom and Tirole, 1997). During banking stress, this heterogeneity creates opposing forces: firms with market access may substitute toward bonds, while marginal issuers may lose financing and adverse market conditions may restrict issuance (Erel et al., 2012; Benmelech and Bergman, 2017).

Crouzet (2018) provides the closest macro-finance benchmark for my analysis. Firms endogenously choose between bank loans and bonds, trading off the restructuring flexibility of bank debt against the lower intermediation costs of market finance. When bank intermediation becomes more costly, large firms with bond-market access substitute toward bonds. At the same time, loans and bonds are complements for firms that use both sources of finance, so tighter bank credit can also reduce bond issuance. Some firms consequently operate at smaller scale even as the largest firms substitute toward market debt. Aggregate borrowing and investment can therefore contract despite firm-level debt substitution. The key implication for this paper is that greater bond issuance by firms retaining market access need not generate an aggregate NFC issuance expansion. My long-run sectoral data ask which of these margins dominates during systemic banking crises.

Investor-side forces reinforce this ambiguity. Crisis-period flight to safety can redi-

rect demand toward public debt and other safer claims (Caballero and Krishnamurthy, 2008), while issuer quality and primary-market access vary over the cycle (Greenwood and Hanson, 2013; Erel et al., 2012). Theory therefore leaves the aggregate sign of NFC bond issuance unresolved. The empirical contribution of this paper is to measure that sign directly over a long historical crisis sample.

Historical Perspectives on Bond Markets A large economic-history literature documents the long-run evolution of bond markets and the shifting balance between bank- and market-based finance, emphasizing the role of political economy, regulation, and institutional change (Rajan and Zingales, 2003; Abraham et al., 2021; Fishlow, 1985). Rich country studies document the development of corporate debt and securities markets in France (Hautcoeur, 1993; Hautcoeur and Riva, 2012), the United States (Hickman, 1953, 1960; Shapiro, 1959), the United Kingdom (Michie, 1999), Scandinavia (Waldenström, 2022a; Skånland, 1967), and Italy (Scheib, 2024). This literature provides extensive evidence on individual markets and institutions, but comparable cross-country series on primary corporate issuance remain scarce. My contribution is to harmonize these primary-market flows by issuer sector and link their long-run evolution to systemic banking crises.

Taken together, existing theory and evidence leave open whether selective firm-level substitution is quantitatively large enough to stabilize aggregate corporate market finance during banking crises. The remainder of the paper addresses this question using long-run sectoral primary issuance data.

3 A New Dataset on Bond Markets

A central contribution of this paper is the construction of a new long-run, cross-country dataset on primary bond issuance by sector. Unlike existing macro-financial datasets that focus on outstanding bank credit, the series compiled here measure the annual volume of newly issued bonds at the sector level. These series capture the allocation of market-based financing across sectors in each country-year and allow a systematic comparison of bond-issuance dynamics with the long-run bank-credit series widely used in the macro-financial literature. Throughout, I use bond issuance as shorthand for bonds and comparable marketable debt securities covered by the underlying primary-market statistics. Appendix A provides the precise instrument-coverage rules and country-specific definitions.

3.1 Construction and Sources

The dataset is compiled from a wide range of sources. For historical periods, generally before 1990 and, for some countries, into the 2000s, I rely primarily on capital-issuance statistics published in statistical yearbooks, central-bank reports, ministry-of-finance publications, and other official sources. Examples include the *Annuaire Statistique de la France*

and the *Bank of Thailand's Annual Economic Reports*. In some cases, I also rely on compilations of individual researchers such as Waldenström (2022b) or special publications of long-term historical data by central banks or statistical offices. In many cases, I contacted archives, central banks, and national statistical offices directly; their staff often helped me identify and digitize relevant source material.

The degree of detail varies across countries and decades, but sources often provide comprehensive breakdowns of securities issuance by different sectors. The necessary distinction between bond and equity issuance became available for some countries only after they had already published general security issuance statistics for quite some time. For the modern period, generally from the 1990s onward, I rely on national or international debt securities statistics by the respective central banks or statistical offices. Where series overlap, I reconcile definitions to create continuous time series. In all cases, I focus on the primary capital market and generally I collect three statistics: gross and net issues of bonds as well as outstanding amounts. The latter two were usually compiled only years, if not decades, after countries started to publish gross issuance statistics. The baseline series are therefore annual flows of newly issued bonds, in contrast to bank-credit datasets that often measure outstanding stocks, as, for example, in Müller and Verner (2023). This comes with advantages but also drawbacks as discussed below. The degree of detail of capital issuance statistics varies, from very high-level aggregations of a single category of corporations to meticulous categorizations of up to ten or more sub-sectors. Therefore, I construct harmonized, yet often necessarily broader, sector aggregates from the sources. As a starting point and for descriptive visualizations displayed below, I construct a broad sector "Total Corporations". This is an aggregate of all types of corporations from financial to non-financial, private and public.[1] Historical sector labels do not always map perfectly into modern institutional sectors: state-owned enterprises, specialized credit institutions, and other public or quasi-public issuers can blur the economic boundary between government, financial, and non-financial corporations. I retain the closest source-consistent classification and document such cases country by country in Appendix A. While allowing me to display long-run patterns of corporate and government bond issues, I use the functional separation between financial and non-financial corporations for the empirical analysis later on. For France in particular, this composition matters: very high issuance volumes in the 1990s–2000s reflect the dominance of financial corporations and the prevalence of short-maturity instruments.

To preserve the economically relevant sectoral distinction for the crisis analysis, I separate issuance by financial corporations, non-financial corporations, and governments through-

[1]For instance, for the Euro Area countries in the sample, after 1990 the constructed sector category "Total Corporations" is the sum of non-financial corporations and financial corporations, whereas the financial-corporation category is itself the sum of bonds issued by 'deposit-taking corporations except central banks' and 'non-MFI financial institutions'.

out the analysis.[2]

The baseline empirical measure is gross bond issuance rather than net issuance or outstanding amounts. Gross issuance captures the flow of securities brought to the primary market and therefore provides a direct measure of primary-market activity. This is especially relevant during systemic banking crises, when refinancing constraints, redemptions and maturity shortening may lead to large divergences between gross and net flows. Adrian et al. (2013), who use changes in outstanding amounts from the Flow of Funds to study U.S. bank lending and corporate bond finance during the GFC, likewise note that direct data on new credit flows would be preferable. The issuance data collected here directly address this measurement issue.

Gross issuance is also the earliest and most consistently available measure across countries. In many cases, official sources began reporting gross issuance decades before net issuance or outstanding amounts. This makes gross flows particularly well suited to a long-run cross-country panel. Gross issuance should therefore be interpreted as a measure of primary-market financing activity rather than as a measure of the outstanding size of the bond market. The next subsection discusses how variation in maturity structure affects this interpretation.

In addition to corporate bond issuance, I collect government bond issuance because government securities provide a natural public-sector benchmark and may attract investor demand toward relatively safe public claims during periods of stress. This allows me to compare issuance dynamics across three macroeconomic sectors: non-financial corporations, financial corporations, and governments. I focus on central or federal government issuance because these series are most consistently reported across countries and periods.[3]

Series are reported in nominal local currency. When a new currency or denomination is introduced, such as the switch to the new franc in France, I rescale earlier observations using the official conversion factor. When overlapping years are reported in different editions of the same source, I use the most recent vintage and extend the series backward using earlier editions.

The harmonized gross-issuance dataset provides broad but uneven long-run coverage across sectors. Table 1 summarizes the series used in the empirical analysis. Total-corporate issuance is available for 19 countries and 1,536 country-year observations, covering 51 banking-crisis onsets. The sectoral split is more restrictive, with 17 countries and

[2]The precise instrument perimeter of the financial-corporation series follows the underlying source and therefore varies across countries and periods. Bonds and debentures issued by banks and other financial institutions are included when they form part of the reported financial-sector debt-security category. Mortgage and covered bonds are included where the source reports them as bond issues and assigns them to financial issuers. Asset-backed securities are not added uniformly; they enter where they are included in the underlying source aggregate and can be allocated consistently. A uniform historical decomposition into unsecured bank bonds, covered bonds, and securitized instruments is therefore not available. Appendix A documents the country-specific instrument coverage.

[3]A complete country-by-country documentation is provided in Appendix A.

Table 1: Coverage of harmonized gross issuance series

Sector	Countries	Observations	First year	Last year	Crisis
Government	15	1055	1850	2024	37
Financial corporations	17	1157	1850	2024	43
Non-financial corporations	17	1251	1855	2024	46
Total corporations	19	1536	1855	2024	51

Notes: The table reports coverage of the harmonized gross issuance series. Crises are defined using the banking-crisis chronology of Baron et al. (2021). Their crisis indicator is binary and equals one in the onset year of a banking crisis and zero otherwise. The final column reports the number of crisis-onset years for which the corresponding sectoral issuance series is observed.

1,157 observations for financial corporations and 17 countries and 1,251 observations for non-financial corporations. Government issuance covers 15 countries and 1,055 observations. Across all sectoral series, the dataset overlaps with 54 distinct banking-crisis onsets, of which 46 are observed in the NFC series.[4]

The relatively large number of covered crisis onsets reflects both the long historical span of the dataset and repeated banking crises within countries. Crisis dates follow Baron et al. (2021). As a benchmark, Appendix Section 8 compares this chronology with the Jordà–Schularick–Taylor crisis chronology on a common country-year sample. The main NFC issuance response is very similar under the alternative crisis dating.

I combine the bond issuance data with macro-financial statistics from the Global Macro Database by Müller et al. (2025), including GDP, inflation, interest rates, current account balances, investment, and bank credit.

3.2 Maturity Structure and Measurement of Issuance

A critical historical dimension for interpreting long-run bond-issuance flows is the evolution of bond maturity. Maturity structures vary substantially across countries and periods and, in many markets, shortened considerably over the twentieth century. As a result, identical levels of outstanding debt can correspond to very different gross issuance volumes: a large outstanding bond market can coexist with comparatively modest annual gross issuance when securities are issued at long maturities (Michie, 2006; Rajan and Zingales, 2003); see also the long-run U.S. data on outstanding corporate bonds in Giesecke et al. (2011, 2014).

Maturities may also shorten temporarily during periods of financial stress. For a given stock of debt, shorter maturities imply more frequent refinancing and therefore tend to raise gross issuance. Maturity compression alone is therefore unlikely to explain the persistent decline in gross NFC issuance documented below. These considerations matter

[4]The United Kingdom is an important limitation in the historical coverage: I was not able to construct issuance series of comparable quality to those available for countries such as France or the Netherlands.

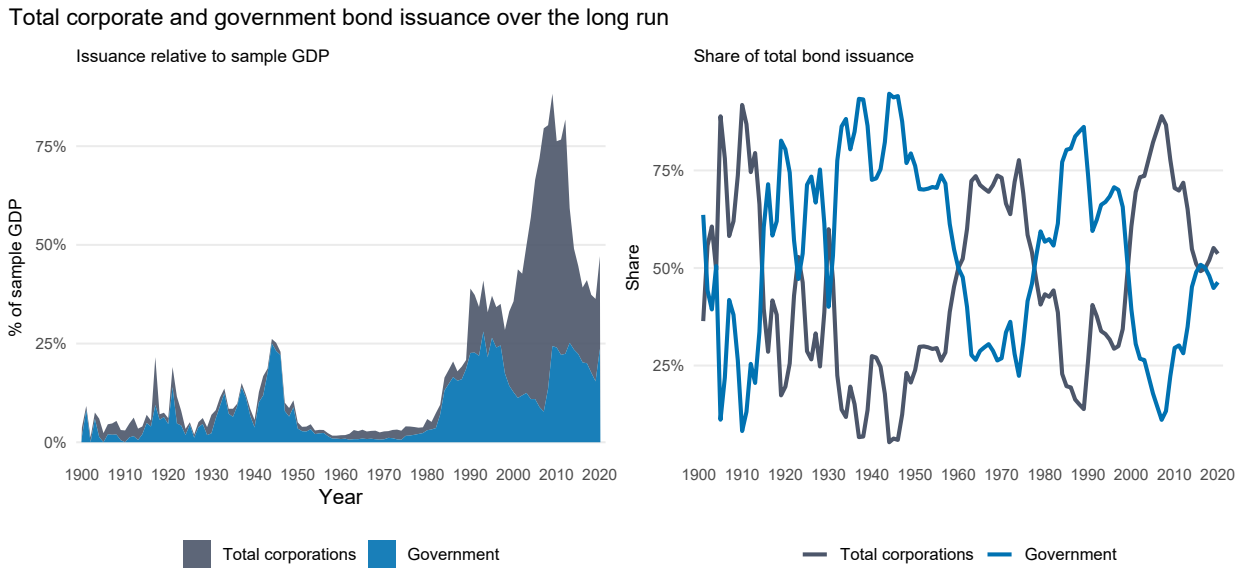


Figure 1: The corporate series is the broad total-corporate aggregate constructed as described in Appendix A. Corporate and government bond gross issues are summed across countries with available data in each year and scaled by the corresponding sum of GDP (ratio-of-sums within the sample). For visualization, the sample is restricted to country-years for which both government and corporate gross issuance are observed. GDP and the gross issues data were converted to USD across countries using the `USDfx` USD exchange rate series from Müller et al. (2025). Series shown are two-year moving averages.

particularly for comparisons of long-run issuance levels. The empirical crisis analysis instead focuses on within-country changes around crisis onsets and interprets gross issuance as a measure of primary-market financing activity and sectoral allocation rather than as a proxy for bond-market size.

3.3 Long-run evolution

To understand how bond markets behave in crises, it is first necessary to understand how primary issuance evolves over the long run. Using newly constructed cross-country data, I document three stylized facts about the size and sectoral allocation of bond markets since 1900. The cross-country descriptive figures begin in 1900, when coverage becomes sufficiently broad for meaningful aggregate comparisons, although individual country series extend further into the nineteenth century. These long-run patterns highlight the time-varying nature of primary issuance and motivate the analysis of how the sectoral composition of primary issuance changes during systemic banking crises.

#1 The size and composition of the primary bond market change substantially over the long run and around major shocks. Figure 1 plots aggregate corporate and government bond gross issuance relative to GDP. Two patterns stand out. First, primary corporate issuance exhibits a pronounced mid-century trough followed by a sharp and sustained post-1980 expansion (Rajan and Zingales, 2003). In this respect, the aggregate flow pattern broadly echoes the post-1980 "big bang" in stock-market capitalization documented by

Kuvshinov and Zimmermann (2022). The sectoral decomposition below, however, shows that this aggregate expansion was highly uneven within the corporate sector: the post-1980 rise was substantially stronger among financial corporations than among NFCs.[5] This pattern is not solely driven by changes in country coverage. Appendix Figure 12 restricts attention to countries observed in the historical, postwar, and modern periods and shows that the postwar trough and subsequent revival remain visible within this fixed long-run cohort.

Second, the composition of issuance varies sharply around major historical shocks. Government bond issuance spikes during wars and systemic crises, temporarily dominating issuance activity and altering the sectoral composition of the primary bond market. This episodic dominance of government issuance highlights that the composition of primary bond market activity changes across sectors during major macro-financial disruptions. Taken together, these patterns indicate that both the size and the sectoral composition of bond issuance are highly time-varying, motivating a closer examination of sector-specific issuance dynamics around financial crises.

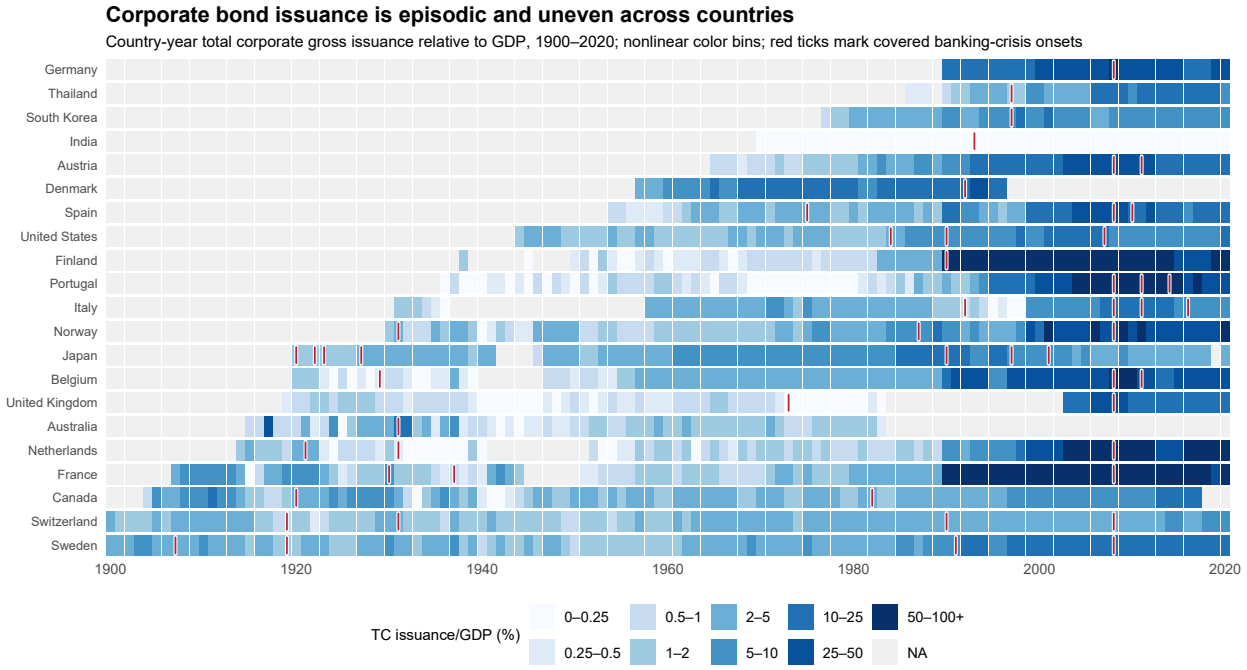


Figure 2: Corporate bond issuance is episodic and uneven across countries. The figure plots total corporate bond gross issuance relative to GDP by country and year from 1900 onward. Darker cells indicate higher issuance intensity. Grey cells denote country-years without observed total corporate gross issuance. Red ticks mark banking-crisis onsets only when total corporate gross issuance is observed for that country-year. The color scale is capped at the 99th percentile for readability. The figure illustrates both the uneven historical coverage of corporate bond issuance data and the concentration of high-issuance episodes in specific countries and periods.

#2 There is substantial and persistent cross-country heterogeneity in the level and

[5]Note that data availability for many countries begins in the interwar period, limiting direct observation of earlier issuance activity.

timing of corporate bond issuance relative to GDP. While aggregate patterns are informative, they mask substantial cross-country heterogeneity. Figure 2 plots total corporate bond gross issuance relative to GDP for individual countries. Issuance levels, timing, and volatility differ markedly across countries, with countries with more recently developed corporate bond markets, such as Thailand and South Korea, reaching issuance intensities comparable to those of long-established bond markets in recent decades. This persistent heterogeneity in issuance levels reinforces the importance of focusing on within-country changes around crises rather than cross-sectional level differences, which may reflect institutional structure, financial development, or regulatory regimes.

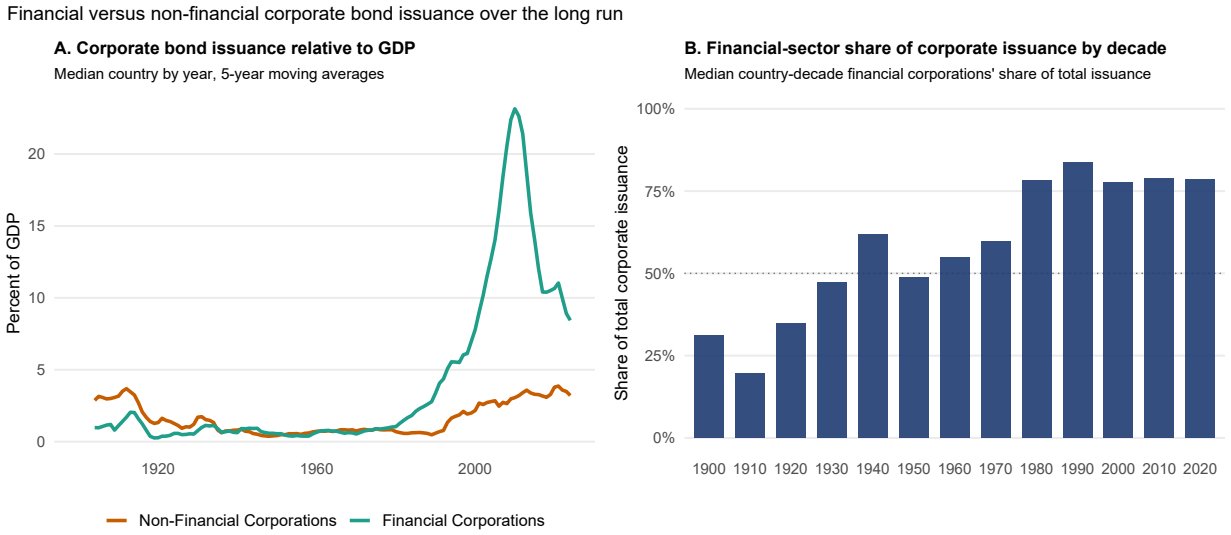


Figure 3: Financial versus non-financial corporate bond issuance over the long run. Panel A plots median-country gross bond issuance relative to GDP for financial corporations (FC) and non-financial corporations (NFC), restricting each year to countries for which both sectoral series are observed. For each year, the figure first computes issuance relative to GDP at the country level and then takes the cross-country median; series are shown as 5-year moving averages. Panel B reports the median country-decade share of FC issuance in total corporate bond issuance, where total corporate issuance is the sum of FC and NFC bond gross issuance. A value above 50 percent indicates that financial corporations account for the majority of corporate bond issuance.

#3 The post-1980 revival in corporate bond issuance was disproportionately concentrated among financial corporations. Aggregate corporate bond issuance masks important heterogeneity within the corporate sector. Figure 3 shows that FC and NFC issuance followed markedly different long-run trajectories. NFC issuance displays a more pronounced U-shaped pattern, with substantial prewar activity, a mid-century contraction, and a subsequent recovery (Rajan and Zingales, 2003). Financial-corporation issuance, by contrast, expands much more strongly after 1980. Panel B shows the same shift in relative terms: the median country-decade FC share of corporate issuance rises above one half in the postwar period and becomes especially large from the 1980s onward. This pattern is consistent with a broader shift toward market-based funding within the financial sector: an increasing part of securities-market financing reflects financial institutions themselves

issuing marketable debt. This distinction matters for the crisis analysis below, because growth in aggregate corporate bond markets need not imply a corresponding expansion in market finance available directly to non-financial firms.

Taken together, the descriptive evidence shows that primary bond issuance is highly time-varying in both scale and composition. This motivates the systematic analysis of banking-crisis episodes below. I focus on within-country dynamics around crisis onsets and ask whether NFC issuance expands to substitute for impaired bank lending or instead contracts. Government issuance provides a public-sector benchmark for how the composition of primary-market activity changes during these episodes.

4 Bond Issuance Dynamics and Financial Crises

This section examines how primary bond issuance evolves around systemic banking-crisis onsets. The central question is whether aggregate non-financial corporate issuance expands when the banking system is under stress, as a broad spare-tire interpretation would suggest. I first document sectoral issuance dynamics around crisis onsets using event-study evidence and then estimate cumulative responses in a long-difference local-projection framework.

The main result is a pronounced divergence across issuing sectors. Non-financial corporate bond issuance contracts persistently following banking crises, while government issuance rises and financial-corporation issuance increases temporarily before declining at medium horizons. These estimates describe aggregate sectoral financing dynamics around realized banking crises; they do not identify within-firm substitution or the causal effect of an exogenous contraction in bank-credit supply.

4.1 Primary Bond Markets Around Financial Crises

First, I study the dynamics of primary bond market activity around systemic banking crises using an event-study design. This allows me to provide a descriptive characterization of how bond issuance evolves in the run-up to, and in the aftermath of, banking crises, separately for non-financial corporations, financial corporations, and governments. I estimate cumulative changes in issuance relative to the year immediately preceding crisis onset and trace these deviations over a symmetric five-year window. This normalization ensures that all estimated responses can be interpreted as deviations from a common pre-crisis baseline, focusing on within-country changes around crises.

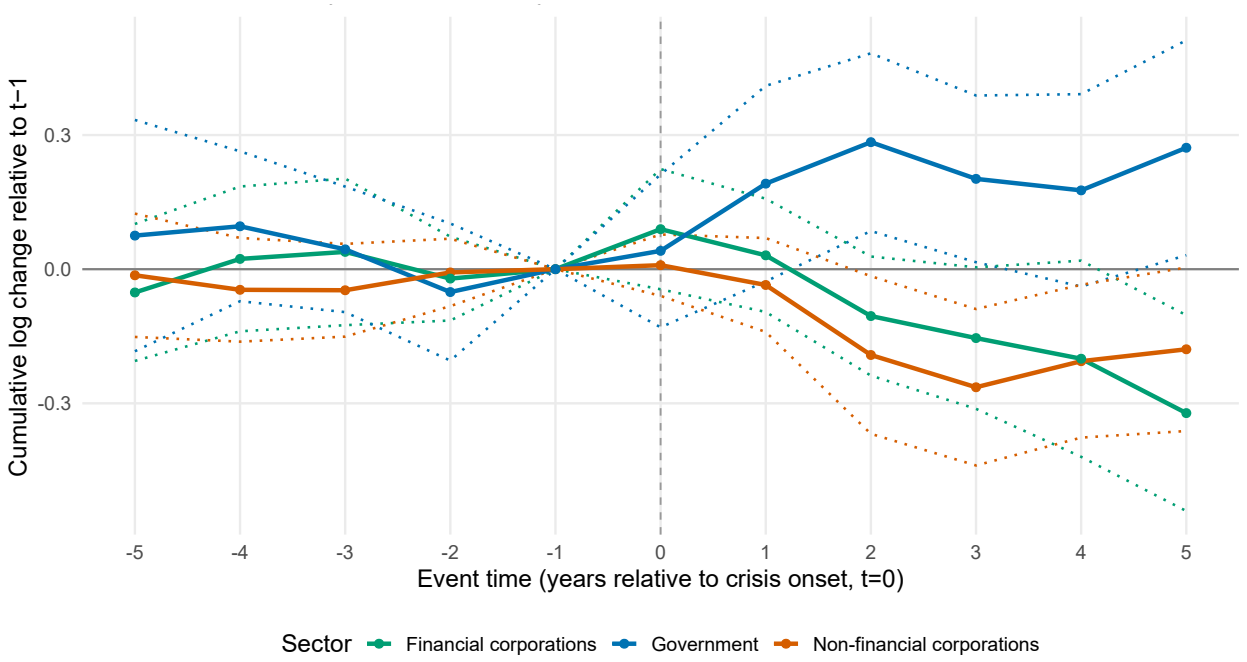
For each horizon $h \in \{-5, \dots, 5\}$, I estimate the following horizon-specific regression:

$$y_{i,t+h} - y_{i,t-1} \;=\; \alpha_i \;+\; \beta_h \, 1\{\text{Banking Crisis}_{i,t} = 1\} \;+\; \varepsilon_{i,t+h}, \tag{1}$$

where $y_{i,t}$ denotes the log of gross bond issuance relative to GDP in country i and year t, and $1\{\text{Banking Crisis}_{i,t}\}$ is an indicator equal to one in the onset year of a systemic banking crisis. Country fixed effects α_i absorb time-invariant cross-country differences in average issuance levels. Each horizon h is estimated separately using a fixed-effects panel regression.

By construction, $(t-1)$ is normalized to zero, so β_h measures the cumulative deviation in issuance h years from crisis onset relative to its pre-crisis level. Standard errors are computed using Driscoll and Kraay (1998) standard errors, allowing for both serial correlation and cross-sectional dependence. Confidence intervals are reported at the 90 percent level.

Figure 4: Bond issuance dynamics around banking crises



Notes: Event study of bond issuance around banking crises as described by the setup in Equation 1. The figure shows cumulative changes in log bond issuance relative to GDP, normalized to zero in the year prior to crisis onset $(t-1)$. Event time $h = 0$ corresponds to the crisis year. Estimates are obtained from horizon-specific regressions with country fixed effects. Dotted lines indicate 90 percent confidence intervals based on Driscoll and Kraay (1998) standard errors with horizon-dependent bandwidths.

Figure 4 summarizes the dynamic response of sectoral bond issuance around systemic banking crises following Equation 1. Government bond issuance rises sharply at the onset of a banking crisis and remains elevated in the years that follow, consistent with increased public borrowing during periods of financial distress. Financial corporations expand issuance around crisis onsets, but the increase is modest relative to governments and fades at medium horizons. In contrast, non-financial corporate bond issuance contracts markedly following crisis onset. Importantly, there is no evidence of an increase in NFC bond issuance even in the crisis year itself ($h = 0$) and issuance remains persistently subdued relative to its pre-crisis level throughout the post-crisis window. Appendix Figure 22 repli-

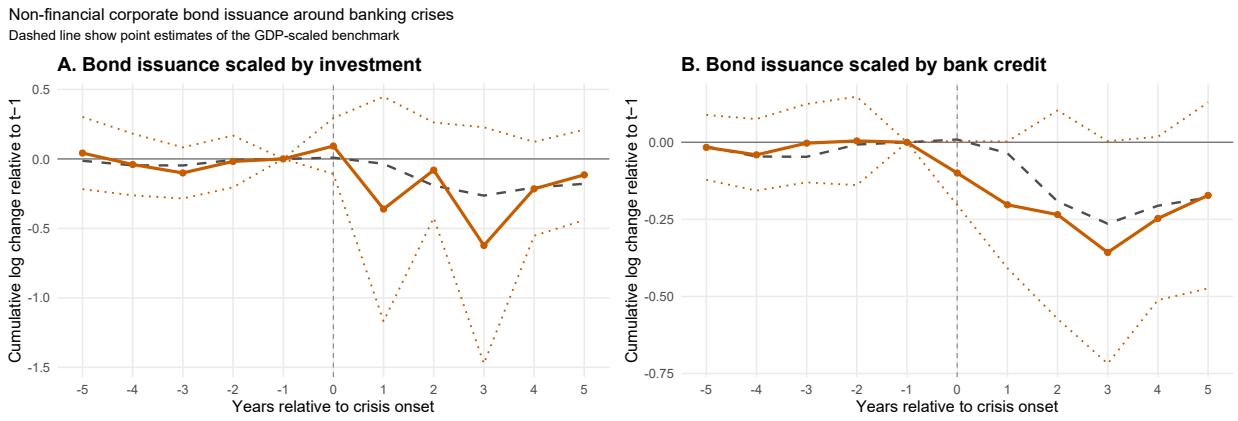


Figure 5: Non-financial corporate bond issuance around banking crises: alternative normalizations. The figure reports event-study estimates for non-financial corporate (NFC) bond issuance using two alternative normalizations. Panel A scales issuance by gross fixed capital formation. Panel B scales issuance by the stock of corporate bank credit, using lagged values to avoid contemporaneous mechanical relationships with issuance. The dashed grey line reproduces the GDP-scaled benchmark. All panels are normalized to zero in the year prior to crisis onset ($t-1$), so the plotted coefficients measure cumulative deviations from the pre-crisis baseline. Event time $t = 0$ denotes the crisis year. Estimates are obtained from horizon-specific country fixed-effects regressions around banking-crisis onsets. Dotted lines indicate 90% confidence intervals based on Driscoll and Kraay (1998) standard errors with horizon-dependent bandwidths.

cates the event study using log issuance in raw units and the sectoral patterns are highly similar in raw units, reinforcing that the NFC contraction is not purely a scaling artifact of GDP movements. Taken together, the event-study evidence shows sharply divergent sectoral responses around banking crises: NFC issuance contracts while government issuance rises and financial-corporation issuance increases initially. The composition of primary bond-market activity therefore shifts substantially following crisis onset.

A natural concern is that lower NFC bond issuance may simply move proportionally with the broader contraction in corporate activity during crises. Figure 5 therefore reports the same event-study design using alternative normalizations of NFC issuance. In addition to issuance relative to GDP, I scale issuance by gross fixed capital formation and by the lagged stock of corporate bank credit. These normalizations do not identify financing demand or market access separately. Rather, they ask whether the fall in bond issuance is approximately proportional to economically relevant measures of the scale of corporate financing activity.

Two results stand out. First, NFC issuance declines not only relative to GDP but also relative to fixed investment, so the contraction is larger than would be implied by a simple proportional decline in investment. Second, issuance also falls relative to the stock of corporate bank credit. Thus, bond issuance does not expand relative to the scale of bank intermediation. These exercises do not establish whether the underlying mechanism operates through financing demand, market access, or issuer composition, but they show that the aggregate NFC issuance contraction is not a mechanical consequence of scaling by contemporaneously falling GDP. Appendix Figure 23 shows that the same qualitative

pattern holds across sectors when issuance is normalized by fixed investment or lagged corporate bank credit.

4.2 Bond issuance following banking crises: systematic evidence

While the event-study evidence provides a first indication of issuance dynamics around banking crises, this section provides more systematic evidence on the issuance patterns following banking crises using long-difference local projections. This approach is particularly well suited for crisis events, where the object of interest is the total deviation from the pre-crisis baseline rather than short-run year-to-year fluctuations.

The long-difference specification of the local projection framework follows Jordà (2005) and in particular Jordà and Taylor (2025), with the dependent variable defined as the cumulative change in bond issuance relative to the year immediately preceding the crisis. For each horizon $h \in \{0, \dots, H\}$, I estimate:

$$100 \times \left(y_{i,t+h} - y_{i,t-1}\right) \;=\; \beta_h \,\text{Crisis}_{i,t} \;+\; \Gamma_h' X_{i,t-1} \;+\; \alpha_i \;+\; \varepsilon_{i,t+h}, \tag{2}$$

where $y_{i,t}$ is either the log of raw issues or the log of issues relative to GDP, depending on the exact specification. The baseline estimation uses log gross issuance rather than issuance scaled by GDP. This choice reflects the object of interest: fluctuations in primary financing activity itself. Scaling by GDP can mechanically mix issuance dynamics with contemporaneous output movements during crises. I therefore use raw issuance as the baseline and then show that the results are robust to normalizing issuance by GDP, fixed investment, and the pre-crisis bank-credit base. The indicator $\text{Crisis}_{i,t}$ equals one in the onset year of a systemic banking crisis, based on the data in Baron et al. (2021). Country fixed effects α_i absorb time-invariant differences in average issuance across countries. The vector $X_{i,t-1}$ includes lags of the dependent variable as well as standard macroeconomic controls, real GDP growth, inflation, the short-term interest rate, and the current account balance, each lagged one and two periods in baseline. The long-difference specification ensures that all estimated responses are interpreted as cumulative deviations relative to the pre-crisis baseline year $t-1$. Estimation is conducted separately for each horizon h. I use Driscoll and Kraay (1998) standard errors, allowing for serial correlation and cross-sectional dependence. To limit the influence of extreme observations, the baseline results apply a symmetric winsorization at the 2% threshold to the issuance data before taking logs.

The crisis indicator should not be interpreted as an exogenous banking-sector shock. Systemic banking crises may themselves arise from broader macroeconomic disturbances that simultaneously affect firms' financing demand and conditions in securities markets. The estimates therefore describe conditional issuance dynamics around realized banking-crisis onsets rather than the causal effect of an exogenous contraction in bank-credit sup-

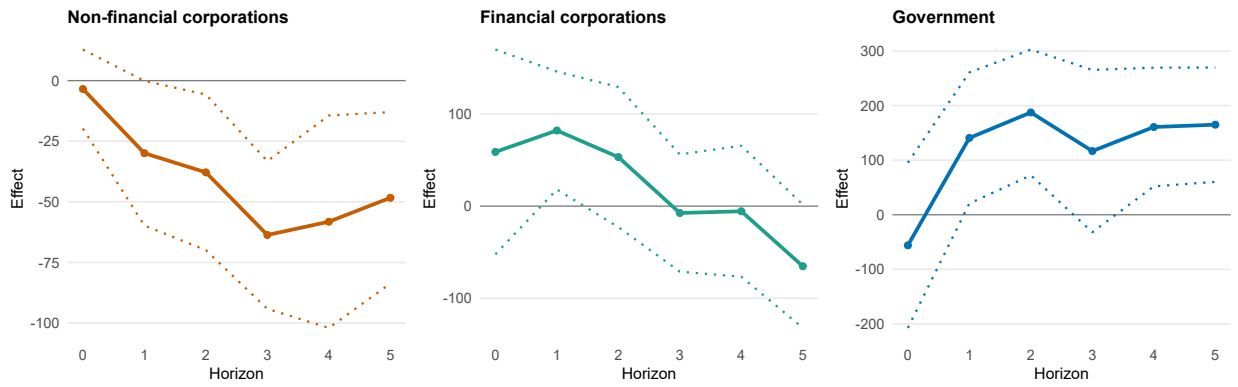


Figure 6: Long-difference local projection estimates of the cumulative response of bond issuance to banking crises. The figure reports the baseline specification based on Equation 2 separately for non-financial corporations, financial corporations, and government issuers. The dependent variable is $100 \times (y_{i,t+h} - y_{i,t-1})$, where $y_{i,t}$ denotes the log of gross bond issuance. Horizon $h = 0$ corresponds to the crisis-onset year, and all responses are measured relative to the pre-crisis year $t - 1$. All specifications include country fixed effects, two lags of the dependent variable, and two lags of macroeconomic controls: real GDP growth, inflation, the short-term interest rate, and the current account balance. The estimation uses Driscoll and Kraay (1998) standard errors, allowing for serial correlation and cross-sectional dependence. Dotted lines denote 90% confidence intervals. Issuance series are winsorized symmetrically at the 2% level prior to log transformation. Vertical scales differ across panels to highlight within-sector dynamics.

ply. The lagged macroeconomic controls, alternative normalizations, and comparisons with non-crisis observations help discipline this interpretation but do not eliminate this identification limitation.

Table 2: Baseline long-difference local projection estimates

Sector	$h = 1$	$h = 3$	$h = 5$
Non-financial corporations	-38.1*	-63.7***	-48.0**
	(19.9)	(18.5)	(23.8)
	N=988, C=16	N=952, C=16	N=918, C=16
Financial corporations	83.9**	-13.3	-80.4*
	(40.0)	(42.9)	(45.0)
	N=945, C=16	N=909, C=16	N=875, C=16
Government	140.8*	116.8	165.1***
	(73.0)	(90.4)	(63.7)
	N=858, C=15	N=822, C=15	N=791, C=15

Notes: The dependent variable is $100 \times [\log(\text{Gross Issuance}_{i,t+h}) - \log(\text{Gross Issuance}_{i,t-1})]$. All specifications include country fixed effects, two lags of the dependent variable, and two lags of macroeconomic controls. Standard errors are Driscoll–Kraay. Parentheses report standard errors. N reports number of observations and C number of countries. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 6 summarizes the baseline long-difference local projection estimates for bond issuance following systemic banking crises, using the log of raw issuance as the dependent variable and the full sample baseline model as presented in Equation 2.

Government bond issuance rises sharply following crisis onset and remains persistently elevated relative to the pre-crisis baseline. This pattern is consistent with increased

public borrowing to stabilize the financial system and support economic activity during and after banking crises. Financial corporations also expand bond issuance in the aftermath of crises, yet to a much smaller extent than governments, and issuance by financial firms declines again at longer horizons. In contrast, non-financial corporate bond issuance contracts significantly following crisis onset. Notably, there is no evidence of increasing NFC bond issuance even in the crisis year itself. Issuance remains persistently below its pre-crisis level throughout the post-crisis window. Aggregate NFC bond issuance therefore does not display the expansion that a broad sector-level spare-tire mechanism would imply. At $h = 1$, NFC issuance is 38 log points below its pre-crisis level; by $h = 3$, the decline reaches 64 log points, corresponding to roughly a one-half reduction in issuance. The contraction is also visible in both pre- and post-1945 crisis episodes (Appendix Section 8.2.8) and is not driven by countries with unusually long historical coverage.

In sum, the event-study and long-difference local projection evidence rejects a broad aggregate "spare tire" pattern in primary corporate bond markets. Instead of a crisis-period expansion in NFC issuance, crises coincide with persistent contraction in NFC primary issuance and a sharp increase in government issuance. The composition of primary-market activity therefore shifts strongly toward the public sector rather than displaying broad corporate substitution.

The results are robust to a wide range of specification and sample checks reported in Section 8, including alternative dependent-variable definitions, lag structures, winsorization choices, and alternative banking-crisis chronologies. Year fixed effects leave the central NFC contraction intact, although the government and FC responses are more sensitive to common global shocks. The medium-run decline in non-financial corporate issuance remains quantitatively similar across these specifications and is not driven by historical sample composition, individual countries, or specific crisis episodes. Allowing responses to differ across pre- and post-WWII crises likewise yields similar qualitative dynamics.

Interpretation. These estimates describe sector-level issuance dynamics and should not be interpreted as within-firm financing choices. Aggregate historical data cannot separate changes in issuance by continuing firms from changes in the set of firms accessing the market. The contraction therefore remains compatible with some firms retaining or increasing bond-market access during crises, as documented in modern firm-level studies. What the aggregate estimates establish is narrower: these margins are not sufficiently large to produce an expansion in NFC primary issuance at the sector level.

As discussed in Section 3.2, crisis-period maturity shortening would, for a given debt stock, tend to increase refinancing-related gross issuance and therefore cannot mechanically account for the observed NFC contraction.

Economic magnitude and crisis specificity. While the local projections establish that banking crises are followed by persistent declines in NFC bond issuance, an important question is whether these responses are economically large relative to normal fluctuations

in issuance activity. Figure 7 addresses this question by comparing cumulative NFC issuance growth at banking-crisis onsets to the same horizon-specific growth object computed over clean non-crisis country-years.

Two patterns emerge. First, NFC issuance growth following banking crises is substantially weaker than outside crises. While cumulative issuance growth in non-crisis periods remains positive on average, crisis episodes exhibit pronounced negative cumulative growth relative to the pre-crisis baseline, particularly at horizons $h = 1$ to $h = 4$. Second, the magnitude of these declines is large relative to ordinary fluctuations in issuance activity and lasts for several years, suggesting that banking crises are associated with unusually weak corporate bond issuance dynamics.

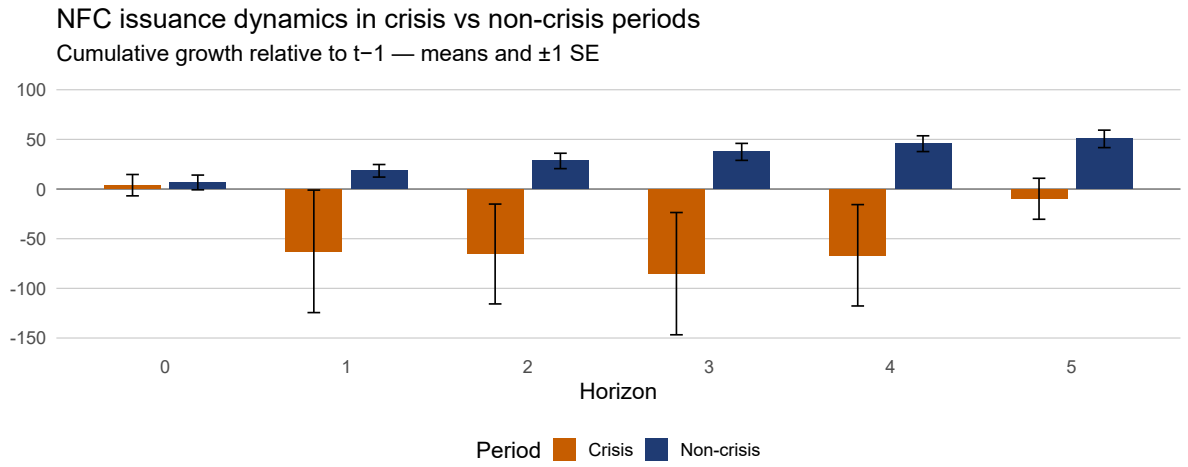


Figure 7: NFC issuance growth in crisis and non-crisis periods. Bars report mean cumulative NFC bond issuance growth relative to the pre-period ($t - 1$) in banking-crisis onset years and in non-crisis country-years. For horizon h, the plotted object is the log of bond issues relative to $t - 1$. Non-crisis observations exclude banking-crisis onsets over horizons $h = 0, \ldots, 5$. Whiskers denote ±1 standard error. The figure is descriptive and does not estimate local projections.

Table 3 reports the corresponding numerical values. By year 3, cumulative NFC issuance growth during banking crises is 112 log points below comparable non-crisis episodes, highlighting the economic significance of the estimated responses.

Table 3: Magnitude of NFC issuance dynamics in crisis and non-crisis periods

Horizon	Crisis mean	Non-crisis mean	Difference	Crisis N	Non-crisis N
1 year	-50.8	20.3	-71.2	39	882
3 years	-74.4	37.3	-111.7	39	846
5 years	-2.0	56.0	-58.0	39	817

Notes: Crisis observations are banking-crisis onset years. Non-crisis observations are clean country-years outside banking-crisis onset windows. Entries report cumulative changes in log NFC gross issuance relative to $t - 1$, multiplied by 100 and therefore expressed in log points.

4.3 Interpreting the aggregate issuance contraction

The aggregate quantity response does not by itself identify why NFC bond issuance contracts after banking crises. Lower issuance may reflect weaker financing demand, tighter access to primary debt markets, changes in the composition of issuers, or combinations of these mechanisms. Nor do the aggregate estimates rule out substitution toward bonds by individual firms that retain market access. They establish a narrower result: whatever substitution occurs at the firm level is not sufficiently broad to generate an expansion in aggregate NFC primary issuance.

This distinction is important when relating the results to evidence from recent crises. Adrian et al. (2013) document substitution from bank loans toward bond finance among U.S. firms with market access during the Global Financial Crisis. Yet this adjustment was neither immediate nor universal. Becker and Benmelech (2025) show that both primary bond issuance and syndicated lending initially contracted sharply after Lehman, with the bond market reopening earlier and disproportionately for higher-quality issuers. International evidence likewise suggests that firms can switch across debt markets when alternative markets remain accessible (Cortina et al., 2021), while aggregate corporate bond issuance declined across many markets during the GFC and expanded much more broadly during the unusually policy-supported COVID-19 episode (Bruno et al., 2024). Firm-level substitution and aggregate market resilience are therefore distinct empirical objects.

I use three pieces of evidence to discipline the interpretation of the aggregate contraction. First, the alternative normalizations presented in Figure 5 show that issuance falls not only relative to GDP but also relative to fixed investment and the pre-crisis stock of corporate bank credit. Appendix Figure 23 documents alternative normalizations for all sectors. t These comparisons do not identify financing demand or market access separately, but they show that the decline is disproportionate to several relevant measures of the scale of corporate financing activity. Second, security-level evidence from Australia illustrates how an aggregate contraction can arise along the extensive margin even when some established issuers continue to place bonds. Third, long-run corporate credit spreads provide a price-side diagnostic of whether deeper issuance contractions coincide with more severe repricing in risky debt markets.

These exercises should therefore be viewed as complementary diagnostics rather than as a structural decomposition of the aggregate response. The Australian evidence documents one concrete historical mechanism, while the spread evidence asks whether quantity contractions are systematically more severe when debt-market conditions deteriorate. Together, they help distinguish a simple aggregate decline in financing activity from a broader contraction that coincides with reduced issuer participation and substantial credit-market repricing.

4.4 Inside the aggregate: security-level evidence from Australia's 1931 banking panic

The aggregate results do not imply that bond markets close completely during banking crises, nor that every firm loses market access. A contraction in sector-level issuance can coexist with continued issuance by a subset of firms. Australia provides a useful diagnostic case because the historical aggregate series can be reconstructed from the underlying individual bond issues compiled by Black et al. (2013). The security-level records identify the issuer, issuer type, issue amount, and maturity, allowing me to separate private non-financial corporations (PNFCs) from non-financial public trading enterprises (PTEs), such as state-owned utilities and transport companies operated on a commercial basis, and to decompose private issuance into the number of active issuers, issuance frequency, issue size, and maturity. In the cross-country dataset, PTEs remain part of the NFC sector, following the source classification applied throughout. Here I instead open up the aggregate to examine how issuer participation and contract terms changed during the financial distress of the early 1930s.

The episode took place against a broader external-financing and depression shock. Australia had relied heavily on overseas borrowing during the 1920s, and the retrenchment of international lending from 1928 onward forced greater reliance on short-term finance. Falling export prices subsequently worsened the balance of payments and intensified domestic financial pressure. Contemporary and later accounts summarized by Baron et al. (2021) describe runs on several banks during this period and date the banking panic to April 1931, when the Government Savings Bank of New South Wales failed. At the same time, Australia did not experience the generalized banking collapse observed in several continental European economies: Baron et al. (2021) record neither a cumulative bank-equity decline above 30 percent nor widespread bank failures, and Australia enters their crisis chronology through the panic criterion.[6] Australia therefore provides a useful but deliberately limited case study. The episode involved a banking panic and severe financial distress, but not a generalized banking collapse, and it should not be interpreted as a clean shock to bank-credit supply. Rather, it allows me to examine how aggregate bond issuance, issuer participation, and maturity evolved within a broader depression and financing shock.

Australia is also informative because the annual NFC aggregate initially appears to provide a striking counterexample to the average cross-country result. NFC gross issuance rises from AUD 60.0 million in 1930 to AUD 188.5 million in 1931. Panel A of Figure 8, however, shows that this annual increase is almost entirely a matter of timing and issuer composition. About 96 percent of all NFC issuance recorded in 1931 occurred in the first

[6] Baron et al. (2021) classify Australia 1931 as a panic banking crisis but not as a bank-equity crisis. Their bank-equity index declines by approximately 23 percent, below their 30 percent threshold, and they do not code widespread bank failures. Their narrative evidence dates the panic to April 1931.

quarter, before the April banking panic. Two issues by the Melbourne Harbour Trust alone account for AUD 179.5 million, or about 95 percent of full-year NFC issuance. The remaining three quarters contain only about AUD 7 million of NFC issuance. An annual observation that therefore appears to show a large crisis-period issuance expansion does not represent a broad expansion in private-firm bond issuance after the banking panic.

The picture is very different for private non-financial corporations. Average annual PNFC issuance falls from AUD 16.6 million in 1927–1930 to AUD 5.2 million in 1931–1933, a decline of 69 percent. Panel B shows that almost the entire contraction occurs along the extensive margin. The average number of active private issuers falls from 8.5 to 2.7 per year, also a decline of about 69 percent. By contrast, the number of issues per active issuer changes little, from 2.26 to 2.13, while average issue size slightly increases. The contraction therefore occurs primarily through a fall in the number of firms issuing bonds rather than through smaller or less frequent issues among continuing issuers.

Continued issuance is also highly concentrated among firms that were already important issuers before the crisis. Of the 16 private firms observed issuing bonds during 1927–1930, only four issue again during 1931–1933. These four firms had already accounted for about 89 percent of pre-crisis PNFC issuance. At the same time, the market was not literally closed: two issuers first observed issuing during 1931–1933 enter the data, although together they account for only about 9 percent of post-crisis private issuance. The crisis period therefore combines continued issuance by a small group of established firms with a sharp decline in the number of active private issuers. Appendix 7.7 documents the issuer distribution and shows that this result is robust to excluding Broken Hill Proprietary, the dominant pre-crisis issuer.

Contract terms also shift sharply. Panel C shows that the share of private issuance with maturity above ten years falls from 59 percent in 1927–1930 to less than 3 percent in 1931–1933. The amount-weighted median maturity declines from 10.6 to 7.9 years, while the share of issuance with maturity of five years or less rises from 21 to 34 percent. The main change is therefore not simply a shift toward shorter securities: long-maturity private issuance nearly disappears. Annual diagnostics in the appendix show that the share of issuance above ten years falls to 6.6 percent in 1931 and to zero in 1932–1933 before rebounding in 1934, suggesting that the maturity compression is concentrated around the crisis period rather than reflecting a smooth secular trend.

The Australian episode should not be interpreted as a causal event study around the exact date of the April 1931 panic. External financing conditions, export prices, and domestic financial conditions had already deteriorated before the panic, and the historical issue dates are observed only at quarterly frequency. The April panic is therefore best viewed as a marker within an unfolding depression rather than as a sharp treatment date.

Its role is diagnostic. The episode shows one concrete way in which continued bond-market activity for some firms can coexist with a sharp contraction in aggregate private

issuance: the number of active issuers falls substantially, while issuance frequency and average issue size among continuing issuers change much less, and long-maturity financing nearly disappears. This pattern is consistent with the aggregate cross-country evidence, but it is not intended as a unique explanation for all banking crises. The next subsection returns to the broader international sample and asks whether larger issuance contractions are systematically associated with more severe credit-market repricing.

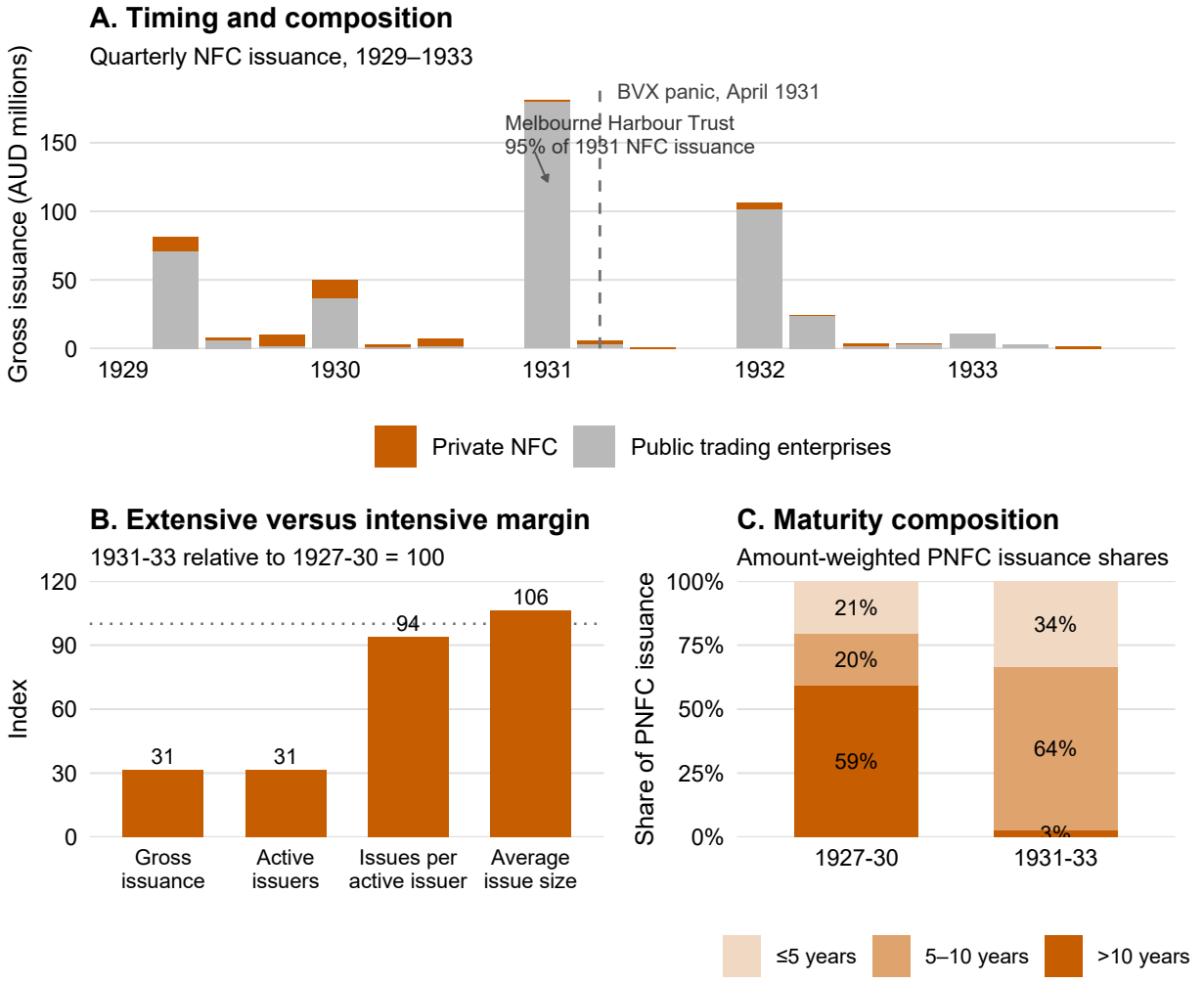


Figure 8: Inside aggregate corporate bond issuance around the Australian banking crisis of 1931. Panel A reports quarterly gross bond issuance by private non-financial corporations (PNFCs) and non-financial public trading enterprises (PTEs). The dashed line marks the banking panic of April 1931. Historical issue dates are recorded at quarterly frequency, so the figure distinguishes issuance recorded in Q1 from issuance in Q2–Q4 but does not assign exact within-quarter timing. Panel B decomposes PNFC issuance in 1931–1933 relative to 1927–1930, with the pre-crisis period normalized to 100, into gross issuance, the number of active issuers, issues per active issuer, and average issue size. Panel C reports the amount-weighted maturity composition of PNFC bond issuance. Maturity calculations are restricted to securities with recorded original maturity of at least one year. PTEs are included in the NFC aggregate used elsewhere in the paper, following the classification of the underlying source, but are separated here to expose the composition of aggregate issuance. Source: Black et al. (2013) underlying security-level bond issuance data and author's calculations.

4.5 Credit-market repricing and the severity of the issuance contraction

The Australian evidence shows one historical episode in which aggregate issuance contracts alongside a sharp decline in issuer participation. I now return to the international sample and ask a complementary question: are larger NFC issuance contractions systematically associated with more severe repricing in risky debt markets?

To measure credit-market repricing, I use the long-run country-level spread data of Krishnamurthy and Muir (2025). Their historical series are constructed from bond prices and yields, while the modern observations combine corporate bond spreads from central banks and commercial data providers. The precise spread object therefore differs across countries and periods. I use the authors' country-normalized spread series and interpret it as a broad measure of risky-debt-market repricing rather than as an issuer-specific NFC borrowing rate.[7] Because historical spread coverage is much narrower than issuance coverage, I treat this exercise as supporting rather than stand-alone evidence.

The economic interpretation of these spreads is deliberately broad. In the structural framework of Merton (1974), risky corporate debt can be viewed as risk-free debt net of a put option on firm assets, so declines in asset values or increases in asset volatility raise default risk and widen credit spreads even if the risk-free rate is unchanged. More generally, observed spreads also incorporate risk premia, liquidity conditions, and investor risk-bearing capacity. Crisis-period spread increases therefore provide a useful measure of the extent of risky-debt repricing, but not a structural measure of the supply of bond finance.

For each horizon, I augment the baseline long-difference local projection with an interaction between the banking-crisis indicator and the normalized spread change at crisis onset:

$$100\big(y_{i,t+h} - y_{i,t-1}\big) = \beta_h \,\mathrm{Crisis}_{i,t} + \theta_h \,\mathrm{Crisis}_{i,t}\big(\Delta s_{i,t} - \overline{\Delta s}_C\big) + \Gamma_h' X_{i,t-1} + \alpha_i + \varepsilon_{i,t+h}. \tag{3}$$

Here $y_{i,t}$ is the log of NFC gross bond issuance, $\Delta s_{i,t}$ is the author-supplied normalized credit-spread change from $t-1$ to t, and $\overline{\Delta s}_C$ is its mean among observed crisis onsets. The control set, two-percent winsorization, country fixed effects, and Driscoll–Kraay inference are identical to the baseline specification. Centering the interaction implies that β_h gives the estimated crisis response at the mean observed spread change, while θ_h measures how the issuance response varies with the extent of contemporaneous credit-market repricing.

Figure 9 summarizes the results. Panel A divides the constant-origin price sample

[7] The strict price sample excludes the Swedish series from 1987 onward because that segment measures the spread on bank loans to non-financial firms rather than a bond-market spread. The historical spread series also cover a broader set of risky securities, including railroads, banks, and sovereign or quasi-sovereign issuers, and are therefore not NFC-only borrowing rates. Appendix D documents the construction, sample restrictions, and observable differences between crisis episodes with and without usable spread data.

into low- and high-repricing terciles. NFC issuance is comparatively resilient following low-repricing crises but contracts sharply following high-repricing crises. At $h = 3$, the estimated response is approximately 31 log points in the low-repricing group and −84 log points in the high-repricing group. The high-minus-low difference is therefore −116 log points, with a p-value of 0.006.

Panel B displays the corresponding crisis-level price–quantity relationship at $h = 3$. Across the 18 crisis onsets with usable spread data, larger spread increases are associated with markedly weaker subsequent NFC issuance: the crisis-level slope is −43.4 (standard error 12.1), and the Spearman rank correlation is −0.67. The scatter is descriptive and does not replace the controlled local projection. The continuous interaction specification in Equation (3) yields the same qualitative result: a one-standard-deviation larger spread increase is associated with an additional 38 log-point contraction in NFC issuance at $h = 3$ and an additional 32 log-point contraction at $h = 5$.

Appendix D shows that the negative spread gradient is also present when issuance is measured relative to GDP, fixed investment, and the lagged stock of corporate bank credit. Thus, the price–quantity association is not specific to the baseline normalization. At the same time, the relationship should not be interpreted as identifying a structural inward shift in bond supply. Wider spreads may reflect weaker borrower fundamentals, higher expected default losses, liquidity deterioration, shifts in investor risk-bearing capacity, or simply a more severe underlying macro-financial crisis. In particular, crises with larger spread increases may also be episodes with deeper output and investment contractions. The evidence therefore establishes a conditional association: the aggregate NFC issuance contraction is substantially larger in crises accompanied by severe risky-debt-market repricing.

4.6 Bond-market depth and post-crisis real recovery

The previous sections show that non-financial corporate bond issuance contracts on average after banking crises. This aggregate contraction does not imply that bond markets are irrelevant for real outcomes. A distinct question is whether countries with deeper NFC bond markets before a crisis subsequently experience stronger real recovery. Motivated by Krishnamurthy and Muir (2025), I examine the relationship between predetermined bond-market depth and post-crisis GDP and investment dynamics across banking-crisis episodes.

A crisis-specific spare-tire mechanism delivers a sharper prediction than a simple positive association between bond-market development and real activity. If deeper bond markets help firms substitute away from impaired bank intermediation, the relationship between bond-market depth and subsequent investment should become more positive when the banking system is under stress. By contrast, a positive depth–investment relationship

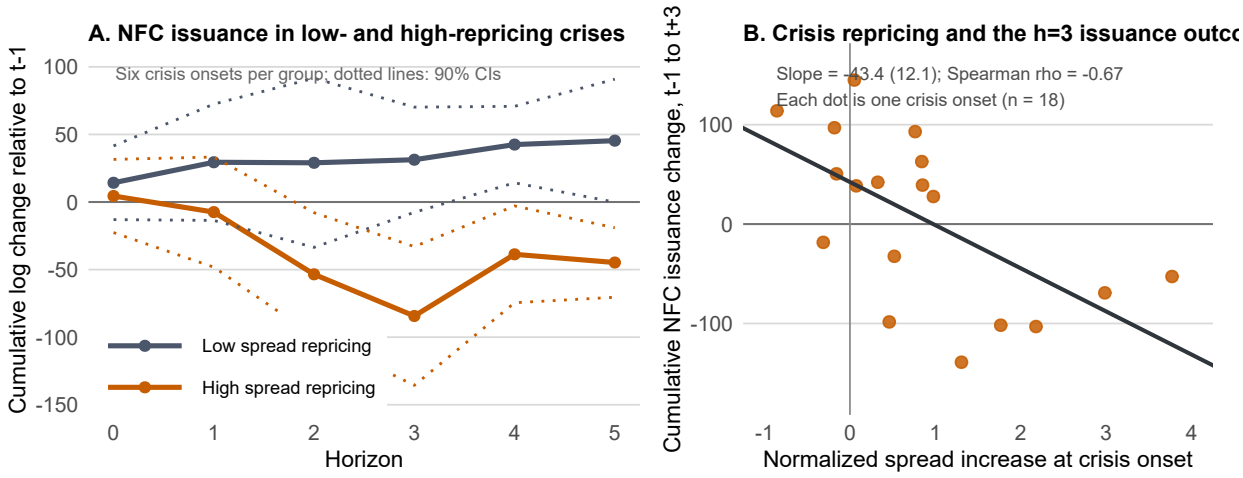


Figure 9: Credit-market repricing and the severity of NFC issuance contractions. Panel A reports jointly estimated baseline LDLP responses for the low- and high-repricing terciles of the exact constant-origin spread sample; each group contains six crisis onsets. Dotted lines denote 90% Driscoll–Kraay confidence intervals. Panel B plots the raw cumulative NFC issuance change from $t-1$ to $t+3$ against the author-supplied normalized spread change for the 18 crisis onsets in the exact price sample. The fitted line reports the crisis-level linear association; standard errors are clustered by country. The raw scatter is descriptive, while the controlled continuous-interaction estimates are reported in the text and Appendix D.

that is equally present in normal times may instead reflect broader financial development, firm composition, institutional quality, or other persistent country characteristics. The empirical exercise therefore asks both whether deeper NFC bond markets are associated with stronger recovery after crises and whether this relationship is unusually strong in crisis states.

Let t_0 denote the onset year of a systemic banking crisis. I define pre-crisis NFC bond-market depth as the average ratio of NFC bond gross issuance to GDP over the five years preceding the crisis:

$$\text{Depth}_{i,t_0} = \frac{1}{5}\sum_{s=1}^{5} \frac{\text{NFC Bond Issues}_{i,t_0-s}}{\text{GDP}_{i,t_0-s}}.$$

The measure is standardized across crisis episodes. Because it is constructed entirely from observations preceding crisis onset, it is not mechanically affected by subsequent issuance or real outcomes. The dependent variable is cumulative real GDP or investment growth from the crisis year t_0 to horizon t_0+h. Measuring outcomes from t_0, rather than from t_0-1, focuses on recovery after the crisis has begun rather than on the initial crisis-year contraction.

For each horizon $h = 1, \ldots, 5$, I estimate:

$$100(\log y_{i,t_0+h} - \log y_{i,t_0}) = \beta_h \text{Depth}_{i,t_0} + \theta_h \text{CreditBoom}_{i,t_0} + \phi_h \text{Shock0}_{i,t_0} + \varepsilon_{i,t_0}. \quad (4)$$

Here $\text{CreditBoom}_{i,t_0}$ is the change in corporate credit-to-GDP from t_0-4 to t_0-1, while $\text{Shock0}_{i,t_0}$ is real GDP growth from t_0-1 to t_0. Both controls are standardized. Standard

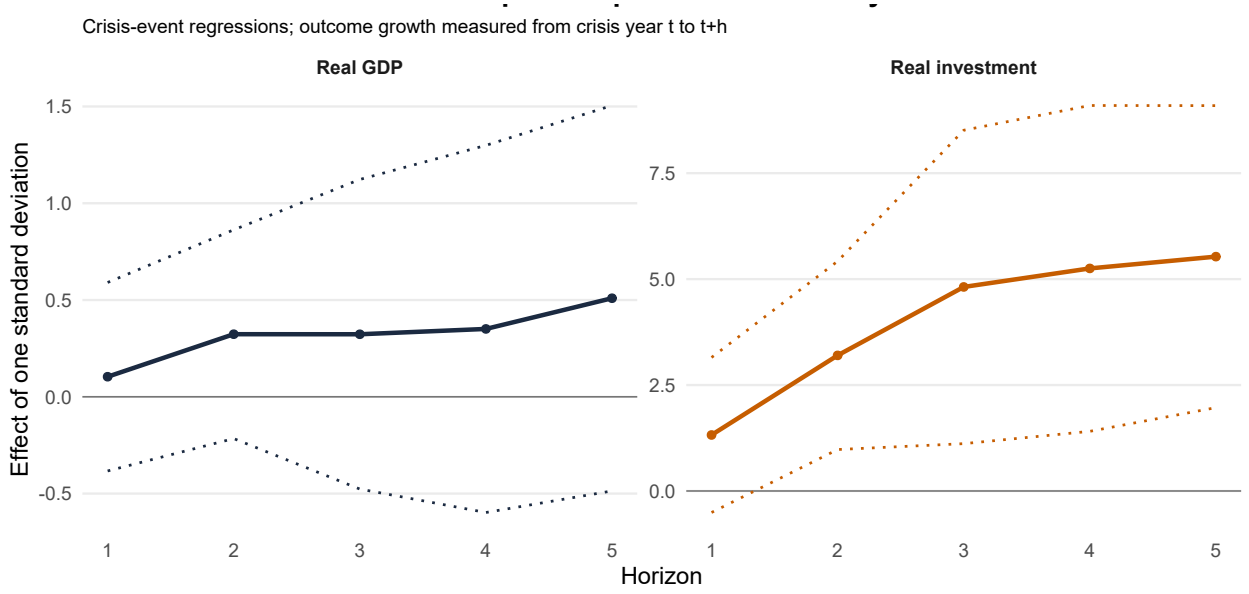


Figure 10: Pre-crisis NFC bond-market depth and post-crisis recovery. The figure reports crisis-event regressions based on Equation 4. The dependent variable is $100 \times (\log y_{i,t_0+h} - \log y_{i,t_0})$, where y denotes real GDP or real investment. Pre-crisis NFC bond-market depth is the standardized five-year average of NFC gross bond issuance relative to GDP over t_0-5 to t_0-1. Regressions control for the pre-crisis corporate credit boom and the crisis-year real GDP shock. Dotted lines denote 90% confidence intervals based on standard errors clustered by country.

errors are clustered by country.

The specification is intentionally cross-crisis and does not include country fixed effects. The coefficient β_h therefore captures the association between differences in pre-crisis bond-market depth across crisis episodes and subsequent recovery. It should not be interpreted causally: bond-market depth may be correlated with other persistent characteristics that also shape recovery, including financial development, industrial composition, institutions, or the prevalence of large firms. The credit-boom and crisis-severity controls help discipline some of these differences but do not eliminate this omitted-variable concern. I therefore interpret the exercise as descriptive evidence on real-outcome heterogeneity across crises rather than as an estimate of the causal effect of deeper bond markets.

Figure 10 shows that pre-crisis NFC bond-market depth is positively associated with investment recovery after banking crises. The association is small at short horizons and becomes larger over the medium run. By five years after crisis onset, crises preceded by deeper NFC bond markets are associated with stronger cumulative real investment growth. The corresponding GDP coefficients are smaller and less precisely estimated. The contrast is consistent with financing conditions being more directly reflected in investment than in aggregate output, although the estimates remain associational.

Appendix Section 8.3 shows that the positive medium-run investment relationship is not specific to the baseline definition of bond-market depth. Similar patterns obtain using alternative pre-crisis windows, the median rather than mean issuance-to-GDP ratio, a log transformation, issuance relative to lagged corporate bank credit, and total-corporate

rather than NFC issuance. The investment slope also remains positive when the pre-crisis credit-boom and crisis-year GDP-shock controls are removed. These exercises show that the baseline correlation is not mechanically driven by a particular depth definition or control set, but they do not resolve the broader cross-sectional identification problem.

A stronger spare-tire interpretation would require this positive depth–investment relationship to become distinctly stronger during banking crises than in normal periods. To examine this, Appendix Table 27 and Figure 37 estimate a full-panel interaction specification. In that exercise, $\text{Depth}_{i,t}$ is a standardized five-year rolling average of NFC bond issuance relative to GDP over $t-5$ to $t-1$. For crisis-onset years this is naturally a pre-crisis measure; for clean non-crisis years it is simply lagged bond-market depth. The specification therefore compares the slope of subsequent real outcomes with respect to the same predetermined depth measure in crisis-onset and non-crisis years.

The resulting investment slopes are positive in both states. NFC bond-market depth is associated with stronger subsequent investment growth in crisis-onset years, but the corresponding non-crisis slope is also positive and of similar magnitude, and the difference between the two is imprecisely estimated. The data therefore do not show a clear state-dependent increase in the relationship between bond-market depth and investment when banking crises occur.

The real-outcome evidence consequently supports a narrower conclusion than a crisis-specific spare-tire interpretation. Deeper NFC bond markets are associated with stronger investment dynamics, including following banking crises, but the same relationship is visible outside crises. Bond-market depth therefore appears to capture a broader investment-resilience margin rather than a financing mechanism that becomes uniquely important when banks are impaired. This complements the issuance evidence above: aggregate NFC issuance contracts after banking crises, while countries with deeper bond markets nevertheless tend to display stronger investment performance more generally.

5 Conclusion

This paper introduces a new long-run dataset on sectoral primary bond issuance across 19 countries and uses it to study how bond markets behave around systemic banking crises. By focusing on gross issuance rather than outstanding stocks or financial-account transactions, the data measure primary-market financing activity directly and distinguish issuance by non-financial corporations, financial corporations, and governments over more than a century of financial history. This sectoral distinction is important: the long-run expansion of corporate bond markets, particularly since the 1980s, increasingly reflects issuance by financial corporations and need not imply a corresponding expansion in market finance available directly to non-financial firms.

The central result is that aggregate NFC bond issuance contracts rather than expands

following banking-crisis onsets. In the baseline estimates, NFC issuance is about 38 log points below its pre-crisis level after one year and 64 log points lower after three years, corresponding to a decline of roughly one half. Government issuance rises strongly on average, while financial-corporation issuance increases initially before declining at longer horizons. The composition of primary bond-market activity therefore shifts markedly around banking crises, but there is no broad expansion in NFC issuance of the kind implied by an aggregate spare-tire interpretation. This result is robust across alternative specifications, normalizations, historical periods, crisis chronologies, and sample restrictions.

These aggregate dynamics do not imply that individual firms never substitute toward bonds when bank intermediation is disrupted. Firm-level substitution and aggregate market resilience are distinct empirical objects. The historical sectoral data cannot separately identify changes in financing demand, market access, or issuer composition. Additional evidence nevertheless helps discipline the interpretation. Security-level data from Australia show one concrete episode in which private issuance contracted primarily through a fall in the number of active issuers, even as a small group of established firms continued to issue and average issue size remained broadly stable. Across the international sample, larger NFC issuance contractions are also associated with more severe repricing in risky debt markets. Neither exercise identifies a structural bond-supply shock, but together they show that aggregate contraction can coexist with continued issuance for some firms and is particularly pronounced when credit markets undergo substantial repricing.

Bond-market depth also matters for real economic outcomes, although not in the simple crisis-specific manner implied by a strong spare-tire hypothesis. Crisis episodes preceded by deeper NFC bond markets are associated with stronger subsequent investment growth, while the corresponding relationship for GDP is weaker. Yet the positive association between bond-market depth and investment is also present in non-crisis years, and the difference between crisis and non-crisis slopes is imprecisely estimated. The evidence therefore points to a broader relationship between bond-market depth and investment resilience rather than to a financing channel that becomes uniquely active when banking crises occur.

Taken together, the results qualify a simple view of market-based finance as an automatic substitute for banking intermediation during financial distress. Deep bond markets can be economically valuable, and individual firms may use them to substitute away from banks, but these margins need not translate into an expansion of aggregate NFC market financing during systemic banking crises. For historical interpretation and contemporary policy alike, the relevant question is therefore not only how large bond markets are, but which sectors issue, which firms retain access, and how primary debt markets themselves behave when the financial system comes under stress.

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6 Appendix A: The Bond Dataset

6.1 Overview

In this section, I provide a general overview and further details of my data coverage, its collection process, including the types of data collected, references to the original archival material, and details on the construction of the time series on a country-by-country basis.

6.2 Dataset Coverage

Instrument coverage. The dataset covers primary issues of bonds and comparable debt securities reported in official capital-issue statistics, central bank publications, statistical yearbooks, and securities-market reports. The baseline object is gross issuance of debt securities by issuing sector. Gross issuance measures the value of debt securities issued during the reporting period, before subtracting redemptions. It therefore measures primary-market activity rather than the change in outstanding debt or net financing. Depending on the historical source, it can include refinancing, conversion or rollover issues. The precise valuation concept may be face/par value, issuing value, effective amount, or new issues delivered; these source-specific definitions are retained and documented country by country.

When sources distinguish bonds from equity, shares, or other securities, I retain only debt securities for the construction of the baseline issuance series. When the source reports a broader debt-security aggregate, I include the series if the category is conceptually close to bond issuance and document the corresponding source definition in the country files. Gross issuance measures the value of debt securities issued during the reporting period, before subtracting redemptions. It therefore measures primary-market activity rather than the change in outstanding debt or net financing. Depending on the historical source, it can include refinancing, conversion or rollover issues. The precise valuation concept may be face/par value, issuing value, effective amount, or new issues delivered; these source-specific definitions are retained and documented country by country.

The treatment of borderline instruments follows the classification in the original source whenever possible. Mortgage bonds and covered bonds are included when they are reported as bond issues and assigned to the issuing sector, typically financial corporations when issued by mortgage-credit institutions. The baseline follows the debt-security perimeter of the underlying source. Short-term marketable debt is therefore included when it forms part of the source's reported debt-security or bond-issuance aggregate, but is not added mechanically when it is reported separately and cannot be allocated to a sector. Separate identification of short-term instruments is not available consistently across countries or historically. In cases where historical sources report bills jointly with government bond issues, the combined series is retained and documented. Publicly backed corporate

issuers, such as railways, utilities, or state-owned enterprises, are classified according to the issuing entity rather than the implicit guarantee. They are therefore included in the corporate sector unless the source explicitly records the issue as central-government debt.

The dataset does not impose a uniform listed/unlisted distinction, because historical sources rarely report this split consistently. Instead, it follows the coverage of the original primary-market statistics. Similarly, foreign-currency or non-domestic issues are included when covered by the source used for a given country. Appendix section 6.5 documents these choices country by country. Historical sources differ somewhat in the valuation and timing convention used for gross issuance. Depending on the source, issues may be reported at face or par value, at the value of the issue or proceeds, or when newly issued securities are delivered or settled. These conventions are sufficiently close for the purpose of measuring annual primary-market issuance, but they are not mechanically identical. I therefore retain the source-specific gross-issuance concept to measure issues in a given year as best as possible.

Time coverage. Coverage varies substantially across countries, sectors, and periods. The dataset expands gradually over time given the uneven availability of long-run primary-market statistics across countries. The sectoral empirical analysis relies on the more restrictive financial-corporation, non-financial-corporation, and government series, whose coverage is summarized in Table 1 and documented in greater detail in Section 6.5. The broader total-corporate series provides an additional valuable robustness check, often extending country, time, and crisis coverage. The qualitative results are similar when this broader aggregate is used.

Historical series were assembled from successive vintages of official statistical publications rather than from a single retrospective database. The main sources are central-bank annual reports and statistical bulletins, national statistical yearbooks, finance-ministry and securities-market publications, and official retrospective statistical compilations. Where publications changed classification over time, I consulted overlapping editions and retrospective tables to identify changes in definitions and avoid double counting. Raw source categories were retained before harmonization, together with their original units and source labels. In several cases, ambiguities could only be resolved through correspondence with national central banks, statistical offices, or archives; these clarifications are recorded in the corresponding country files. In many cases I contacted archives, central bank and national statistical offices directly and they generously supported me in identifying and digitizing relevant sources, often for the first time. Figure 11 shows two typical sources I encountered during the process of gathering relevant sources.

Source selection gives priority to official primary statistics and official retrospective compilations. Researcher-constructed historical series are used when they provide uniquely detailed or otherwise unavailable historical coverage, as in Australia and Sweden. Where

multiple sources overlap, overlaps are used primarily for validation and for identifying definitional breaks rather than mechanically forcing levels to coincide.

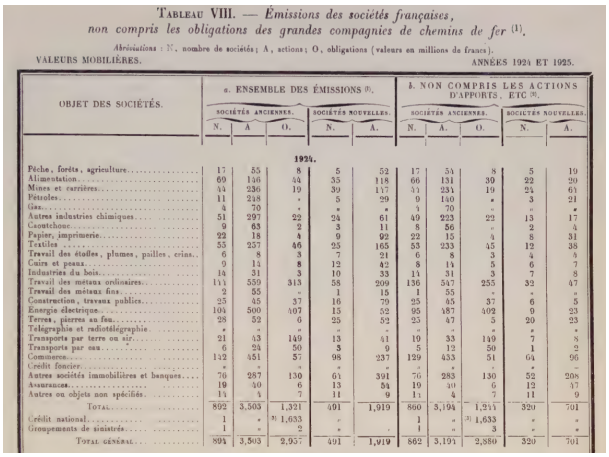


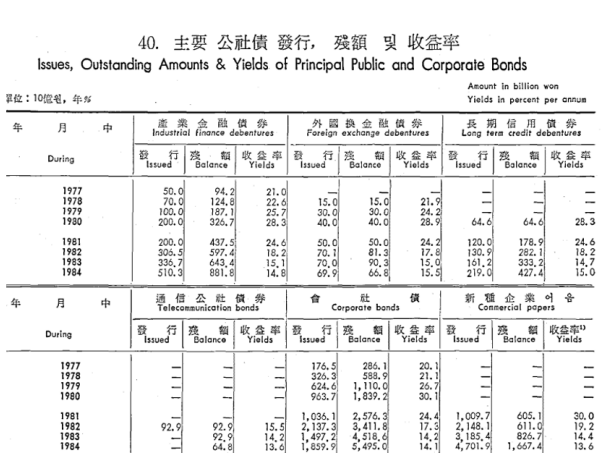
TABLEAU VIII. — *Émissions des sociétés françaises, non compris les obligations des grandes compagnies de chemins de fer* (1).

Abréviations : N., nombre de sociétés; A, actions; O, obligations (valeurs en millions de francs).

VALEURS MOBILIÈRES. ANNÉES 1924 ET 1925.

OBJET DES SOCIÉTÉS.	a. ENSEMBLE DES ÉMISSIONS (2).			b. NON COMPRIS LES ACTIONS D'APPORTS, ETC (3).					
	SOCIÉTÉS ANCIENNES.			SOCIÉTÉS NOUVELLES.		SOCIÉTÉS ANCIENNES.			SOCIÉTÉS NOUVELLES.	
	N.	A.	O.	N.	A.	N.	A.	O.	N.	A.
1924.										
Pêche, forêts, agriculture	17	55	8	5	52	17	54	8	5	19
Alimentation	69	146	44	35	118	66	131	30	22	20
Mines et carrières	44	236	19	39	117	41	231	19	24	61
Pétroles	11	218	»	5	29	9	140	»	3	21
Gaz	4	70	»	»	»	4	70	»	»	»
Autres industries chimiques	51	297	22	24	61	49	223	22	13	17
Caoutchouc	9	63	2	3	11	8	56	»	2	4
Papier, imprimerie	22	18	4	9	22	22	15	4	8	31
Textiles	55	257	46	25	165	53	233	45	12	38
Travail des étoffes, plumes, paille, crins	6	8	3	7	21	6	8	3	4	4
Cuirs et peaux	9	14	8	12	42	8	14	5	6	7
Industries du bois	14	31	3	10	33	14	31	3	7	8
Travail des métaux ordinaires	111	559	315	58	209	136	547	255	32	67
Travail des métaux fins	2	55	»	1	15	1	55	»	»	»
Construction, travaux publics	25	45	37	16	79	25	45	37	6	5
Énergie électrique	104	500	407	15	52	95	487	402	9	23
Terres, pierres au feu	28	52	6	25	52	23	47	5	20	23
Télégraphie et radiotélégraphie	»	»	»	»	»	»	»	»	»	»
Transports par terre ou air	21	53	149	13	41	19	33	149	7	8
Transports par eau	6	24	50	3	9	5	12	50	1	2
Commerce	142	451	57	98	237	129	433	51	64	96
Crédit foncier	»	»	»	»	»	»	»	»	»	»
Autres sociétés immobilières et banques	76	287	130	61	391	76	283	130	52	208
Assurances	19	40	6	13	84	19	40	6	12	47
Autres ou objets non spécifiés	14	4	7	11	9	11	4	7	11	9
TOTAL	892	3,503	1,321	491	1,919	860	3,194	1,211	320	701
Crédit national	1	»	(3) 1,633	»	»	1	»	(3) 1,633	»	»
Groupements de sinistrés	1	»	2	»	»	1	»	3	»	»
TOTAL GÉNÉRAL	894	3,503	2,957	491	1,919	862	3,194	2,889	320	701

Source: Annuaire Statistique de la France (1925)



40. 主要 公社債 發行, 殘額 및 收益率

Issues, Outstanding Amounts & Yields of Principal Public and Corporate Bonds

單位 : 10億원, 年%

Amount in billion won
Yields in percent per annum

年 月 中 During	産業金融債券 Industrial finance debentures			外國換金融債券 Foreign exchange debentures			長期信用債券 Long term credit debentures		
	發行 Issued	殘額 Balance	收益率 Yields	發行 Issued	殘額 Balance	收益率 Yields	發行 Issued	殘額 Balance	收益率 Yields
1977	50.0	94.2	21.0	—	—	—	—	—	—
1978	70.0	124.8	22.6	15.0	15.0	21.9	—	—	—
1979	100.0	187.1	25.7	30.0	30.0	24.2	—	—	—
1980	200.0	326.7	28.3	40.0	40.0	28.9	64.6	64.6	28.3
1981	200.0	437.5	24.6	50.0	50.0	24.2	120.0	178.9	24.6
1982	305.5	597.4	18.2	70.1	81.3	17.8	130.9	282.1	18.2
1983	336.7	643.4	15.1	70.0	90.3	15.0	161.2	333.2	14.7
1984	510.3	881.8	14.8	69.9	66.8	15.5	219.0	427.4	15.0

年 月 中 During	通信公社債券 Telecommunication bonds			會社債 Corporate bonds			新種企業어음 Commercial papers		
	發行 Issued	殘額 Balance	收益率 Yields	發行 Issued	殘額 Balance	收益率 Yields	發行 Issued	殘額 Balance	收益率[1] Yields
1977	—	—	—	176.5	286.1	20.1	—	—	—
1978	—	—	—	326.3	588.9	21.1	—	—	—
1979	—	—	—	624.6	1,110.0	26.7	—	—	—
1980	—	—	—	963.7	1,839.2	30.1	—	—	—
1981	—	—	—	1,036.1	2,576.3	24.4	1,009.7	605.1	30.0
1982	92.9	92.9	15.5	2,137.3	3,411.8	17.3	2,148.1	611.0	19.2
1983	—	92.9	14.2	1,497.2	4,518.6	14.2	3,185.4	826.7	14.4
1984	—	64.8	13.6	1,859.9	5,495.0	14.1	4,701.9	1,667.4	13.6

Source: Korea Statistical Yearbook (1985)

Figure 11: Examples of historical source material. The figure shows examples of original statistical publications used to construct the long-run bond issuance dataset. The left panel reports French capital-issue statistics from the *Annuaire Statistique de la France*; the right panel reports bond-market statistics from the *Korea Statistical Yearbook*.

Worked construction example: France, 1924. Figure 11 illustrates the construction using the *Annuaire statistique de la France*. The original table reports both equity (*actions*) and bond (*obligations*) issues by type of company, in millions of francs. For the bond series I retain only the *obligations* columns. Importantly, the first reported "Total" of 1,321 million francs does not constitute total corporate bond issuance: Crédit National (1,633 million) and the *Groupements de sinistrés* (2 million) are reported separately below this subtotal. I therefore use the source's *Total général*, giving TC issuance of 2,956 million francs. FC is constructed from the identifiable financial categories,

$$130_{\text{banks/real estate}} + 1,633_{\text{Crédit National}} + 6_{\text{insurance}} + 2_{\text{groupements}} = 1,771.$$

The remaining 1,185 million francs are assigned to the NFC sector given their historical categories. The source categories, subtotals, and final mapping are retained in the underlying country file together with the exact edition and page reference. Finally, values originally expressed in old francs are converted to new francs using the 1960 redenomination rate. Thus, for example, TC issuance of 2,956 million old francs becomes 29.56 million new francs in the final harmonized file.

6.3 Construction of sector classifications and harmonization choices

The historical sources differ substantially in their issuer classifications, terminology, and level of sectoral detail. I therefore do not impose a mechanical modern institutional-sector

classification on all historical observations. Instead, the harmonization proceeds from the issuer categories reported in the original source and distinguishes three types of mapping.

First, when the historical category maps clearly into one of the harmonized sectors, I use a direct mapping. Examples include banks and insurance companies assigned to financial corporations (FC), industrial and manufacturing firms assigned to non-financial corporations (NFC), and central-government issues assigned to government (GOV).

Second, some historical sources provide categories that are broader than the corresponding modern institutional sector but nevertheless constitute the closest available historical proxy. In these cases I retain a documented historical proxy. Examples include source categories that jointly combine several activities (e.g. real estate and finance), or historical issuer classifications whose institutional boundaries do not coincide exactly with modern national-accounting definitions. Such mappings are retained when they provide a substantively informative approximation and are explicitly described in the country documentation.

Third, where an issuer category cannot be assigned defensibly to either NFC or FC, I leave it unallocated at the sectoral level. When the category clearly belongs to the broader corporate or non-government issuer universe, it is retained in the total-corporate aggregate (TC) but excluded from the finer NFC/FC decomposition. This preserves the information contained in the historical source without manufacturing sectoral precision that the source does not support.

The resulting sector classifications should therefore be interpreted as harmonized historical issuer concepts rather than retrospective applications of a single modern sector standard. The objective is to preserve economically meaningful distinctions – in particular between financial corporations, non-financial corporations, and central government – while remaining transparent about cases in which historical reporting does not permit an exact mapping.

Total corporations. TC is a broad corporate or non-government corporate-issuer aggregate whose exact perimeter is necessarily source-dependent. Where sufficiently clean NFC and FC series are available, TC is generally constructed as their sum. In some historical sources, however, the reported corporate aggregate also contains residual issuer categories that cannot be allocated reliably between NFC and FC; these are retained in TC only. In other cases the source itself defines a broader corporate universe whose legal or statistical perimeter differs from the modern NFC+FC concept. TC should therefore not be interpreted mechanically as NFC+FC in every country-year. Its purpose is to preserve the broadest consistently identifiable corporate bond-issuance measure available in each source, with deviations from the harmonized NFC/FC perimeter documented country by country. When TC contains source-defined residual categories, the empirical sectoral analysis relies on the separately harmonized NFC and FC series rather than assuming an exact

accounting identity between the three aggregates.

Government. The baseline GOV concept aims to capture central or federal government gross debt-security issuance. Subnational governments, municipalities, provinces, cantons, and other public-law bodies are generally excluded when they can be identified separately. Historical source definitions are not always perfectly aligned with a modern central-government sector, however. In some periods the published category may include closely related public entities, jointly report territories or dependencies, or change its instrument perimeter over time. Such source-defined exceptions are retained when they cannot be separated consistently and are documented in the corresponding country notes if further information was found.

6.4 Treatment of breaks, zeros, missing years, denomination changes and non-annual data

The historical sources used in this project exhibit unavoidable gaps, definitional breaks, and changes in denomination over time. I address these issues using a transparent and conservative harmonization strategy that prioritizes traceability over mechanical continuity.

Missing years are left explicitly missing and coded as `NA`. No interpolation or extrapolation is applied to fill gaps in the underlying source material. As a result, observed changes in issuance reflect only information contained in the original sources.

Structural breaks, especially when arising from changes in publication formats, reporting standards, or source coverage, are not mechanically adjusted or smoothed. Instead, such breaks are preserved in the data and explicitly flagged on a case-by-case basis in the country files, particularly for years in which the underlying source or statistical publication changes.

Currency and denomination changes are harmonized using official conversion rates. For euro area countries, all final series are expressed in pre-euro national currencies, with post-1999 data originally reported in euros converted back using the irrevocable conversion rates set at euro adoption. Analogous procedures are applied in other cases of denomination changes, such as the conversion from old to new francs in France. All conversions are applied consistently within country over time and documented in the corresponding country files.

A value is coded as zero only when the source explicitly indicates nil/no issuance, or when a complete micro-level/source table establishes that the relevant category had no issues. Blank or unavailable observations are otherwise coded missing. Historical typographical symbols are interpreted according to each publication's legend rather than under a universal rule.

Monthly and quarterly gross flows are summed to calendar years. For monthly modern series, incomplete calendar years are excluded where appropriate. Historical fiscal-year observations are retained according to the source convention and their calendar-year assignment is documented country by country.

Table 4: Coverage of constructed gross bond issuance series

Country	Sector	Start	End	Missing years	Obs.	Banking crisis onset years covered
Australia	Government	1915	1983	None	69	1931
	Financial corporations	1915	1983	None	69	1931
	Non-financial corporations	1915	1983	None	69	1931
	Total corporations	1915	1983	None	69	1931
Austria	Government	1953	2021	None	69	2008, 2011
	Financial corporations	1965	2021	None	57	2008, 2011
	Non-financial corporations	1953	2021	None	69	2008, 2011
	Total corporations	1965	2021	None	57	2008, 2011
Belgium	Government	1990	2021	None	32	2008, 2011
	Financial corporations	–	–	–	0	–
	Non-financial corporations	1947	2021	None	75	2008, 2011
	Total corporations	1920	2021	1940–1946	95	1929, 2008, 2011
Denmark	Government	1957	1996	None	40	1992
	Financial corporations	–	–	–	0	–
	Non-financial corporations	–	–	–	0	–
	Total corporations	1957	1996	None	40	1992
Finland	Government	1938	2021	1939–1944, 1946–1949	74	1990
	Financial corporations	1938	2021	1939–1944, 1946–1949	74	1990

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Table 4: Coverage of constructed gross bond issuance series (continued)

Country	Sector	Start	End	Missing years	Obs.	Banking crisis onset years covered
	Non-financial corporations	1938	2021	1939–1944, 1946–1949	74	1990
	Total corporations	1938	2021	1939–1944, 1946–1949	74	1990
France	Government	1921	2021	1945–1950	95	1930, 1937, 2008
	Financial corporations	1907	2021	1945–1950	109	1930, 1937, 2008
	Non-financial corporations	1907	2021	1945–1956	103	1930, 1937, 2008
	Total corporations	1907	2021	1945–1950	109	1930, 1937, 2008
Germany	Government	1990	2021	None	32	2008
	Financial corporations	1990	2021	None	32	2008
	Non-financial corporations	1990	2021	None	32	2008
	Total corporations	1990	2021	None	32	2008
Italy	Government	–	–	–	0	–
	Financial corporations	1931	2021	1937–1957	70	1992, 2008, 2011, 2016
	Non-financial corporations	1931	2021	1937–1957	70	1992, 2008, 2011, 2016
	Total corporations	1931	2021	1937–1957	70	1992, 2008, 2011, 2016
Japan	Government	1920	2024	1942–1945, 2019	100	1920, 1922, 1923, 1927, 1990, 1997, 2001
	Financial corporations	1920	2024	1942–1945, 2019	100	1920, 1922, 1923, 1927, 1990, 1997, 2001
	Non-financial corporations	1920	2024	1942–1945, 2019	100	1920, 1922, 1923, 1927, 1990, 1997, 2001

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Table 4: Coverage of constructed gross bond issuance series (continued)

Country	Sector	Start	End	Missing years	Obs.	Banking crisis onset years covered
	Total corporations	1920	2024	1942–1945, 2019	100	1920, 1922, 1923, 1927, 1990, 1997, 2001
Netherlands	Government	1914	2021	1941–1951	97	1921, 1931, 2008
	Financial corporations	1914	2021	1941–1951	97	1921, 1931, 2008
	Non-financial corporations	1914	2021	1941–1951	97	1921, 1931, 2008
	Total corporations	1914	2021	1941–1951	97	1921, 1931, 2008
Norway	Government	–	–	–	0	–
	Financial corporations	1984	2024	None	41	1987, 2008
	Non-financial corporations	–	–	–	0	–
	Total corporations	1930	2024	None	95	1931, 1987, 2008
Portugal	Government	1969	2021	None	53	2008, 2011, 2014
	Financial corporations	1990	2021	None	32	2008, 2011, 2014
	Non-financial corporations	1990	2021	None	32	2008, 2011, 2014
	Total corporations	1936	2021	None	86	2008, 2011, 2014
South Korea	Government	1977	2024	1983–1985, 1992–1993	43	1997
	Financial corporations	2008	2024	None	17	None
	Non-financial corporations	1993	2024	None	32	1997
	Total corporations	1977	2024	None	48	1997
Spain	Government	1954	2021	None	68	1975, 2008, 2010

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Table 4: Coverage of constructed gross bond issuance series (continued)

Country	Sector	Start	End	Missing years	Obs.	Banking crisis onset years covered
	Financial corporations	1962	2021	None	60	1975, 2008, 2010
	Non-financial corporations	1962	2021	None	60	1975, 2008, 2010
	Total corporations	1954	2021	None	68	1975, 2008, 2010
Sweden	Government	1854	2021	31	137	1878, 1907, 1919, 1991, 2008
	Financial corporations	1850	2021	None	172	1878, 1907, 1919, 1991, 2008
	Non-financial corporations	1855	2021	None	167	1878, 1907, 1919, 1991, 2008
	Total corporations	1835	2021	None	187	1878, 1907, 1919, 1991, 2008
Switzerland	Government	1850	2024	None	175	1870, 1919, 1931, 1990, 2008
	Financial corporations	1900	2024	None	125	1919, 1931, 1990, 2008
	Non-financial corporations	1900	2024	None	125	1919, 1931, 1990, 2008
	Total corporations	1900	2024	None	125	1919, 1931, 1990, 2008
Thailand	Government	1979	2023	1981–1985	40	1979, 1997
	Financial corporations	1998	2023	None	26	None
	Non-financial corporations	1998	2023	None	26	None
	Total corporations	1986	2023	1989	37	1997
United Kingdom	Government	–	–	–	0	–
	Financial corporations	2003	2024	None	22	2008
	Non-financial corporations	2003	2024	None	22	2008

Continued on next page

Table 4: Coverage of constructed gross bond issuance series (continued)

Country	Sector	Start	End	Missing years	Obs.	Banking crisis onset years covered
United States	Total corporations	1919	2024	1984–2002	87	1973, 2008
	Government	1934	2023	1936–1938, 1942, 1944, 1946, 1974–1989	68	1990, 2007
	Financial corporations	1970	2023	None	54	1984, 1990, 2007
	Non-financial corporations	1900	2023	1944–1969	98	1907, 1930, 1984, 1990, 2007
	Total corporations	1944	2023	None	80	1984, 1990, 2007

Notes: The table reports coverage of constructed annual gross bond issuance series. Start and end years refer to the first and last non-missing observations in the final harmonized dataset. Missing years denote internal gaps between the start and end year. Banking crisis onsets covered refer to crisis onset years from Baron et al. (2021) for which the corresponding sector series is observed at $t = 0$. Sources, series-construction details, breaks, switches, classifications, and other documentation are provided in Section 6.5.

6.5 Country-Specific Data and Source Documentation

Notes on abbreviations and sectors in the country table:

- **GOV**: central or federal government gross debt-security issuance, subject to documented historical source-specific exceptions;
- **TC**: broad total-corporate or non-government corporate-issuer aggregate; where clean sectoral series are available, typically the sum of NFC and FC, possibly plus unallocated corporate residuals;
- **NFC**: non-financial corporations and source-based historical proxies for non-financial corporate issuers;
- **FC**: financial corporations and source-based historical proxies for financial-sector issuers.

6.5.1 Australia

Table 5: Source construction for Australia: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1915–1983	Black et al. (2013) and underlying issue-level data obtained from the authors/Reserve Bank of Australia	Private non-financial corporations; non-financial public trading enterprises (PTEs).	No aggregate NFC issuance series is taken directly from the source. Annual NFC gross issuance is reconstructed from the underlying individual bond issues by summing issues classified as private non-financial corporations and non-financial public trading enterprises. The inclusion of PTEs follows the issuer classification in Black et al. (2013), which treats commercially operated public enterprises as non-financial corporations.
FC	Gross issues	1915–1983	Black et al. (2013) and underlying issue-level data obtained from the authors/Reserve Bank of Australia	Banks; non-bank financial institutions.	Annual FC gross issuance is reconstructed from the issue-level data as the sum of bonds issued by banks and non-bank financial institutions. The bank category includes both privately and publicly owned banks.
TC	Gross issues	1915–1983	Black et al. (2013) and underlying issue-level data obtained from the authors/Reserve Bank of Australia	Private non-financial corporations; non-financial public trading enterprises; banks; non-bank financial institutions.	Annual total-corporate issuance is constructed as the sum of the reconstructed NFC and FC series. Government issuers and non-resident issuers are excluded from TC.
GOV	Gross issues	1915–1983	Black et al. (2013) and underlying issue-level data obtained from the authors/Reserve Bank of Australia	Government.	Annual GOV gross issuance is reconstructed by summing the individual issues assigned to the Government issuer category in the underlying data.

Notes. Australian gross-issuance series are reconstructed from individual bond issues rather than transcribed from published aggregate issuance statistics. The underlying unit-record data were compiled by Black et al. (2013) from historical stock-exchange publications and Reserve Bank of Australia sources and were obtained from the authors/Reserve Bank of Australia. Each issue is assigned an issuer type in the source data, and annual sectoral aggregates are constructed by summing the face value of issues within the corresponding issuer categories. Years in which a sector records no individual issues are coded as zero. The sector mapping follows the institutional classification used in the underlying data. NFC combines private non-financial corporations and non-financial public trading enterprises (PTEs); FC combines banks and non-bank financial institutions; and TC is the sum of these two sectors. Public ownership by itself does not lead an issuer to be classified as government: Black et al. (2013) treat commercially operated PTEs and government-owned banks as corporations, thereby maintaining a consistent issuer classification through later privatisations. Non-resident issuers are excluded from the domestic corporate aggregates. The underlying Black et al. dataset defines bonds as debt securities with an original maturity of at least twelve months and excludes hybrid securities; the present reconstruction follows this source definition. Issues are measured at face value. The historical data primarily cover onshore issuance; the underlying study notes that comprehensive offshore issuance is not available before 1983 and that over-the-counter issuance prior to the early 1980s may be incompletely captured. The supplied issue-level/grouped

data used for the present reconstruction contain observations from 1915 through 1983, marginally extending the 1917 starting date emphasized in the paper. The current dataset retains the coverage supplied in these underlying data.

6.5.2 Austria

Table 6: Source construction for Austria: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1953–1989	Oesterreichische Nationalbank, Mitteilungen, various editions	Bund.	GOV consists of bond issuance by the federal government. Issues by Länder and cities, public-law funds and corporations, and other public-sector entities are excluded from the central-government aggregate.
NFC	Gross issues	1953–1989	Oesterreichische Nationalbank, Mitteilungen, various editions	Electricity enterprises; industry and other domestic corporate issuers.	NFC is constructed from the source categories corresponding to electricity enterprises and industrial issuers. In earlier publications, industry and other domestic issuers are reported jointly and the combined category is retained. Where other domestic issuers become separately identifiable, they are not mechanically assigned to NFC if their sector cannot be established.
FC	Gross issues	1965–1980	Oesterreichische Nationalbank, Mitteilungen, various editions	Inländische Kreditunternehmungen.	FC is constructed from gross bond issuance by domestic credit institutions. A comparable gross-issuance series is not available in the earlier source material and FC is therefore left missing before 1965.
FC	Gross issues	1981–1989	Oesterreichische Nationalbank, *Statistisches Monatsheft*, various editions	Bank debt-security issues.	From 1981, the source provides a more detailed breakdown of bank issuance. FC is constructed using the reported aggregate of bank debt-security issues, combining the relevant bank bond and other bank debt-security categories. The change in source classification is retained as a documented coverage break rather than mechanically adjusted.
TC	Gross issues	1965–1989	Oesterreichische Nationalbank, statistical monthly publications, various editions	Financial corporations, non-financial corporations, and other domestic corporate issuers where separately reported.	TC is constructed from the underlying corporate issuer categories. It includes the NFC and FC components and, where the source reports an additional domestic corporate residual that cannot be reliably assigned to either sector, this category is retained in TC without forcing it into the finer NFC/FC decomposition. TC can therefore exceed NFC plus FC in such years.

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Table 6: Source construction for Austria: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. The historical Austrian series are constructed from successive vintages of Oesterreichische Nationalbank statistical publications, specifcally capital market information found in the *Mitteilungen des Direktoriums der Oesterreichischen Nationalbank*, after 1991 called *Statistisches Monatsheft*. FC gross issuance is unavailable before 1965, while the bank-issuance classification becomes substantially more detailed from 1981. Other financial sector issues then from banks is not reported as a category in the source historically. Residual domestic corporate issuance that cannot be assigned reliably to NFC or FC is retained in TC where separately reported. From 1990 onward, the historical OeNB construction is replaced by harmonized series constructed from the underlying ECB issuer-sector data. NFC corresponds to ECB sector 1100; FC combines sectors 1220 and 1235; TC is constructed as NFC plus FC; and GOV corresponds to central government sector 1311. Monthly ECB gross issuance is aggregated to annual frequency by summing the twelve monthly observations

6.5.3 Belgium

Table 7: Source construction for Belgium: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1920–1926	Baudhuin (1943), *Placements: Principes Permanents d'Economie Privée*, p. 267	Gross bond issues (*obligations*) of joint-stock companies.	The earliest Belgian TC observations are taken from a contemporary private statistical compilation Its continuity with the subsequent official series is supported by the close correspondence between Baudhuin's figures and the Banque Nationale de Belgique (BNB) statistics in the overlapping years.
TC	Gross issues	1927–1936	Banque Nationale de Belgique, *Rapport annuel*, various editions	*Émissions des sociétés industrielles et commerciales belges et congolaises: obligations*.	Gross bond issuance is taken directly from the BNB capital-issue tables. The contemporary reporting perimeter includes both Belgian and Congolese industrial and commercial companies. This historical coverage is retained as reported.
TC	Gross issues	1937–1939	Banque Nationale de Belgique, *Rapport annuel*, various editions	*Entreprises privées: obligations*.	A change in the presentation of the BNB statistics relabels the earlier industrial-and-commercial-company category as "private enterprises."
TC	Gross issues	1940–1946	—	—	No annual gross corporate bond-issuance observations are available in the material collected for these years. The observations are therefore left missing and no interpolation is applied.
TC	Gross issues	1947–1957	*Annuaire Statistique de la Belgique*, various editions	Total bond issues of Belgian companies.	TC follows the complete reported corporate bond-issuance aggregate.
FC	Gross issues	1947–1957	*Annuaire Statistique de la Belgique*, various editions	Banks; insurance companies; financial and real-estate operations.	FC is constructed by summing the separately identifiable financial-sector categories reported in each year. For some year-sector combinations no issuance are reported, hence taken as numerical zero .
NFC	Gross issues	1947–1957	*Annuaire Statistique de la Belgique*, various editions	Residual non-financial corporate issuance.	NFC is constructed as the TC aggregate excluding FC components.

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Table 7: Source construction for Belgium: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1958–1979	*Annuaire Statistique de la Belgique*, various editions	Total Belgian joint-stock companies.	TC uses the reported total of new bond and cash-bond issues by Belgian joint-stock companies, including continuous issues. The source total is preferred to summing the detailed industry rows. The historical reporting perimeter can include Belgian companies whose activities were located abroad.
FC	Gross issues	1958–1979	*Annuaire Statistique de la Belgique*, various editions	Banks; financial and real-estate companies.	FC is constructed as the sum of bank issuance and issuance by the source category "financial and real-estate companies." Banks are added separately because they are not contained in the latter category.
NFC	Gross issues	1958–1979	*Annuaire Statistique de la Belgique*, various editions	Residual non-financial corporate issuance.	NFC is constructed as the TC aggregate excluding FC components.
TC	Gross issues	1980–1989	*Annuaire Statistique de la Belgique*, various editions	Belgian-franc issues and foreign-currency/ECU issues; continuous and non-continuous issues.	TC is constructed from the reported aggregate new issues in Belgian francs together with issues denominated in foreign currencies and ECU.
FC	Gross issues	1980–1989	*Annuaire Statistique de la Belgique*, various editions	Credit institutions, insurance, business services, and rental.	FC follows the broad source category *Institutions de crédit*, *assurances*, *services fournis aux entreprises*, *location*. This category is broader than the modern financial-corporation sector and may include some business-service, real-estate, and rental activities that cannot be separated consistently from the financial activities in the historical source.
NFC	Gross issues	1980–1989	*Annuaire Statistique de la Belgique*, various editions	Residual non-financial corporate issuance.	NFC is constructed as the TC aggregate excluding FC components.
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the underlying ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.

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Table 7: Source construction for Belgium: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. The earliest TC observations, for 1920–1926, come from the private statistical compilation of Baudhuin (1943); official BNB statistics are used from 1927 onward. The official pre-war series explicitly covers Belgian and Congolese companies through 1936, reflecting the contemporary statistical reporting perimeter. A reporting break occurs in 1937 when the earlier category of Belgian and Congolese industrial and commercial companies is replaced by the broader heading "private enterprises." For the post-war period, source categories such labelled "industrial and commercial companies" include banks, insurance companies, and financial and real-estate activities. From 1947 onward, TC is therefore taken from the complete reported corporate total; FC is constructed from the identifiable financial categories; and NFC is obtained from excluding FC issues. From 1980–1989, the available financial-sector aggregate becomes broader and includes credit institutions, insurance, business services, and rental activities. From 1990 onward, the historical national-source construction is replaced by harmonized series constructed from the underlying ECB issuer-sector data. NFC corresponds to ECB sector 1100; FC combines sectors 1220 and 1235; TC is constructed as NFC plus FC; and GOV corresponds to central government sector 1311. Monthly ECB gross issuance is aggregated to annual frequency by summing the twelve monthly observations.

6.5.4 Canada

Table 8: Source construction for Canada: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1904–1937	*Canada Year Book*, various editions	Dominion government bonds.	The Dominion government category is assigned to GOV. From 1934, the source notes that the Dominion series includes also Treasury-bill financing.
GOV	Gross issues	1938–1947	*Canada Year Book*, various editions	Federal government bonds.	The reported Federal bond-issue series is used as published. The source explicitly excludes Treasury bills and other financing with a maturity of less than one year. These instruments cannot be added separately from the available source material.
GOV	Gross issues	1948–1958	*Canada Year Book*, various editions	Federal government bonds.	The reported Federal government bond-issue series is used as published.
GOV	Gross issues	1959–1964	Bank of Canada, *Statistical Summary*	Government of Canada direct and guaranteed bonds; gross new issues delivered.	The baseline switches to the Bank of Canada series on gross new issues delivered. The reported total across domestic and foreign placements is used. The switch in 1959 is retained as a source and definition break rather than mechanically adjusted to the preceding *Canada Year Book* series.
GOV	Gross issues	1965–1973	—	—	No observations are available in the archival material underlying the country file. These years are left missing and no interpolation is applied.
GOV	Gross issues	1974–2017	Bank of Canada, *Statistical Summaries* and *Banking and Financial Statistics*, various editions	Government of Canada direct and guaranteed bonds; gross new issues delivered.	The reported total is used.
TC	Gross issues	1904–1935	*Canada Year Book*, various editions	Railway bonds; corporation bonds.	Total corporate issuance is constructed as the sum of railway and corporation bond issuance.
TC	Gross issues	1936–1958	*Canada Year Book*, various editions	Corporation bonds.	From 1936, the source no longer reports railways as a separate class and notes that remaining railway bond issues are included under "Corporation." The reported corporation aggregate is therefore used as TC.

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Table 8: Source construction for Canada: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1959–1973	Bank of Canada, *Statistical Summaries*, various editions	Corporate bonds; gross new issues delivered.	The baseline switches to the Bank of Canada series on corporate gross new issues delivered. The reported total is used.
TC	Gross issues	1974–2000	Bank of Canada, archival *Statistical Summaries*, various editions	Corporate bonds; gross new issues delivered.	The reported total is used, including foreign placements where they are included in the published total.
TC	Gross issues	2001–2017	Bank of Canada, *Banking and Financial Statistics*, Table F9	Corporate bonds; gross new issues delivered.	The reported total is used.

Notes. Canada provides long-run gross issuance series for the federal government and the total corporate sector. GOV refers to the federal Government of Canada; provincial and municipal bond issues are excluded. Railway issues are included in TC when reported separately and, from 1936, are incorporated in the source's broader corporation category. In 1959, the construction switches from the *Canada Year Book* series to Bank of Canada statistics on gross new issues delivered. The source switch is retained without a mechanical level adjustment. Instrument coverage follows the categories reported in the underlying historical sources. In particular, the Dominion government series includes Treasury-bill financing from 1934, while the Federal bond series for 1938–1947 explicitly excludes Treasury bills, deposit certificates, and other financing with maturity below one year.

6.5.5 Denmark

Table 9: Source construction for Denmark: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1957–1996	Danmarks Nationalbank, *Reports and Accounts*, various editions	Government paper payable in Danish kroner.	GOV follows the gross-issuance series reported by Danmarks Nationalbank for government paper payable in Danish kroner.
TC	Gross issues	1957–1970	Danmarks Nationalbank, *Reports and Accounts*, various editions	Mortgage-credit bonds; special-series bonds issued by the Ship Credit Fund; other bonds.	TC is constructed as the sum of the reported non-government bond categories. Mortgage-credit bonds constitute the largest component. "Other bonds" capture the remaining non-government bond issues reported by the source. The Ship Credit Fund was established in 1961; no separate gross issuance from the Fund is present before its establishment, and the category enters the aggregate when gross issues are first reported.
TC	Gross issues	1971–1995	Danmarks Nationalbank, *Reports and Accounts*, various editions	Mortgage-credit bonds; special-series bonds issued by the Ship Credit Fund; other bonds.	TC continues to be constructed as the sum of mortgage-credit bond issuance, bonds issued in special series by the Ship Credit Fund, and other bond issuance. A change in the presentation of mortgage-credit institutions around 1971 reorganizes the underlying mortgage categories, but the broader mortgage-credit aggregate is retained consistently for the construction of TC.
TC	Gross issues	1996	Danmarks Nationalbank, *Reports and Accounts*	Mortgage-credit bonds; other bonds.	The Ship Credit Fund is no longer reported as a separate issuance category in the source. TC therefore follows the revised source presentation and is constructed as mortgage-credit bond issuance plus other bond issuance.

Notes. The Danish historical statistics do not provide a sufficiently consistent issuer decomposition to construct separate NFC and FC gross-issuance series. The final dataset therefore uses GOV and a broader total-corporate/non-government bond aggregate only. TC is constructed directly from the non-government bond categories reported by Danmarks Nationalbank rather than by assigning each underlying issuer to a modern institutional sector. In particular, the aggregate contains the large Danish mortgage-credit bond market, bonds issued by the Ship Credit Fund when separately reported, and a residual "other bonds" category. The latter is retained because its underlying issuers cannot be divided consistently into financial and non-financial corporations over the full sample. The Ship Credit Fund was established in 1961 and subsequently appears as a separate bond-issuance category; by 1996 it is no longer separately reported. The historical source explicitly restricts the government gross-issuance series to paper payable in Danish kroner. This source definition is preserved and documented.

6.5.6 France

Table 10: Source construction for France: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1907–1921	*Annuaire statistique de la France*, various editions	Total bond issues of French companies.	TC as total of all of corporate bond issues.
FC	Gross issues	1907–1921	*Annuaire statistique de la France*, various editions	*Commerce, banque, assurances*.	The early source jointly reports commerce, banking and insurance and does not permit these activities to be separated. The complete joint category is used as the closest available historical FC proxy. Consequently, FC in this early period is broader than the usual financial-corporation concept because it also contains commercial firms.
NFC	Gross issues	1907–1921	*Annuaire statistique de la France*, various editions	Remaining corporate issuer categories.	NFC is constructed from the remaining identifiable non-financial corporate sectors, excluding the source category *Commerce, banque, assurances*. The early NFC/FC decomposition should therefore be interpreted as a source-constrained historical approximation rather than an exact institutional-sector split.
GOV	Gross issues	1921–1944	*Annuaire statistique de la France* and *Bulletin de la Statistique générale de la France*, various editions	State issues.	GOV follows the central-government bond-issuance series reported by the historical statistical publications. A comparable gross-issuance series is not retained before 1921.
FC	Gross issues	1922–1939	*Annuaire statistique de la France*, various editions	Banks and financial/real-estate companies; Crédit Foncier; Crédit National where separately identifiable; insurance companies.	From 1922 the source provides a substantially finer issuer classification. FC is constructed from the separately identifiable financial issuer categories. Because some historical headings combine real-estate and banking companies, this remains a source-based approximation to the modern FC sector.
NFC	Gross issues	1922–1939	*Annuaire statistique de la France*, various editions	Industrial, commercial, utility, transport and other non-financial company categories.	NFC is constructed from the remaining identifiable non-financial corporate sectors. The greater detail after 1921 permits a substantially cleaner NFC/FC decomposition than in the preceding period.
TC	Gross issues	1922–1939	*Annuaire statistique de la France*, various editions	Corporate bond issues across all reported company sectors.	TC follows the source total and is used as a check on the constructed NFC and FC components.

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Table 10: Source construction for France: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
FC	Gross issues	1940–1944	*Bulletin de la Statistique générale de la France*, table "Émissions des sociétés françaises"	Crédit Foncier; other real-estate companies and banks; insurance companies.	FC is the sum of *Crédit foncier*, *Autres sociétés immobilières et banques*, and *Assurances*. Crédit National is additionally shown by the source as a "dont" item of total issuance and is retained only as an auxiliary series; it is not added separately to FC because its issuance is already represented in the detailed issuer categories.
NFC	Gross issues	1940–1944	*Bulletin de la Statistique générale de la France*	Agriculture; mining; petroleum; manufacturing industries; utilities; construction; communications; transport; commerce; and other identifiable non-financial industries.	NFC is constructed by summing the detailed non-financial issuer categories. A small residual category, *Autres objets non spécifiés*, is not mechanically assigned to either NFC or FC.
TC	Gross issues	1940–1944	*Bulletin de la Statistique générale de la France*	Total company bond issues.	TC follows the published source total. Because unclassified issuance is retained in TC but not necessarily assigned to NFC or FC, TC may differ slightly from NFC+FC in years in which the residual category is nonzero.
TC	Gross issues	1951–1958	Conseil national du crédit (CNC), *Rapports annuels*, especially the retrospective 1951–1960 securities-issuance tables	Organismes collecteurs d'épargne; national enterprises; Postes et Télécommunications; competitive-sector enterprises and other identifiable corporate issuers.	TC is constructed from the broad corporate issuer blocks reported by the CNC. It includes both the identifiable financial-sector issuance and the non-financial corporate blocks, including national enterprises, PTT and competitive-sector enterprises. A clean NFC series is not imposed for 1951–1956 because the detailed private-sector classification is insufficiently consistent.
FC	Gross issues	1951–1966	Conseil national du crédit, *Rapports annuels*, various editions	Organismes collecteurs d'épargne; Caisse nationale des autoroutes where separately reported; investment and regional-development companies; portfolio and other financial companies.	FC is constructed by summing the financial issuer categories identifiable in the detailed CNC tables. The composition of the financial block changes across editions as institutions become separately reported.
NFC	Gross issues	1957–1958	Conseil national du crédit, *Rapports annuels*	National enterprises; Postes et Télécommunications; industrial and commercial companies; and other identifiable private non-financial companies.	A separate NFC series becomes feasible from 1957 using the detailed issuer tables. It is constructed from the identifiable public and private non-financial company categories.

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Table 10: Source construction for France: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1959–1966	Conseil national du crédit, *Rapports annuels*, various retrospective securities-issuance tables	National enterprises and PTT; industrial and commercial companies; real-estate and other non-financial company categories.	NFC is reconstructed from the corresponding detailed non-financial issuer categories. The exact underlying source headings change across editions, so the construction follows the sector content rather than mechanically requiring identical category names in every year.
FC	Gross issues	1967–1971	Conseil national du crédit, *Rapports annuels*	Institutions financières; investment and regional-development companies; portfolio and other financial companies.	FC combines the source's financial-institution block with additional financial-company categories that remain separately reported as statistical classification changes. Once the Caisse nationale des autoroutes can be identified separately it is included.
NFC	Gross issues	1967–1971	Conseil national du crédit, *Rapports annuels*	Industrial and commercial companies; real-estate and other non-financial companies; national enterprises and PTT.	NFC is constructed from the corresponding non-financial issuer categories.
FC	Gross issues	1972–1980	Conseil national du crédit, *Rapports annuels*, retrospective securities-market tables	Institutions financières.	From 1972 a broader source-reported financial-institution aggregate is available and is used directly. Components such as individual savings-collecting institutions or the Caisse nationale des autoroutes are not added again because they are already contained in the broader financial aggregate.
NFC	Gross issues	1972–1980	Conseil national du crédit, *Rapports annuels*, retrospective securities-market tables	Sociétés non financières.	NFC follows the source-based non-financial-corporation aggregate. The detailed public- and private-enterprise components are retained in the underlying country file for validation and source reconciliation.
FC	Gross issues	1981–1986	French securities-market statistics, successive CNC/statistical-yearbook tables	Institutions financières.	FC follows the revised source aggregate for financial institutions. The reporting structure changes relative to the 1970s, but the broader financial-sector perimeter is retained.
FC	Gross issues	1987–1989	French statistical securities-market tables	Établissements de crédit et assimilés.	The source changes to a headline category of credit institutions and assimilated institutions, which is used as the FC proxy for the final pre-ECB years.

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Table 10: Source construction for France: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1981–1989	French statistical securities-market tables	Sociétés non financières.	NFC follows the source-reported non-financial-corporation aggregate. Public national enterprises are included when classified with non-financial corporations. Up to 1984 PTT issuance can be identified separately and incorporated consistently; from 1985 PTT is merged into the public-administration reporting structure and can no longer be isolated, producing a documented perimeter break in the NFC series.
TC	Gross issues	1959–1989	Constructed from the French historical issuer-sector series	Harmonized NFC and FC components.	TC is constructed as the sum of NFC and FC.
GOV	Gross issues	1951–1989	Conseil national du crédit and subsequent French securities-market statistics	État / Trésor public.	GOV follows the State-level gross bond-issuance series, labelled *État* or *Trésor public* in different source vintages. Local authorities and other public-sector entities are not included.
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the underlying ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as NFC+FC rather than using a separately reported total-corporations series, ensuring that the modern corporate aggregate has the same sectoral perimeter as the underlying harmonized NFC and FC series.

Notes. France combines a particularly large number of historical source classifications. For 1907–1921, the principal limitation is the joint reporting of *Commerce*, *banque*, *assurances*. This entire category is assigned to FC. The resulting early FC series necessarily contains some non-financial commercial activity and should be interpreted as a broad source-based financial proxy. From 1922 onward the historical statistical publications provide much more detailed issuer classifications, allowing banking, credit, insurance and non-financial company issuance to be separated more reliably.

The 1940–1944 tables provide particularly detailed industry information. Crédit National is shown separately by the source as an "of which" item of aggregate issuance and therefore is not added to the financial-sector categories a second time. Small amounts reported under *Autres objets non spécifiés* remain within TC but are left unallocated between NFC and FC.

There is no annual corporate issuance series in the current compilation for 1945–1950, and these observations are left missing.

Post-war data are compiled primarily from successive *Rapports annuels* of the Conseil national du crédit and later French securities-market publications. The detailed composition of FC changes with the source. Through the mid-1960s it is reconstructed from savings-collecting institutions and other identifiable financial companies. During 1967–1971, the emerging *institutions financières* aggregate is supplemented by financial-company categories that remain separately reported. From 1972 the broader source-reported financial-institution aggregate is sufficiently comprehensive and is used directly; the individual institutions contained within it are therefore not added a second time. The source classification changes again in the 1980s, culminating in the use of *établissements de crédit et assimilés* in the final pre-ECB years.

Publicly owned market enterprises are treated according to their economic function rather than ownership. National enterprises and PTT are included with NFC when separately identifiable. The Caisse nationale des autoroutes is reassigned to the financial sector once the source permits it to be identified separately and its later treatment within the financial-institution statistics becomes clear. From 1985 onward PTT can no longer be separated from the public-administration reporting block, producing a documented break in the perimeter of the historical NFC series.

The baseline government series is restricted to the central State, reported as *État* or *Trésor public* depending on source vintage. Local authorities and other public-sector entities are not added to GOV. Foreign-sector security issues, where reported by the historical sources, are retained separately and excluded from the domestic corporate aggregates.

All franc-denominated observations are expressed in new francs in the final country file. Historical observations originally reported in old francs are converted using the 1960 redenomination rate of 100 old francs to one new franc.

A major source and coverage break occurs in 1990. The final dataset deliberately switches from the historical French securities-market statistics to harmonized ECB Securities Issues Statistics. The ECB framework covers a broader and differently organized universe of debt securities—particularly relevant for France, where modern gross issuance includes large volumes of shorter-maturity marketable debt. The resulting level increase is therefore treated as a statistical source and instrument-coverage break rather than mechanically level-adjusted. From 1990 onward NFC corresponds to ECB sector 1100; FC combines sectors 1220 and 1235; GOV corresponds to central government sector 1311; and TC is constructed as NFC+FC. Monthly gross issuance is summed to annual frequency.

6.5.7 Finland

Table 11: Source construction for Finland: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1938, 1945	*Statistical Yearbook of Finland*, retrospective bond-loan tables	State, sales of bonds in marks.	Isolated historical gross-issuance observations are available for 1938 and 1945 and are retained. GOV corresponds to the State category; municipal and parish issuance is excluded.
NFC	Gross issues	1938, 1945	*Statistical Yearbook of Finland*, various editions, retrospective bond-loan tables	Other issuers.	The source's residual "Other" issuer category is assigned to NFC. In 1945 the source reports no sales for this category and the observation is coded as zero. Other pre-1950 years for which gross issues are not reported are left missing.
FC	Gross issues	1938, 1945	*Statistical Yearbook of Finland*, various editions, retrospective bond-loan tables	Credit institutions.	FC corresponds to bond sales by credit institutions. In 1945 the source reports no issuance for this category and the observation is coded as zero.
TC	Gross issues	1938, 1945	*Statistical Yearbook of Finland*, various editions, retrospective bond-loan tables	Other issuers; credit institutions.	TC is constructed as NFC plus FC.
GOV	Gross issues	1950–1982	*Statistical Yearbook of Finland*, various editions	State, sales of markka-denominated bonds.	GOV uses the reported State gross bond-sales series. Municipal and parish bonds are excluded. The historical source excludes the Treasury Bond Loan of 1953 and indemnity bonds distributed by the State. For 1982–1987 the later source classification includes Treasury notes, creating a documented change in instrument coverage.

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Table 11: Source construction for Finland: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1950–1982	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Other issuers.	The source reports gross bond sales for the State, municipalities and parishes, credit institutions, and "Others." The "Others" category is used as the gross-issuance counterpart of the harmonized `Industry And Other` NFC series. This interpretation is supported by the historical outstanding-amount tables, which separately identify industrial corporations and other issuers while also reporting credit institutions as a distinct financial category, and by clarification from the Bank of Finland that the "Others" category includes non-financial corporations.
FC	Gross issues	1950–1982	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Credit institutions.	FC corresponds directly to gross bond sales by credit institutions.
TC	Gross issues	1950–1982	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Credit institutions; other issuers.	TC is constructed as the sum of FC and the harmonized NFC proxy based on the source's "Others" category. Municipal and parish issuance is excluded.
GOV	Gross issues	1983–1989	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Central government, total issues.	From 1983 the source reorganizes bond statistics by main institutional sector. GOV uses total central-government issues, combining public issues and private placements. The source notes that Treasury notes are included during 1982–1987.
NFC	Gross issues	1983–1989	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Enterprises, total issues.	NFC uses total bond issues by enterprises, defined as the sum of public issues and private placements.
FC	Gross issues	1983–1989	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Financial institutions, total issues.	FC uses total bond issues by financial institutions, combining public issues and private placements.

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Table 11: Source construction for Finland: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1983–1989	*Statistical Yearbook of Finland*, various editions; underlying source Bank of Finland	Enterprises; financial institutions.	TC is constructed as NFC plus FC. The transition from the earlier "credit institutions" and "other issuers" classification to explicit enterprises and financial institutions in 1983 produces a substantial level break, which is retained rather than mechanically adjusted.
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the underlying ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. For Finland, older source tables separately report industrial corporations, other issuers, and credit institutions. For the harmonized series, industrial corporations and other issuers are combined into an "Industry and Other" category, which provides the closest historical counterpart to the NFC sector. For gross issuance, however, the corresponding historical tables report only the broader residual category "Others" alongside credit institutions. This category is therefore used as the gross-issuance NFC proxy; clarification from the Bank of Finland confirms that it includes non-financial corporations. A substantial reporting break occurs in 1983. From that year onward, the source explicitly distinguishes enterprises and financial institutions and reports both public issues and private placements. The resulting increase in measured NFC and TC issuance is therefore treated as a source and coverage break rather than mechanically adjusted. The post-1983 enterprise category replaces the earlier historical proxy based on "Others." From 1990 onward, the national historical construction is replaced by harmonized series constructed from the underlying ECB issuer-sector data despite the large level difference in the overlap with the Finnish historical source. NFC corresponds to ECB sector 1100; FC combines sectors 1220 and 1235; TC is constructed as NFC plus FC; and GOV corresponds to central government sector 1311. Monthly ECB gross issuance is aggregated to annual frequency by summing the twelve monthly observations.

6.5.8 Germany

Table 12: Source construction for Germany: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed directly from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized NFC and FC gross-issuance series.

Notes. The current German series relies on ECB Securities Issues Statistics from 1990 onward. The sector aggregates used in the dataset are constructed from the underlying ECB issuer-sector series rather than taken from pre-assembled ECB NFC, FC, and TC aggregates. NFC corresponds to ECB sector 1100; FC combines sectors 1220 and 1235; and TC is subsequently constructed as NFC plus FC. Monthly gross issuance is aggregated to annual frequency by summing the twelve monthly observations; years without complete January–December coverage are not retained. Historical German issuance data from the Deutsche Bundesbank and historical compilations by Hoffmann and Rahlf are retained in the underlying country file but are not currently incorporated into the harmonized dataset. Their coverage and definitions do not permit a sufficiently consistent decomposition of corporate issuance into financial and non-financial corporations.

6.5.9 Italy

Table 13: Source construction for Italy: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1931–1936	Banca d'Italia, *L'Economia Italiana nel Sessennio 1931–1936*, Table XXIX	Stato, nuove emissioni.	The central-government category "Stato" is assigned to GOV. Municipal and provincial issues are excluded. The observation assigned to 1935 requires particular caution: the source reports an aggregate covering operations between June 1934 and December 1936, which is assigned to 1935 in the country file.
NFC	Gross issues	1931–1936	Banca d'Italia, *L'Economia Italiana nel Sessennio 1931–1936*, p. 178, Table XXIX	Obbligazioni industriali, emesse direttamente, nuove emissioni.	NFC issuance includes industrial bonds issued directly by the underlying industrial entities.
FC	Gross issues	1931–1936	Banca d'Italia, *L'Economia Italiana nel Sessennio 1931–1936*, p. 178, Table XXIX	Land and building credit institutions; agricultural-improvement credit institutions.	FC issuance is constructed as the sum of bond issues by land and building credit institutions and agricultural-improvement credit institutions.
TC	Gross issues	1931–1936	Banca d'Italia, *L'Economia Italiana nel Sessennio 1931–1936*, p. 178, Table XXIX	Directly issued industrial bonds; industrial bonds issued through parastatal credit institutions; financial-corporation bond issues.	TC is constructed as the sum of NFC, FC, and industrial bonds reported as issued through parastatal credit institutions. The latter category is retained in TC but is not assigned to either NFC or FC because the source groups together industrial financing undertaken through institutions whose issuer-sector classification is ambiguous for the purposes of the harmonized sector split. Consequently, TC need not equal NFC plus FC during 1931–1936.
GOV	Gross issues	1958–1989	Banca d'Italia, *Bollettino Statistico*, various editions	Titoli di Stato and underlying government debt-security categories.	The series is constructed from reported gross issues of central-government debt securities. For 1969–1988, ordinary Treasury bills (BOT) are reported separately from the main *Titoli di Stato* aggregate and are added back. Territorial authorities and public enterprises are not included in GOV.

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Table 13: Source construction for Italy: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1958–1989	Banca d'Italia, *Bollettino Statistico*, various editions	Public non-financial enterprises and private enterprises.	NFC issuance combines bond issues by public non-financial enterprises with those of private enterprises. Public-enterprise categories include the state railway and, depending on the reporting period, ENEL, ENI, IRI/EFIM, and Autostrade IRI.
FC	Gross issues	1958–1968	Banca d'Italia, *Bollettino Statistico*, various editions	Special credit institutions: mobiliare and immobiliare.	FC issuance is constructed from bond issues of special credit institutions classified as financial intermediaries, principally the *istituti speciali mobiliari* and *istituti speciali immobiliari*.
FC	Gross issues	1969–1998	Banca d'Italia, *Bollettino Statistico*, various editions	Special credit institutions, including public-works, mobiliare, and immobiliare sections.	FC combines the financial-intermediary categories reported in the national source. Banca d'Italia data are retained through 1998 because the ECB sector components used to construct the modern FC aggregate do not provide observations for Italy over 1990–1998.
TC	Gross issues	1958–1989	Banca d'Italia, *Bollettino Statistico*, various editions	Financial and non-financial corporate bond issuance.	TC is constructed as the sum of the corresponding NFC and FC gross-issuance series.
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1999–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200). The underlying ECB components do not provide usable FC observations for Italy over 1990–1998, so the national Banca d'Italia FC series is retained through 1998.

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Table 13: Source construction for Italy: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1990–1998	ECB Data Portal and Banca d'Italia	Constructed NFC and FC series.	Because the ECB sector components used to construct FC are unavailable over 1990–1998, TC is constructed as the sum of the final sectoral series: ECB-based NFC gross issuance and the retained Banca d'Italia FC gross-issuance series.
TC	Gross issues	1999–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. Italian historical data are unavailable in the current country file for 1937–1957 and these years are left missing. For 1931–1936, industrial bonds issued through parastatal credit institutions are retained in TC but excluded from the NFC/FC decomposition because their classification by issuer sector is not sufficiently clear from the source. Hence, TC exceeds NFC plus FC whenever this category has entries. For the modern period, the sector aggregates are constructed from the underlying ECB issuer-sector series rather than taken from pre-assembled ECB NFC, FC, and TC aggregates. NFC corresponds to ECB sector 1100; the constructed FC aggregate combines sectors 1220 and 1235; and TC is constructed from the final NFC and FC series. For Italy, the national Banca d'Italia FC series is retained through 1998 because the ECB components required for the constructed FC aggregate are unavailable before 1999. Monthly ECB gross issuance is aggregated to annual frequency by summing the twelve monthly observations.

6.5.10 India

Table 14: Source construction for India: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1970–2022	Reserve Bank of India, *Handbook of Statistics on the Indian Economy*, Table 70, "New Capital Issues by Non-Government Public Limited Companies"; underlying source SEBI	Debentures/Bonds; separately reported Bonds where applicable.	TC follows debt-security issuance by non-government public limited companies. The source's "Debentures/Bonds" category is used throughout and the separately reported "Bonds" category is added where applicable. In 2007–08, for example, the construction combines both reported debt categories. Source values reported in crore rupees are converted to rupees using 1 crore = 10^7 rupees.

Notes. The Indian series is taken from the Reserve Bank of India's *Handbook of Statistics on the Indian Economy*, Table 70, which reports new capital issues by non-government public limited companies. The debt component is used as a source-based total-corporate bond-issuance proxy. The source does not provide a sufficiently consistent financial versus non-financial issuer decomposition, and separate NFC and FC series are therefore not constructed. The coverage is determined by the source's legal and statistical population of non-government public limited companies and should therefore not be interpreted as issuance by all Indian corporations. The time convention changes within the source. Observations through 1980 are reported on a January–December basis, while observations from 1981 onward are reported for April–March fiscal years. Fiscal-year observations are assigned to the calendar year in which the fiscal year begins. Where the source reports a separate "Bonds" category in addition to "Debentures/Bonds", both debt categories are included.

6.5.11 Japan

Table 15: Source construction for Japan: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1919–1941	Bank of Japan, Institute for Monetary and Economic Studies (IMES), Historical Statistics, "Outstanding Amounts of Government Securities and Corporate Bonds"	Government bonds and other government securities.	Annual gross issuance is constructed by aggregating the monthly issue amounts in the Bank of Japan historical database. The underlying database covers April 1919–December 1941 and reports amounts issued, redeemed, and outstanding separately for government securities and corporate bonds. For 1919 the data starts in April, i.e. it is a partial-year observation, and therefore coded as NA for the final compilation.
FC	Gross issues	1919–1941	Bank of Japan, Institute for Monetary and Economic Studies (IMES), Historical Statistics, "Outstanding Amounts of Government Securities and Corporate Bonds"	Bank bonds.	FC is constructed by summing monthly issues of bank bonds to annual frequency. For 1919 the data starts in April, i.e. it is a partial-year observation, and therefore coded as NA for the final compilation.
NFC	Gross issues	1919–1941	Bank of Japan, Institute for Monetary and Economic Studies (IMES), Historical Statistics, "Outstanding Amounts of Government Securities and Corporate Bonds"	Corporation bonds.	NFC corresponds to the non-bank corporation-bond component of the historical database. Equivalently, it is consistent with the harmonized construction of TC less the separately identified bank-bond component. For 1919 the data starts in April, i.e. it is a partial-year observation, and therefore coded as NA for the final compilation.
TC	Gross issues	1919–1941	Bank of Japan, Institute for Monetary and Economic Studies (IMES), Historical Statistics, "Outstanding Amounts of Government Securities and Corporate Bonds"	Bank bonds and corporation bonds.	TC is constructed as the sum of annualized bank-bond and corporation-bond gross issuance. The underlying monthly data distinguish internal and external bonds; both components are retained where reported. For 1919 the data starts in April, i.e. it is a partial-year observation, and therefore coded as NA for the final compilation.

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Table 15: Source construction for Japan: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1946–1984	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-22-a	Government bonds and other government securities.	GOV uses the reported total issues of government securities. The historical table is reported on a fiscal-year basis and includes the government-security categories covered by the contemporary source.
FC	Gross issues	1946–1984	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-23-a	Bank debentures.	Bank debentures are assigned to FC. These are securities issued by specialized financial institutions under the institutional arrangements prevailing in the period.
NFC	Gross issues	1946–1984	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-23-a	Industrial/corporate bonds.	NFC is constructed as TC minus bank debentures. Over this period this residual corresponds closely to the separately reported industrial/corporate-bond component. Convertible bonds are included within the historical industrial-bond category in years in which the source reports them jointly.
TC	Gross issues	1946–1984	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-23-a	Bank debentures and industrial/corporate bonds.	TC combines the corporate debt-security categories used for the harmonized private corporate sector. Bonds issued by public corporations and local governments, which are separately identified in the source, are excluded from TC.
GOV	Gross issues	1985–1997	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-22-b	Government securities, total issues.	GOV follows the source-reported government total. The observations are reported on a fiscal-year basis.
FC	Gross issues	1985–1997	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-23-b	Bank debentures.	FC corresponds to bank debentures, which the source defines as securities issued by financial institutions authorized to issue such instruments under special legislation.
NFC	Gross issues	1985–1997	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-23-b	Straight bonds; convertible bonds; bonds with subscription rights.	NFC is constructed residually as TC minus bank debentures. Consequently, the harmonized NFC concept is broader than the source's separately reported "straight bonds" series and also incorporates the other non-bank corporate bond instruments included in TC.

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Table 15: Source construction for Japan: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1985–1997	Statistics Bureau of Japan, *Historical Statistics of Japan*, Table 14-23-b	Bank debentures; straight bonds; convertible bonds; bonds with subscription rights.	TC is constructed from the corporate debt-security categories reported by the source. Yen-denominated bonds issued by non-residents are excluded and retained separately in the country file.
GOV	Gross issues	1998–2018	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Government bonds.	GOV uses the JSDA government-bond issuance aggregate. The JSDA definition includes government bonds issued through the market and through designated public-sector subscriptions, subject to the instrument exclusions specified in the source documentation.
FC	Gross issues	1998–2018	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Bank debentures.	The separately reported bank-debenture series is retained as the historical FC proxy. This is not an exhaustive institutional-sector measure of all financial-corporation bond issuance: under the JSDA classification, some straight bonds issued by banks are included within the broader corporate-bond category.
NFC	Gross issues	1998–2018	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Corporate straight bonds.	The JSDA corporate-bond series is used as the closest available NFC proxy after the source transition. The classification is not identical to a modern institutional-sector NFC concept: JSDA corporate straight bonds may include bonds issued by banks as well as selected public or quasi-public corporations. The resulting series is therefore interpreted as a source-based proxy rather than an exact NFC sector aggregate.
TC	Gross issues	1998–2018	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Corporate bonds, asset-backed bonds, convertible bonds, privately placed corporate bonds, and bank debentures.	TC retains the broader corporate debt-security universe available in the JSDA statistics. Consequently, during the JSDA period TC need not equal the sum of the narrower NFC proxy and bank debentures.
GOV	Gross issues	2020–2024	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Government bonds.	The same JSDA construction is continued using the later source vintage.

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Table 15: Source construction for Japan: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
FC	Gross issues	2020–2024	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Bank debentures.	The JSDA bank-debenture series is retained as the FC proxy.
NFC	Gross issues	2020–2024	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Corporate straight bonds.	The JSDA corporate-bond series is retained as the NFC proxy, subject to the same institutional-classification limitation as in 1998–2018.
TC	Gross issues	2020–2024	Japan Securities Dealers Association (JSDA), *Issuing, Redemption and Outstanding Amounts of Bonds*	Corporate debt-security categories and bank debentures.	TC follows the same broad JSDA corporate debt-security construction as in 1998–2018.

Notes. For 1919–1941, the pre-war series are constructed from the Bank of Japan IMES Historical Statistics. The annual gross-issuance series are obtained by summing the monthly issue flows. The data begins in April 1919 and runs through December 1941. Its underlying source is the Bank of Japan Research Department's *Statistics of Loans and Debentures*, subsequently published under several revised titles. Through 1997, the corporate-sector split is constructed from a broad TC aggregate and the separately identifiable bank-debenture series. FC is assigned to bank debentures and NFC is obtained residually as TC minus FC. For 1946–1984 this residual is close to the source's industrial/corporate-bond category. From 1985, the broader TC aggregate additionally contains several corporate instruments, including straight bonds and convertible bonds. A more substantial classification break occurs with the switch to JSDA statistics in 1998. JSDA categories are organized primarily by security type rather than by institutional sector. In particular, "Corporate Straight Bonds" may include straight bonds issued by banks, while "Bank Debentures" cover a specific class of securities issued by designated financial institutions. The post-1998 NFC and FC series should therefore be interpreted as source-based sector proxies given source limitations. TC is broader and includes additional corporate debt instruments such as asset-backed and convertible bonds and, where covered by the source, privately placed corporate bonds. These classification differences are retained and documented. The change in statistical source and classification in 1998 is therefore treated as a source break. 2019 is left missing due to source limitations.

6.5.12 Sweden

Table 16: Source construction for Sweden: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
FC	Gross issues	1835–2021	Waldenström (2022), *The Swedish Bond Market, 1835–2020*, Historical Monetary and Financial Statistics for Sweden, Vol. III, Sveriges Riksbank; companion data	Financial institutions, domestic/SEK issues; financial corporations, foreign-currency issues	FC is constructed as the sum of gross issuance by financial institutions in local currency and financial-corporation issuance in foreign currency. The source financial sector includes mortgage institutions and later the source classifications banks, credit institutions and other financial-market institutions.
NFC	Gross issues	1855–2021	Waldenström (2022), *The Swedish Bond Market, 1835–2020*, Historical Monetary and Financial Statistics for Sweden, Vol. III, Sveriges Riksbank; companion data	Non-financial corporations; railway companies; industrial corporations; other corporations; foreign-currency non-financial corporate issues	NFC combines the contemporaneously available non-financial corporate categories. Railway companies are classified as NFC. The historical source separately reports railway, industrial and other corporate borrowers in periods in which this finer breakdown is available; later source aggregates are used directly. Foreign-currency non-financial corporate issuance is added to the corresponding domestic series.
TC	Gross issues	1835–2021	Waldenström (2022), *The Swedish Bond Market, 1835–2020*, Historical Monetary and Financial Statistics for Sweden, Vol. III, Sveriges Riksbank; companion data	Financial institutions and non-financial corporate borrowers, including railways; domestic and foreign-currency issues	TC is constructed as the sum of FC and NFC. It therefore includes financial corporations as well as railway, industrial and other non-financial corporate issuers.
GOV	Gross issues	1854–2021	Waldenström (2022), *The Swedish Bond Market, 1835–2020*, Historical Monetary and Financial Statistics for Sweden, Vol. III, Sveriges Riksbank; companion data	State/central government, domestic/SEK issues and foreign-currency issues	GOV is the sum of central-government bonds issued in local and foreign currency.

Notes. The Swedish series are based on the long-run primary-bond-market database constructed by Waldenström (2022a). The issuance concept is the value of newly issued bond loans. Short-term money-market instruments, including Treasury bills and bank certificates with maturities of at most one year, are outside the source perimeter.The source distinguishes four broad borrower groups: the central government, municipalities, financial institutions and corporations. Financial institutions include mortgage associations and, in later periods, banks, credit institutions, insurance companies and other financial-market actors. The source corporate category is non-financial historically consists primarily of manufacturing firms and railway companies, together with other private non-financial institutions. Accordingly, railway-company issuance is included in NFC in the harmonized dataset.

The baseline aggregates combine domestically and externally issued bonds. Domestic issues are generally identified as SEK-denominated bonds, while external issues are identified as bonds issued abroad and/or denominated in foreign currency. The final country series therefore adds the domestic and foreign-currency components by issuer sector. GOV is restricted to the central government. TC is defined consistently with the rest of the cross-country dataset as the sum of FC and NFC.

6.5.13 Switzerland

Table 17: Source construction for Switzerland: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1848–2006	Swiss National Bank, *Historical Time Series 2: Capital and Stock Markets* (2007), Table 2.1 and supplementary online historical series	Confederation (*Bund*); publicly and, where available, non-publicly floated CHF bonds	GOV follows the source category for the Confederation. The historical issuing-value series is available back to 1848. The SNB notes that the Confederation category also includes bonds issued directly by the Swiss Federal Railways (SBB/CFF); this source-defined historical perimeter cannot be separated consistently and is therefore retained. Cantonal and municipal issuance is excluded.
TC	Gross issues	1900–2006	Swiss National Bank, *Historical Time Series 2: Capital and Stock Markets* (2007), Tables 2.1 and 2.1a	Utilities; manufacturing; commerce; banks; mortgage bond institutions; holding companies; transport companies; finance companies; other	TC is constructed by summing the corporate issuer categories reported by the SNB. Publicly floated bonds are available from 1900. Non-publicly floated bonds are additionally included where reported, principally for 1971–2003. The residual category *Other* is retained in TC but is not allocated mechanically to either NFC or FC.
FC	Gross issues	1900–2006	Swiss National Bank, *Historical Time Series 2: Capital and Stock Markets* (2007), Tables 2.1 and 2.1a	Banks; mortgage bond institutions; holding companies; finance companies	FC is constructed as the sum of the identifiable financial issuer categories, using both publicly and non-publicly floated issues where available. Holding companies are assigned to FC as a harmonization choice because they are separately identified as entities whose principal function is the holding and control of participations or assets.
NFC	Gross issues	1900–2006	Swiss National Bank, *Historical Time Series 2: Capital and Stock Markets* (2007), Tables 2.1 and 2.1a	Utilities; manufacturing; commerce; transport companies	NFC is constructed from the identifiable non-financial issuer categories. The residual *Other* category is excluded from NFC and retained only in TC.
GOV	Gross issues	2007–2024	Swiss National Bank Data Portal, *Capital market borrowing of CHF bond issues*	Domestic issuers – Confederation	GOV follows the SNB Confederation series. Cantons and municipalities remain separately reported by the source and are excluded from the baseline GOV aggregate. Quarterly gross issuance is summed to calendar years.

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Table 17: Source construction for Switzerland: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	2007–2024	Swiss National Bank Data Portal, *Capital market borrowing of CHF bond issues*	Utility providers (electricity, gas, water); Industry	NFC is constructed as the sum of utility providers and industry. The modern SNB sector classification differs from the historical issuer classification and does not separately preserve categories such as commerce or transport.
FC	Gross issues	2007–2024	Swiss National Bank Data Portal, *Capital market borrowing of CHF bond issues*	Mortgage bond institutions; Banks; Insurance; Other services	FC is constructed as the sum of mortgage bond institutions, banks, insurance companies, and the SNB category *Other services*. The inclusion of *Other services* is a harmonization choice in the compiled dataset. The SNB capital-market statistics do not provide a direct institutional-sector mapping or a bridge linking this category to the earlier historical issuer categories; it should therefore not be interpreted as an SNB classification of all *Other services* issuers as financial corporations.
TC	Gross issues	2007–2024	Swiss National Bank Data Portal, *Capital market borrowing of CHF bond issues*	Mortgage bond institutions; utility providers; industry; banks; insurance; other services; others	TC is constructed from all domestic corporate issuer categories, excluding the Confederation, cantons and municipalities. The residual category *Others* is included in TC but is not allocated to NFC or FC. TC can therefore exceed NFC+FC.

Notes. The historical series are taken primarily from the Swiss National Bank's 2007 *Historical Time Series 2: Capital and Stock Markets*. The corporate issuer breakdown begins in 1900, while the Confederation issuing-value series is available back to 1848.

For the historical period, the dataset uses the SNB's *issuing value* as the gross-issuance concept. This measure includes conversion issues. Publicly floated bonds are supplemented by non-publicly floated bonds where the latter are available. The SNB supplementary Table 2.1a reports non-publicly floated bonds by issuer for 1971–2003.

The historical issuer classification allows a relatively detailed sectoral decomposition. Banks, mortgage bond institutions, finance companies and holding companies are assigned to FC. Utilities, manufacturing, commerce and transport companies are assigned to NFC. The residual *Other* category is retained in TC only. This preserves the observed corporate issuance without forcing an uncertain sectoral allocation, and implies that TC does not necessarily equal the sum of NFC and FC.

The treatment of holding companies is a harmonization decision. Because holding companies are separately identified by the historical source and primarily exist to hold and control participations or assets, they are assigned to FC throughout the historical construction. The historical GOV series follows the SNB's *Confederation* category. The SNB explicitly notes that this category also includes bonds issued directly by the Swiss Federal Railways. Because these issues cannot be separated consistently over the long historical sample, the source-defined category is retained.

The final dataset switches after 2006 to the modern data from SNB's Data Portal, where quarterly gross issues are summed to annual frequency. In the modern construction, NFC consists of utility providers and industry. FC consists of mortgage bond institutions, banks, insurance companies and *Other services*.The remaining *Others* category is retained in TC only. The switch in 2007 from the historical SNB compilation to the current Data Portal entails a change in issuer classification and source construction. The resulting series are linked without level adjustment; the break is preserved as part of the underlying historical statistical record.

6.5.14 South Korea

Table 18: Source construction for South Korea: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1977–2004	Bank of Korea, *Economic Statistics Yearbook*, various editions	Treasury bonds.	The baseline GOV series uses gross issuance of Treasury bonds. Other public-sector securities reported separately by the Bank of Korea are not added to the baseline series. Where a sufficiently comparable Treasury-bond observation is unavailable in the historical source, the observation is left missing rather than reconstructed.
GOV	Gross issues	2005–2024	Bank of Korea statistical database	Treasury bonds.	The historical Treasury-bond series is continued using the corresponding modern Bank of Korea series.
TC	Gross issues	1977–1992	Bank of Korea, *Economic Statistics Yearbook*, various editions	Corporate bonds.	TC follows the source-reported "Corporate bonds" series. The source separately reports several financial-institution debt instruments, including Industrial Finance Debentures and, in some vintages, Foreign Exchange and Long-term Credit Debentures. The historical "Corporate bonds" category should therefore not be interpreted as an exhaustive institutional-sector aggregate of NFC and FC issuance.
TC	Gross issues	1993–2024	Bank of Korea, *Economic Statistics Yearbook* and data supplied by the Bank of Korea	Corporate bonds: non-financial firms, non-bank financial institutions, and, from 2000, asset-backed securities.	The series extends the historical Bank of Korea Statistical Yearbook concept. Bank of Korea clarification confirms that this aggregate excludes bonds issued by banks. From 2000 onward it includes asset-backed securities issued by either NFCs or FCs. A separate broader series supplied by the Bank of Korea includes bank bond issuance, but is not used for the long-run TC series in order to preserve continuity with the historical Yearbook definition.
NFC	Gross issues	1993–2024	Bank of Korea, data supplied upon request	Bonds issued by non-financial firms.	NFC is taken directly from the sector-specific series supplied by the Bank of Korea. This is preferred to treating the broader source category "Corporate bonds" as equivalent to NFC issuance.

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Table 18: Source construction for South Korea: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
FC excluding banks	Gross issues	2000–2024	Bank of Korea, data supplied upon request	Bonds issued by financial institutions excluding banks.	The series identifies bond issuance by non-bank financial corporations separately from NFC issuance. It is retained as an intermediate component for constructing the harmonized FC aggregate once bank issuance also becomes available.
Banks	Gross issues	2008–2024	Bank of Korea, data supplied upon request	Bonds issued by banks.	Bank bond issuance becomes separately available in the supplied sectoral series from 2008.
FC	Gross issues	2008–2024	Bank of Korea, data supplied upon request	Financial institutions excluding banks; banks.	The harmonized FC series is constructed as the sum of bond issuance by financial institutions excluding banks and bond issuance by banks. Earlier observations are not backcast because a complete financial-sector decomposition is not available.
Auxiliary FC	Gross issues	1977–2024	Bank of Korea, *Economic Statistics Yearbook* and modern statistical database	Industrial Finance Debentures.	Industrial Finance Debentures are issued by the Korea Development Bank (KDB). Following clarification from the Bank of Korea, they are classified as financial-sector securities. Although KDB is government-owned, these debentures are not treated as central-government debt. The series is retained separately and is not interpreted as an exhaustive measure of historical FC bond issuance.

Notes. The Korean bond statistics contain several categories whose definitions do not map directly into the harmonized institutional sectors used elsewhere. The TC series follows the Bank of Korea's historical "Corporate bonds" aggregate. Bank of Korea clarification indicates that this aggregate should not be equated with either NFC issuance or with a complete NFC-plus-FC sector. The corresponding Statistical Yearbook concept covers bonds issued by non-financial firms and non-bank financial institutions and, from 2000 onward, includes asset-backed securities that may be issued by either NFCs or FCs. Bank bond issuance are only available after 2007.

Bank of Korea clarification provided with the post-1992 data establishes that the Statistical Yearbook "Corporate bonds" concept excludes banks: it comprises issuance by non-financial firms and financial institutions excluding banks and, from 2000 onward, asset-backed securities issued by either NFCs or FCs. A separate series supplied by the Bank of Korea includes bank issuance. This precise decomposition is documented for the modern continuation; for the earlier historical observations, the Yearbooks separately report several financial-institution debenture categories, so the "Corporate bonds" series is likewise not interpreted as an exhaustive NFC-plus-FC aggregate.

Industrial Finance Debentures are reported separately over a much longer period. According to clarification provided by the Bank of Korea, these securities are issued by the Korea Development Bank to finance industrial development and are appropriately classified with financial corporations. The baseline GOV series is deliberately narrower than the full set of Korean public-sector securities and consists of Treasury bonds. Bank of Korea clarification identifies Grain Securities, National Housing Bonds, Foreign Exchange Stabilization Fund Bonds, and Seoul Metropolitan Subway Bonds as additional public-sector securities. These categories are retained separately in the underlying country file but are not added to the baseline GOV series. Monetary Stabilization Bonds are issued by the Bank of Korea for monetary-policy operations and are excluded from both GOV and the corporate aggregates.

6.5.15 Spain

Table 19: Source construction for Spain: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1954–1961	Banco de España, *Boletín Estadístico*, December 1965, Table VI-2	Private issues: bonds (*Emisiones privadas: Obligaciones*).	TC follows the source-reported aggregate of private bond issues, reported in effective amounts. The underlying table does not provide a sufficiently consistent issuer-sector decomposition for constructing separate NFC and FC series over these years. NFC and FC are therefore left unavailable rather than allocated retrospectively.
GOV	Gross issues	1954–1961	Banco de España, *Boletín Estadístico*, December 1965, Table VI-2	Central administration (*Administración central*).	GOV corresponds to gross public bond issuance by the central administration. INI, other official bodies, municipalities and provincial authorities, and official credit institutions are separately reported by the source and are excluded from GOV. Source-reported absence of central-government issuance in 1958 and 1959 is retained as zero.
NFC	Gross issues	1962–1964	Banco de España, *Boletín Estadístico*, December 1975, Tables VI-1 and VI-3	Private non-financial corporations; Instituto Nacional de Industria (INI).	NFC is constructed as the sum of gross bond issuance by private non-financial corporations and separately reported INI issuance. INI is classified as a public non-financial enterprise rather than central government. This provides a broader NFC concept than the private-sector series alone.
FC	Gross issues	1962–1989	Banco de España, *Mercado de Valores: Instituciones Financieras*, *Renta Fija*, *Series Históricas 1962–1982*, Table VI-5; subsequent *Boletines Estadísticos*	Financial institutions (*Instituciones financieras*).	FC follows the source-reported gross issuance of financial institutions. The underlying historical tables distinguish the credit system from other financial institutions, including mortgage-credit institutions where separately reported; the published financial-institution total is used.
NFC	Gross issues	1965–1989	Banco de España, *Boletín Estadístico*, historical and subsequent editions	Non-financial enterprises, total sector.	NFC follows the broader Banco de España aggregate for non-financial enterprises. The series incorporates the relevant public non-financial enterprises, including INI, so INI issuance is not added separately to avoid double counting.

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Table 19: Source construction for Spain: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1962–1989	Banco de España, historical securities-market statistics and subsequent *Boletines Estadísticos*	Harmonized NFC and FC components.	TC is constructed as the sum of NFC and FC. The expansion from the pre-1962 private-issues aggregate to separately identified non-financial and financial issuer sectors produces a pronounced coverage break in 1962, which is retained rather than adjusted.
GOV	Gross issues	1962–1989	Banco de España, *Mercado de Valores: Administraciones Públicas, Series Históricas 1962–1981*, Document No. 8212, Table VI-1; subsequent *Boletines Estadísticos*	State, negotiable debt (*Estado: emisiones de deudas negociables*).	GOV uses gross issues of negotiable debt by the State. The State category is interpreted as central government. Other public administrations and public-sector institutions are excluded. The historical series is deliberately restricted to negotiable debt; separately reported non-negotiable liabilities are not added.
INI	Gross issues	1962–1997	Banco de España, historical securities-market tables and subsequent *Boletines Estadísticos*	Instituto Nacional de Industria.	INI issuance is retained separately as an auxiliary public-enterprise series. It enters the harmonized NFC construction only where NFC must be reconstructed from components; once the source provides a comprehensive non-financial-enterprise total, it is not added separately. Source-reported zeros at the end of the series are retained as genuine zeros.
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the underlying ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. For 1954–1961, the source reports an aggregate of private bond issues but does not provide a sufficient decomposition to construct separate NFC and FC series. TC is therefore retained directly from the source while the sectoral series remain unavailable. A substantial reporting change occurs in 1962, when securities issuance becomes available by institutional issuer category. From this point onward TC is constructed as sum of NFC and FC.

For 1962–1964, private non-financial-corporation bond issuance is reported separately from issuance by the Instituto Nacional de Industria (INI). NFC is therefore constructed as the sum of these two components. INI is treated as a public non-financial enterprise. From 1965 onward, Banco de España reports a broader non-financial-enterprise aggregate that already incorporates the relevant public enterprises; the total is used directly and INI is not added again.

The financial-sector series begins in 1962. Banco de España's historical financial-institution tables distinguish the credit system and other financial institutions and report a corresponding gross-issuance total. This source-defined financial-institution aggregate is used as FC.

The government construction is restricted throughout to central government. In the earlier public-issues tables, *Administración central* is explicitly distinguished from INI, other official bodies, municipalities and provincial authorities, and official credit institutions. From 1962 onward the corresponding concept is the *Estado* series for negotiable government debt.

A major source and coverage break occurs in 1990. The final dataset deliberately switches from the historical Banco de España securities-market series to harmonized ECB Securities Issues Statistics. The ECB debt-securities framework has a substantially broader instrument and issuer coverage, producing a large level increase relative to the national historical series in the overlap. The overlap is therefore not used to level-adjust or mechanically splice the two series. For the ECB data, NFC corresponds to sector 1100; FC combines sectors 1220 and 1235; GOV corresponds to central government sector 1311; and TC is constructed as sum of NFC and FC. Monthly ECB gross issuance is summed to calendar-year frequency.

6.5.16 Netherlands

Table 20: Source construction for the Netherlands: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1914–1918	*Jaarcijfers voor Nederland*, various editions	Dutch state loans (*Ned. Staatsschuld*).	GOV uses the separately reported Dutch state-loan category. Provincial and municipal issues are excluded.
GOV	Gross issues	1919–1936	*Jaarcijfers voor Nederland*, various editions	Netherlands and Netherlands Indies state loans (*Ned. en N.-I. Staatsleeningen*).	The historical source reports Netherlands and Netherlands Indies state loans jointly. The reported aggregate is retained because the two components cannot be separated consistently. Provincial and municipal bonds are excluded.
GOV	Gross issues	1937–1940	*Jaarcijfers voor Nederland*, various editions	State of the Netherlands (*Staat der Nederlanden*).	GOV follows the source's revised state-loan category. Provincial, municipal and water-board bonds are excluded.
FC	Gross issues	1914–1940	*Jaarcijfers voor Nederland*, various editions	Bank and credit-institution bonds; mortgage-bank *pandbrieven*.	FC is constructed as the sum of bonds issued by banks and credit institutions and mortgage-bank *pandbrieven*. The terminology changes slightly across successive editions, but these two financial issuer groups can be identified throughout the pre-war sample.
NFC	Gross issues	1914–1940	*Jaarcijfers voor Nederland*, various editions	Industry; trade; mining; petroleum; shipping; railways and tramways; cultivation companies and, where separately reported, rubber, sugar, tobacco, tea and other cultivation companies.	NFC is constructed by summing all separately identifiable non-financial corporate issuer categories. The precise disaggregation varies across source vintages. Railways and tramways are included in NFC and are also retained separately in the country file as an auxiliary series.
TC	Gross issues	1914–1940	*Jaarcijfers voor Nederland*, various editions	FC and NFC categories; miscellaneous bonds (*Diverse/Diversen obligatiën*).	TC includes all identified financial and non-financial corporate bond categories as well as the source's residual "miscellaneous bonds" category. Because the latter cannot be assigned reliably to FC or NFC, TC can exceed FC+NFC. The residual category is retained in TC rather than allocated mechanically to either sector.
GOV	Gross issues	1952–1980	Centraal Bureau voor de Statistiek (CBS), *Statistisch zakboek*, various editions	State loans (*Rijk/Staatsleningen*).	GOV uses central/state government bond issuance. Provinces, municipalities, polders, water boards and other public-law bodies are excluded from the baseline GOV series.

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Table 20: Source construction for the Netherlands: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
FC	Gross issues	1952–1957	CBS, *Statistisch zakboek*, various editions	Banking and lending institutions.	The earliest post-war FC observations use the consistently identifiable banking and lending-institution category available in the source. Other financial subcategories are not sufficiently consistently reported over these first years to construct the later broader financial-sector aggregate.
FC	Gross issues	1958–1967	CBS, *Statistisch zakboek*, various editions	Banking, credit and insurance services (*Bank-, krediet- en verzekeringswezen*).	FC uses the source-reported aggregate for banking, credit and insurance activities. Where Bank voor Nederlandsche Gemeenten is already contained as a subcategory of this aggregate, it is not added separately, avoiding double counting.
FC	Gross issues	1968–1980	CBS, *Statistisch zakboek*, various editions	Banking, credit and insurance services; Bank voor Nederlandsche Gemeenten / Nederlandse Waterschapsbank; real-estate funds.	FC combines the source's banking, credit and insurance aggregate with Bank voor Nederlandsche Gemeenten and, subsequently, Nederlandse Waterschapsbank when these institutions are reported separately from the financial aggregate. Real-estate funds (*Vastgoedfondsen*) are also assigned to FC where separately reported. The treatment therefore follows the changing source presentation while avoiding double counting when the public-sector banks are already included within the broader financial category.
NFC	Gross issues	1952–1954	CBS, *Statistisch zakboek*, various editions	Trade; industry; electricity, gas and water; mining; petroleum; shipping; railways and tramways; religious and charitable institutions; sugar where separately reported.	NFC is constructed by summing the separately reported non-financial private issuer categories. Polders and water boards, which appeared in an earlier working construction, are excluded because they are public-law bodies rather than corporations.
NFC	Gross issues	1955–1956	CBS, *Statistisch zakboek*, various editions	Industry; trade; electricity, gas and water; cultures; shipping and air transport; railways and tramways; mining; religious and charitable institutions.	The components are summed to obtain a broad private non-financial issuer aggregate. The available sector detail varies with the source table.

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Table 20: Source construction for the Netherlands: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1957–1960	CBS, *Statistisch zakboek*, various editions	Trade, industry and miscellaneous; cultures; shipping and air transport; railways and tramways; mining and petroleum; religious and charitable institutions.	NFC is constructed from the broader trade/industry/miscellaneous aggregate plus the separately reported non-financial issuer categories. Components already contained in the broader aggregate are not added separately.
NFC	Gross issues	1961–1962	CBS, *Statistisch zakboek*, various editions	Trade, industry and miscellaneous; shipping and air transport; railways and tramways; religious and charitable institutions.	The construction follows the sector detail available in these editions; categories not separately reported are not imputed.
NFC	Gross issues	1963–1969	CBS, *Statistisch zakboek*, various editions	Trade, industry and miscellaneous; cultures; shipping and air transport; railways and tramways; mining and petroleum; religious and charitable institutions.	NFC is the sum of the corresponding source categories. This remains a broad historical private non-financial issuer concept rather than a retrospective ESA-sector classification.
NFC	Gross issues	1970–1980	CBS, *Statistisch zakboek*, various editions	Trade, industry and miscellaneous; cultures; shipping and air transport; railways and tramways; mining and petroleum; health care and related institutions.	From 1970, the earlier religious/charitable category is replaced by the source's health-care and related-institutions category. Real-estate funds are not included in NFC and are instead assigned to FC. The remaining source categories are summed to form NFC.
TC	Gross issues	1952–1980	CBS, *Statistisch zakboek*, various editions	Harmonized FC and NFC components.	TC is constructed as FC+NFC. Provincial and municipal bodies, polders and water boards, and foreign issuers are excluded.
GOV	Gross issues	1981–1989	CBS statistical yearbooks, various editions	Government bonds (*Overheid: obligaties*).	A new source presentation begins in 1981. GOV follows the source-reported government-bond category.
FC	Gross issues	1981–1989	CBS statistical yearbooks, various editions	Financial institutions: ordinary bonds, subordinated bonds and *pandbrieven*; banks for lower government.	FC is constructed as the sum of ordinary bonds, subordinated bonds and *pandbrieven* issued by financial institutions, plus bonds issued by the separately reported "banks for lower government" category.
NFC	Gross issues	1981–1989	CBS statistical yearbooks, various editions	Other private sector: ordinary bonds and subordinated bonds.	NFC is constructed as the sum of ordinary and subordinated bond issuance reported for the "other private sector" (*Overige private sector*).
TC	Gross issues	1981–1989	CBS statistical yearbooks, various editions	Harmonized FC and NFC components.	TC is constructed as FC+NFC. The change in source presentation in 1981 is retained rather than retrospectively reallocating the earlier detailed categories.

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Table 20: Source construction for the Netherlands: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the underlying ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. The Dutch historical construction relies on unusually detailed issuer classifications, rather than imposing a single modern institutional classification retrospectively, the construction maps the source categories into the closest consistent GOV, FC, NFC and TC concepts. For 1914–1940, the data are compiled from successive editions of *Jaarcijfers voor Nederland*. FC consists of bank and credit-institution bonds and mortgage-bank *pandbrieven*. NFC combines the identifiable non-financial issuer categories, including industrial, commercial, extractive, transport and colonial cultivation enterprises. A residual "miscellaneous bonds" category is retained in TC but not assigned to either FC or NFC. Consequently, pre-war TC can exceed the sum of FC and NFC. From 1919 through the mid-1930s, government bond statistics jointly report the Netherlands and Netherlands Indies. No annual gross-issuance series is currently available for 1941–1951, coded as NA For 1952–1980, the exact set of categories changes repeatedly. The NFC series therefore sums the non-financial private issuer categories available in each source vintage. In the early post-war tables these include combinations of trade, industry, utilities, mining and petroleum, shipping, railways, cultivation activities, and religious/charitable institutions. The latter are retained because the historical category combines private non-commercial and health-related institutions that cannot be separated consistently. From 1970 the source instead identifies health care and related institutions more directly.

Fo financial issuers, Bank voor Nederlandsche Gemeenten, and later Nederlandse Waterschapsbank, can either appear inside the broader banking/credit/insurance category or as separately reported issuers. They are classified with FC because they operate as financial intermediaries. They are added separately only when they are not already contained in the reported financial aggregate. Real-estate funds (*Vastgoedfondsen*) are classified with FC where separately reported. Polders and water boards are excluded from NFC and, because the baseline GOV concept is restricted to the central/state government, are not added to GOV. A further reporting change occurs in 1981. The source focuses on broader issuer group reporting. FC is then constructed from ordinary and subordinated bonds and *pandbrieven* of financial institutions together with bonds of banks for lower government, while NFC consists of ordinary and subordinated bonds of the "other private sector." TC is constructed as their sum. From 1990 onward, the final dataset switches deliberately to the harmonized ECB Securities Issues Statistics. This creates a pronounced level break relative to the historical CBS series because the modern debt-securities statistics use a substantially broader and differently organized instrument and issuer universe. The overlap is therefore not used to level-adjust or splice the historical series. The ECB construction is preferred from 1990 for consistency with the modern series used for the other euro-area countries. NFC corresponds to sector 1100; FC combines sectors 1220 and 1235; GOV corresponds to central government sector 1311; and TC is the sum of NFC and FC. Monthly gross issuance is aggregated to annual frequency by summing the twelve monthly observations.

6.5.17 Norway

Table 21: Source construction for Norway: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1930–1952	Statistics Norway, *Historical Statistics 1994*, Table 24.25	Central government.	GOV uses the source-reported central-government bearer-bond issues in Norwegian kroner. The table reports foreign-currency issuance only as an aggregate without a consistent issuer decomposition, so no foreign-currency amount is allocated to GOV in years in which the issuer is not identified. In 1952 the source notes that the central-government category includes NOK 48 million issued by state enterprises; this amount cannot be separated consistently and is left within the reported source category.
TC proxy	Gross issues	1930–1952	Statistics Norway, *Historical Statistics 1994*, Table 24.25	Local government including municipal enterprises; loan associations; state banks where separately reported.	Before 1953 the historical source does not permit a clean NFC/FC decomposition. A broad source-based corporate-sector proxy is therefore retained. It combines loan associations, the joint category "local government including municipal enterprises," and, from 1946, separately reported state banks. The residual "Others" category is excluded because it cannot be allocated reliably; before 1946 it also contains state banks. The pre-1953 TC series should consequently not be interpreted as a precise institutional-sector NFC+FC aggregate.
GOV	Gross issues	1953–1992	Statistics Norway, *Historical Statistics 1994*, Table 24.26	Central government.	GOV follows the central-government issuer category. Issues denominated in Norwegian kroner and, where separately attributed to central government, foreign-currency issues are combined. No allocation of unclassified foreign-currency issuance is imposed.

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Table 21: Source construction for Norway: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1953–1982	Statistics Norway, *Historical Statistics 1994*, Table 24.26	Private non-financial enterprises; state enterprises.	NFC is constructed as the sum of private non-financial enterprises and state enterprises. This treatment follows Statistics Norway's institutional-sector classification, under which market-producing public enterprises owned by central government are part of the non-financial-corporation sector rather than general government. Foreign-currency issues are added when the source identifies the issuing sector.
NFC	Gross issues	1983–1992	Statistics Norway, *Historical Statistics 1994*, Table 24.26	Private non-financial enterprises; state enterprises; municipal enterprises.	From 1983, municipal enterprises are separately reported and are added to NFC together with private non-financial enterprises and state enterprises. Statistics Norway's institutional-sector classification places public enterprises owned by both central and local government in the NFC sector. During 1972–1978 municipal-enterprise issues are included within the broader municipalities-and-counties category and cannot be separated; they are therefore not imputed to NFC for those years.
FC	Gross issues	1953–1992	Statistics Norway, *Historical Statistics 1994*, Table 24.26	State banks; commercial banks; savings banks; loan associations; financial companies; life- and non-life-insurance companies.	FC is constructed by summing the separately reported financial issuer categories whenever available. State banks are classified with FC despite public ownership because they are financial intermediaries. The same functional treatment is applied to commercial and savings banks, loan associations, financial companies, and insurance companies. Foreign-currency issuance is included whenever an issuer-sector breakdown is available and is not otherwise imputed.
TC	Gross issues	1953–1992	Statistics Norway, *Historical Statistics 1994*, Table 24.26	Harmonized NFC and FC components.	TC is constructed as NFC+FC. Central and local government are excluded. Because the historical source does not always allocate total foreign-currency issuance across issuer sectors, TC includes foreign-currency issuance only where the corresponding NFC or FC issuer can be identified.

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Table 21: Source construction for Norway: gross bond issuance – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1993–1998	Statistics Norway, *Statistical Yearbook*, various editions	Central government including social security.	GOV follows the corresponding central-government gross bond-issuance category in the Statistical Yearbook tables.
NFC	Gross issues	1993–1998	Statistics Norway, *Statistical Yearbook*, various editions	State enterprises etc.; municipal enterprises; other Norwegian sectors.	NFC is constructed as the sum of state enterprises, municipal enterprises, and the residual "other Norwegian sectors" category. The treatment of public enterprises as NFCs is consistent with Statistics Norway's institutional-sector classification.
FC	Gross issues	1993–1998	Statistics Norway, *Statistical Yearbook*, various editions	State lending institutions; commercial banks including Postal Bank; savings banks; other credit institutions.	FC is constructed as the sum of the four separately reported financial-intermediary categories.
TC	Gross issues	1993–1998	Statistics Norway, *Statistical Yearbook*, various editions	Harmonized NFC and FC components.	TC is constructed as NFC+FC. Local and central government bond issuance is excluded.
GOV	Gross issues	1999–2024	Statistics Norway (SSB), modern securities statistics, successive StatBank series	Central government.	The historical government series is continued using the harmonized modern SSB issuer-sector data assembled in the country source file.
NFC	Gross issues	1999–2024	Statistics Norway (SSB), modern securities statistics, successive StatBank series	Public and private non-financial corporations.	NFC follows the modern SSB non-financial-corporation classification. The construction includes public enterprises owned by central and local government together with private non-financial corporations, consistent with SSB's institutional-sector definition.
FC	Gross issues	1999–2024	Statistics Norway (SSB), modern securities statistics, successive StatBank series	Financial corporations.	FC follows the harmonized modern SSB financial-corporation universe, including the relevant banking, credit-intermediation, insurance and other financial issuer categories represented in the successive source tables.
TC	Gross issues	1999–2024	Statistics Norway (SSB), modern securities statistics, successive StatBank series	Harmonized NFC and FC components.	TC is constructed as NFC+FC.

Notes. The Norwegian historical data are unusually detailed by issuer sector but undergo several changes in classification and instrument coverage. For 1930–1952, Table 24.25 reports central government, a combined local-government-and-municipal-enterprise category, loan associations, state banks, and a residual "Others" category. Before 1946 state banks are included within "Others." These categories do not permit a defensible historical NFC/FC decomposition, so only a broad source-based TC proxy is retained for this period. In particular, the joint local-government/municipal-enterprise category cannot be split retrospectively, while "Others" is left unallocated rather than assigned mechanically to the corporate sector.

From 1953 onward, Table 24.26 provides substantially finer issuer information and allows separate NFC and FC series to be constructed. The historical NFC series combines private non-financial

enterprises and state enterprises and, from 1983, separately reported municipal enterprises. This classification follows Statistics Norway's institutional-sector treatment: publicly owned enterprises engaged in market production of non-financial goods and services belong to the NFC sector, whereas public financial intermediaries such as state lending institutions belong to FC. Municipal enterprises cannot be separately identified during 1972–1978 because the source includes them within municipalities and counties.

FC combines state banks, commercial banks, savings banks, loan associations, financial companies, and life- and non-life-insurance companies whenever separately reported. TC is then constructed directly as the sum of NFC and FC.

The source separately reports local and foreign-currency issuance, but sector classification here appears only later. The harmonized sector series therefore add foreign-currency issues only where the issuing sector is explicitly identified; unallocated foreign-currency issuance is not distributed proportionally across sectors.

A source transition occurs after 1992. For 1993–1998, sectoral bond issuance is reconstructed from successive Statistics Norway *Statistical Yearbook* tables. The NFC aggregate combines state enterprises, municipal enterprises, and other Norwegian sectors; FC combines state lending institutions, commercial and postal banks, savings banks, and other credit institutions (see sector classification in *SSB (1991)Bank and Credit Statistics 1990–91*). From 1999 onward, the final series use harmonized modern Statistics Norway securities statistics assembled from successive StatBank series. The modern classification continues the same institutional principle, distinguishing non-financial corporations, including publicly owned market producers, from financial corporations and general government.

6.5.18 Portugal

Table 22: Source construction for Portugal.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1936–1989	Instituto Nacional de Estatística (2001), *Portuguese Historical Statistics*, Table 5J.5, pp. 434–436	Private bond issues	Total-corporate gross bond issuance follows the reported private-bond-issues series. The historical source does not provide an FC/NFC decomposition. From the underlying source, *Estatísticas Monetárias e Financeiras*, is it clear that this concerns corporate issues.
TC	Gross issues	1990–2021	ECB Data Portal, Securities Issues Statistics	Total corporations	Total-corporate gross bond issuance constructed as the sum of FC and NFC .
FC	Gross issues	1990–2021	ECB Data Portal, Securities Issues Statistics	Financial corporations	Financial-corporation gross bond issuance follows the ECB financial-corporations series.
NFC	Gross issues	1990–2021	ECB Data Portal, Securities Issues Statistics	Non-financial corporations	Non-financial-corporation gross bond issuance follows the ECB non-financial-corporations series.
GOV	Gross issues	1969–1989	Instituto Nacional de Estatística (2001), *Portuguese Historical Statistics*, Table 5J.5, pp. 434–436	Public bonds issues	The reported "public bonds issues" series is used as government gross bond issuance, the category is interpreted as public-debt bond issuance, consistent with the terminology of the underlying Portuguese public-debt statistics.
GOV	Gross issues	1990–2021	ECB Data Portal, Securities Issues Statistics	Government	Government gross bond issuance follows the ECB government series. As for corporate issuance, the ECB series is preferred over the overlapping INE observations from 1990 onward.

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Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Central government (ECB sector 1311), debt securities.	GOV is constructed from the underlying ECB central-government debt-security series. Monthly gross issuance is summed to annual frequency.
NFC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Non-financial corporations (ECB sector 1100), debt securities.	NFC is constructed from the ECB non-financial-corporation issuer-sector series. Monthly gross issuance is summed to annual frequency.
FC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Banks (ECB sector 1220) and other financial institutions (ECB sector 1235).	FC is a harmonized aggregate constructed for this dataset rather than a directly reported ECB series. It combines debt-security issuance by banks (sector 1220, instrument F33000) with long-term debt-security issuance by other financial institutions (sector 1235, instrument F33200).
TC	Gross issues	1990–2021	European Central Bank, Securities Issues Statistics, ECB Data Portal	Constructed NFC and FC aggregates.	TC is constructed as the sum of the harmonized ECB-based NFC and FC gross-issuance series.

Notes. The historical Portuguese issuance data are taken from Table 5J.5, "Issue and transaction of securities," of *Portuguese Historical Statistics*, compiled by the Instituto Nacional de Estatística. Its underlying source is *Estatísticas Monetárias e Financeiras*. The private-bond series is used as the historical total-corporate series because of information in the underlying source. No sectoral split between financial and non-financial corporations is available. From 1990 onward, the data comes from the harmonized ECB series. For the ECB data, NFC corresponds to sector 1100; FC combines sectors 1220 and 1235; GOV corresponds to central government sector 1311; and TC is constructed as sum of NFC and FC. Monthly ECB gross issuance is summed to calendar-year frequency.

6.5.19 United Kingdom

Table 23: Source construction for the United Kingdom: gross corporate bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
TC	Gross issues	1919–1949	Midland Bank Monthly Review	Aggregate corporate debt issues.	Total-corporate gross bond issuance follows the historical aggregate debt-issues series. For 1940–1945, the source reports only a cumulative total for the six-year period rather than annual observations. As an explicit exception to the general no-interpolation rule, this cumulative amount is distributed equally across the six calendar years in order to preserve the total issuance volume reported by the source. The resulting annual observations are flagged as imputed and contain no information about the within-period timing of issuance.
TC	Gross issues	1950–1959	UK Macroeconomic Dataset; Ryland Thomas compilation	Aggregate corporate debt issues.	Total-corporate gross bond issuance is taken directly from the reported aggregate debt-issues series.
TC	Gross issues	1960–1983	UK Macroeconomic Dataset; Ryland Thomas compilation	Convertible debt; other debt.	Total-corporate gross bond issuance equals the sum of convertible debt and other debt.
TC	Gross issues	2003–2024	Bank of England Capital Issuance Statistics, Bankstats Table E3.1	Monetary financial institutions; other financial corporations; private non-financial corporations.	Quarterly gross bond issuance is summed to annual totals. Total-corporate issuance equals financial-corporation plus non-financial-corporation bond issuance. The BoE "Other sectors" category is excluded because it covers public corporations and non-profit institutions serving households.
FC	Gross issues	2003–2024	Bank of England Capital Issuance Statistics, Bankstats Table E3.1	Monetary financial institutions; other financial corporations.	Financial-corporation gross bond issuance equals the sum of monetary financial institutions and other financial corporations. Quarterly gross issues are summed within each calendar year.
NFC	Gross issues	2003–2024	Bank of England Capital Issuance Statistics, Bankstats Table E3.1	Private non-financial corporations.	Non-financial-corporation gross bond issuance follows the private non-financial-corporation bond-issues series. Quarterly gross issues are summed within each calendar year.

Notes. The Bank of England capital-issuance statistics cover non-government primary-market issuance by UK-resident entities. The baseline modern series uses the bond-issuance category only. Commercial paper is retained separately in the country file and can be combined with bonds to construct a broader debt-securities measure, but it is not included in the baseline bond-issuance series. The historical data do not provide a consistent FC/NFC split before 2003. No bond-issuance series is currently available for 1984–2002, coded as missing without interpolation. Missing observations are otherwise left explicitly missing with one documented exception. The Midland Bank source reports corporate debt issuance over 1940–1945 only as a cumulative six-year total. Rather than discard the reported issuance volume entirely, I allocate this total equally across the six years. These observations are explicitly flagged in the underlying data and should be interpreted as a temporal allocation of a known cumulative amount, not as observed annual issuance.

6.5.20 Thailand

Table 24: Source construction for Thailand: gross bond issuance.

Series	Measure	Period	Source	Original categories	Construction and notes
GOV	Gross issues	1979–1991	Bank of Thailand Annual Reports	Government bonds; Treasury bills.	Annual government gross issuance equals the sum of government bond and Treasury bill issuance reported in the annual reports.
GOV	Gross issues	1992–2023	BoT FM_CM_002	Government bonds; Treasury bills.	Monthly flows are summed to annual totals. The source switch from annual reports to the modern BoT statistical portal occurs around 1992.
FC	Gross issues	1998–2004	BoT FM_CM_003	Banking, finance, insurance and leasing.	Monthly flows are summed to annual totals. This category is assigned to financial corporations.
FC	Gross issues	2005–2023	BoT FM_CM_004	Financial intermediation.	Monthly flows are summed to annual totals. The 2005 source switch introduces a classification break relative to the pre-2005 financial-sector category.
NFC	Gross issues	1998–2004	BoT FM_CM_003	Non-financial sectors excluding financial sector and others.	Monthly flows are summed to annual totals. The category "Others" is left unallocated and is not included in the NFC aggregate.
NFC	Gross issues	2005–2023	BoT FM_CM_004	All non-financial business categories.	Monthly flows are summed to annual totals. The NFC aggregate excludes the category Financial Intermediation.
TC	Gross issues	1986–1991	Bank of Thailand Annual Reports	State-owned enterprises; other non-government, non-SOE sectors.	Total corporate gross issuance is constructed as the sum of state-owned enterprise bond issuance and other corporate-sector issuance.
TC	Gross issues	1992–2023	BoT FM_CM_002	State enterprise bonds; private debt securities.	Monthly flows are summed to annual totals. Total corporate issuance is constructed as the sum of state enterprise bond issuance and private debt securities.

Notes. Early data for Thailand was available in the Bank of Thailand Annual Reports and the more recent data comes from the Bank of Thailand statistical data portal. Monthly gross issuance is aggregated to annual frequency by summing monthly flows within each calendar year. The main source break occurs in 1992 for government and total corporate issuance, when the construction switches from annual reports to the modern Bank of Thailand statistical portal. For financial and non-financial corporations, the main classification break occurs in 2005, when the source changes from BoT FM_CM_003 to BoT FM_CM_004. Ambiguous "Others" categories are left unallocated when they cannot be assigned cleanly to financial or non-financial corporations.

6.5.21 United States

Table 25: Source construction for the United States.

Series	Measure	Period	Source	Original categories	Construction and notes
NFC	Gross issues	1900–1943	Hickman (1953), Appendix A, Table A-1	Total of Railroads; public utilities; industrial corporations.	Non-financial-corporate gross bond issuance follows Hickman's "Offered during year" column for the aggregate "All Industries Combined" of railroads, public utilities, and industrial corporations. The measure records par amounts of bond offerings and includes both cash and noncash offerings.
NFC	Gross issues	1970–1994	Federal Reserve Bulletin, annual new-security-issues tables	Manufacturing; commercial and miscellaneous; transportation; public utility; communication.	Non-financial-corporate gross issuance is constructed as the sum of the reported non-financial industry groups.
NFC	Gross issues	1995–2023	Federal Reserve Bulletin; Board of Governors, *New Security Issues, U.S. Corporations*	Bonds: non-financial corporations.	Non-financial-corporation gross bond issuance follows the directly reported non-financial-corporate bond-issues series.
TC	Gross issues	1944–1969	Board of Governors, *Banking and Monetary Statistics, 1941–1970*, Table 12.32	Corporations: bonds and notes, new issues.	Total-corporate gross bond issuance follows the reported new-issues series for corporate bonds and notes.
TC	Gross issues	1970–2023	Federal Reserve Bulletin; Board of Governors, New Security Issues, U.S. Corporations	Corporate bonds, total.	Total-corporate gross bond issuance follows the annual total-bonds category.
FC	Gross issues	1970–1994	Federal Reserve Bulletin, annual new-security-issues tables	Real estate and financial.	Financial-corporation gross issuance follows the reported "Real estate and financial" industry group. This category includes real-estate issuers and is therefore broader than a pure financial-corporation measure.

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Table 25: Source construction for the United States – continued.

Series	Measure	Period	Source	Original categories	Construction and notes
FC	Gross issues	1995–2023	Federal Reserve website; Board of Governors, Data section, New Security Issues Table 1.46., U.S. Corporations	Bonds: financial corporations.	Financial-corporation gross bond issuance follows the directly reported financial-corporate bond-issues series. From 1995 onwards real estate is not part of the financial category anymore due to a change in reporting.
GOV	Gross issues	1934–1973	*Statistical Abstract of the United States*, selected annual tables	U.S. Government public debt issues or new issue volume.	Federal-government gross issuance is transcribed from historical *Statistical Abstract* tables. Available observations cover 1934–1935, 1939–1941, 1943, 1945, and 1947–1973. Missing years are retained as missing.
GOV	Gross issues	1990–2023	U.S. Treasury Fiscal Data	U.S. Treasury securities, new issue volume.	Federal-government gross issuance follows the Treasury-security issuance series retained in the country file. The interval 1974–1989 remains missing without interpolation or splicing.

Notes. The United States series combine historical reconstructions of corporate bond financing, Board of Governors capital-market statistics, Federal Reserve Bulletin tables, the Federal Reserve Financial Accounts, and U.S. Treasury data. The final compiled dataset distinguishes between a long historical non-financial-corporate bond series, a later broad total-corporate series, and sectoral financial-corporation/non-financial-corporation series from 1970 onward.

The historical Hickman data cover railroads, public utilities, and industrial corporations. These issuer groups are classified as non-financial corporations. They are therefore used to construct the early NFC series and are not treated as a complete total-corporate aggregate.

The main historical sectoral distinction is between the NFC series available from 1900 and the broader TC series available from 1944. The pre-1944 Hickman aggregate should therefore not be interpreted as complete total-corporate issuance, because it does not include financial-corporate issuers. NFC gross issuance has an internal gap between 1944 and 1969; FC issuance begins in 1970. The 1970–1994 FC category combines financial and real-estate issuers, whereas the source reports financial corporations directly from 1995 onward.

7 Appendix B: Additional Results

This appendix presents additional results that complement the main text in various ways. The aim is descriptive: to visualize data structure, coverage, and distributional patterns that motivate the specifications used in the paper. Details are described in the respective figure notes.

7.1 Corporate bond issuance in a fixed long-run country cohort

Figure 12 assesses whether the long-run evolution of corporate bond issuance reflects changing country composition in the underlying sample. It restricts attention to countries with total-corporate issuance coverage in the historical, postwar, and modern eras. The fixed-cohort comparison confirms the broad pattern documented in the main text: corporate primary-market activity was economically meaningful before World War II, declined during the postwar decades, and revived after 1980. While the level and composition of issuance differ substantially across countries—most notably in the modern period—the postwar trough is visible within the same set of long-run bond-market economies.

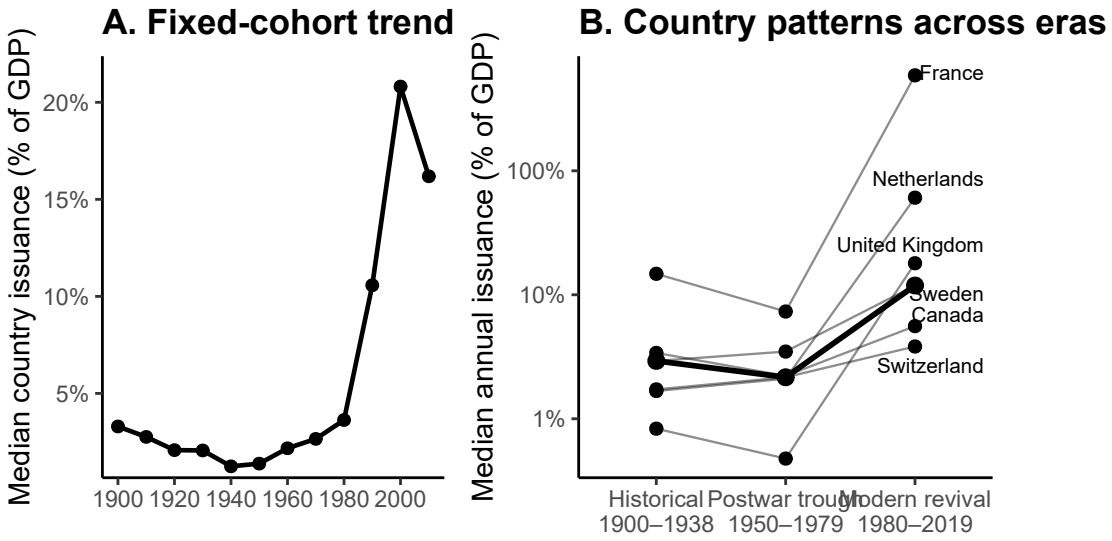


Figure 12: Corporate bond issuance in a fixed long-run country cohort. The figure restricts the sample to countries with total-corporate bond-issuance coverage in the historical, postwar, and modern periods. Panel A reports the median across countries of country-decade median gross bond issuance relative to nominal GDP. Panel B reports country-specific median annual issuance relative to GDP in three broad eras; the vertical axis is logarithmic. The fixed-cohort comparison shows that the postwar decline and subsequent revival in corporate primary-market activity are not mechanically generated by changes in country coverage.

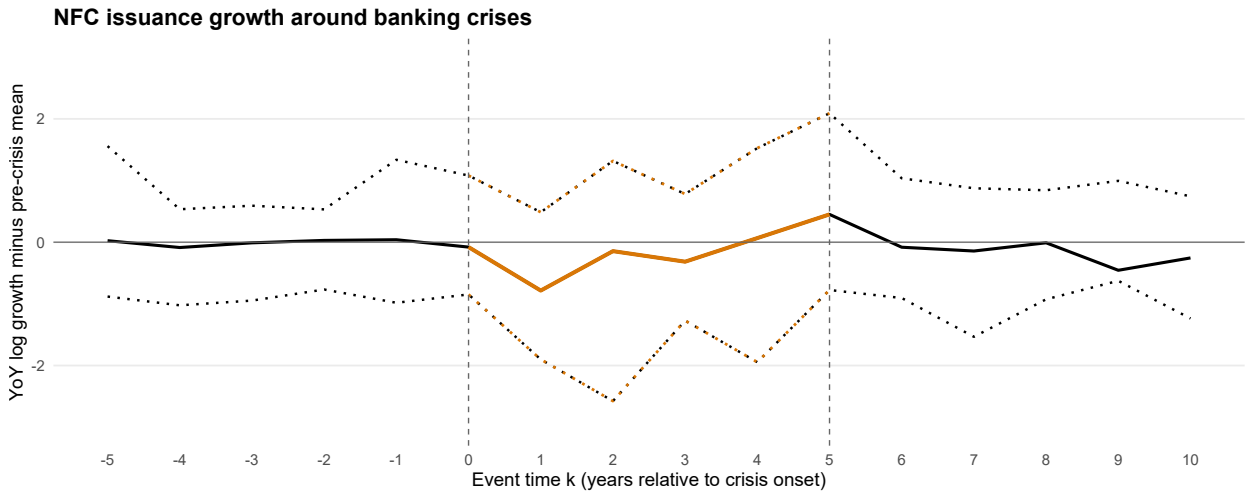


Figure 13: Corporate bond issuance growth before, during and after banking crises. The figure plots the mean year-over-year growth rate of NFC bond issuance before, during and after banking crisis, in log differences. For each crisis episode, the plotted series is demeaned by subtracting the average issuance growth observed in the pre-crisis window of that same country-episode, defined as years $t_0 - K_{\text{pre}}$ to $t_0 - 1$ (here $K_{\text{pre}} = 5$, i.e. a five year pre-crises window). The solid line reports the cross-episode mean, and the shaded area reports the 5th to 95th percentile range across episodes. The vertical dashed line marks crisis onset (t_0), and the dotted line marks the end of the crisis window (t_5). Positive values indicate issuance growth above the episode-specific pre-crisis average; negative values indicate issuance growth below the pre-crisis average. Issuance growth is stable prior to crises but declines sharply in the years following crisis onset, indicating a substantial disruption in corporate bond market activity during banking crises.

7.2 Corporate bond issuance growth around banking crises vs non-crises

Figure 13 provides an alternative visualization of issuance dynamics around banking crises. Unlike the cumulative growth statistic shown in Figure 7 in the main text, the figure plots year-over-year NFC issuance growth demeaned by the pre-crisis average within each crisis episode. he figure illustrates that average issuance growth is stable prior to crises but weakens sharply following crisis onset, motivating the conditional analysis below.

7.3 From descriptive crisis patterns to the conditional issuance response

The baseline local projections estimate a sizable decline in non-financial corporate bond issuance following banking crises. Because these estimates summarize an average conditional response across a relatively small number of historical crisis episodes, a natural question is how closely this result maps into the underlying country-level observations. In particular, a negative average response does not imply that issuance must decline in every individual banking crisis. This subsection therefore provides a deliberately descriptive link between the crisis-level data and the regression estimates.

Panel A of Figure 14 starts from the simplest possible statistic: the fraction of observations for which cumulative NFC issuance rises relative to year $t - 1$. Around one half of banking-crisis observations display positive cumulative changes in gross issuance, especially at short horizons. Taken in isolation, this could suggest substantial heterogeneity in

the response of bond markets to banking distress. However, positive issuance growth is also common in normal times. In the exact estimation sample used for the baseline local projections, the fraction of non-crisis observations with positive cumulative gross issuance rises from around 60% at short horizons to around 70% at longer horizons. For issuance relative to GDP, the corresponding non-crisis fraction is lower, but remains above the crisis fraction at medium horizons. Hence, the sign of the raw change alone does not provide the relevant benchmark for evaluating the crisis response.

This distinction follows directly from the estimand in the baseline local projections. The coefficient on the banking-crisis indicator does not ask whether issuance is literally lower than in year $t-1$. Rather, it measures whether issuance evolves differently following a banking crisis than would normally be expected conditional on country fixed effects, lagged issuance, and lagged macroeconomic conditions. An episode can therefore display positive cumulative issuance growth and at the same time contribute to a negative estimated crisis effect if issuance would otherwise have risen more strongly.

Panel B makes this comparison explicit. For every crisis observation, I calculate the difference between the observed long-difference outcome and the fitted outcome from the corresponding baseline local projection after setting the crisis indicator to zero, while leaving all other regressors unchanged. This provides a model-implied non-crisis benchmark for the same country-year observation. Across horizons, roughly 60–70% of banking crises lie below this fitted non-crisis path. For NFC issuance relative to GDP, the share reaches slightly above 70% around the three-year horizon, where the baseline contraction is also most pronounced. Thus, while positive raw issuance changes are far from uncommon during crises, issuance is nevertheless unusually weak relative to normal dynamics in a clear majority of episodes.

Panel C shows the full distribution of these crisis-specific deviations rather than only their signs. There is substantial heterogeneity across historical banking crises, including a meaningful set of observations with issuance above the fitted non-crisis path. At the same time, the distributions are shifted toward negative values, particularly over the medium run. The baseline local-projection coefficients, shown by the diamonds, summarize these crisis-level deviations and lie within rather than outside the underlying distribution. The negative average response is therefore not equivalent to a claim that every banking crisis produces a decline in bond issuance, nor does it rely on treating particular prominent crises as representative of the full historical sample.

Taken together, these descriptive exercises clarify the interpretation of the main results. Banking crises need not be associated with an absolute decline in NFC bond issuance in every country and every episode. Instead, the systematic result is that issuance is weaker than its normal conditional path following banking crises. This distinction is particularly relevant when interpreting prominent episodes in which bond markets appear to substitute for impaired bank credit. Such cases are present in the historical sample, but they

coexist with a broader distribution in which below-normal issuance is the more common response. The local-projection estimates provide a disciplined way of summarizing this cross-episode heterogeneity without assigning disproportionate weight to individual crisis experiences.

7.4 Issuance dynamics in banking crises and non-financial recessions

A natural question is whether the issuance patterns documented in the main text are specific to systemic banking crises or instead reflect more general downturn dynamics. To address this, Figure 15 compares the event-study path of non-financial corporate (NFC) bond issuance around banking-crisis onsets to the corresponding path around non-financial recession onsets.

The recession comparison is constructed using a simple dating rule based on annual real GDP growth. A recession onset is defined as the first year of negative real GDP growth following a year with non-negative growth.[8] To isolate non-financial recessions, I exclude all recession onsets that occur within a ±2-year window around a systemic banking-crisis onset. Thus, the recession sample excludes downturns that are contemporaneous with, or very close to, systemic banking crises. The event-study specification mirrors the baseline design in Equation 1. In both cases, the dependent variable is log NFC bond issuance relative to GDP, normalized to zero in the year prior to the event ($t-1$).

Figure 15 shows that NFC bond issuance weakens in both cases, but the decline around systemic banking crises is substantially larger and more persistent than around non-financial recessions. This suggests that the contraction in corporate bond issuance documented in the main text is not merely a generic cyclical feature of downturns, but is particularly pronounced in episodes associated with financial-sector distress.

7.5 Changing relative compositions around banking crises

The baseline results show that banking crises are associated with falling non-financial corporate issuance and rising government issuance. A natural complementary question is whether crises also change the relative composition of bond market activity. To examine this, Figure 16 plots sectoral issuance shares around banking crises, where the denominator is total bond market issuance defined as the sum of total corporate and government bond gross issuance. For each sector, the left column shows the average level share around crises, while the right column shows the change in that share relative to its pre-crisis value in year $t-1$.

[8] Formally, a recession onset in country i and year t is defined as an observation with $g_{i,t} < 0$ and $g_{i,t-1} \geq 0$, where $g_{i,t}$ denotes annual real GDP growth computed from the log difference in real GDP. This is a pragmatic approximation rather than a full business-cycle chronology.

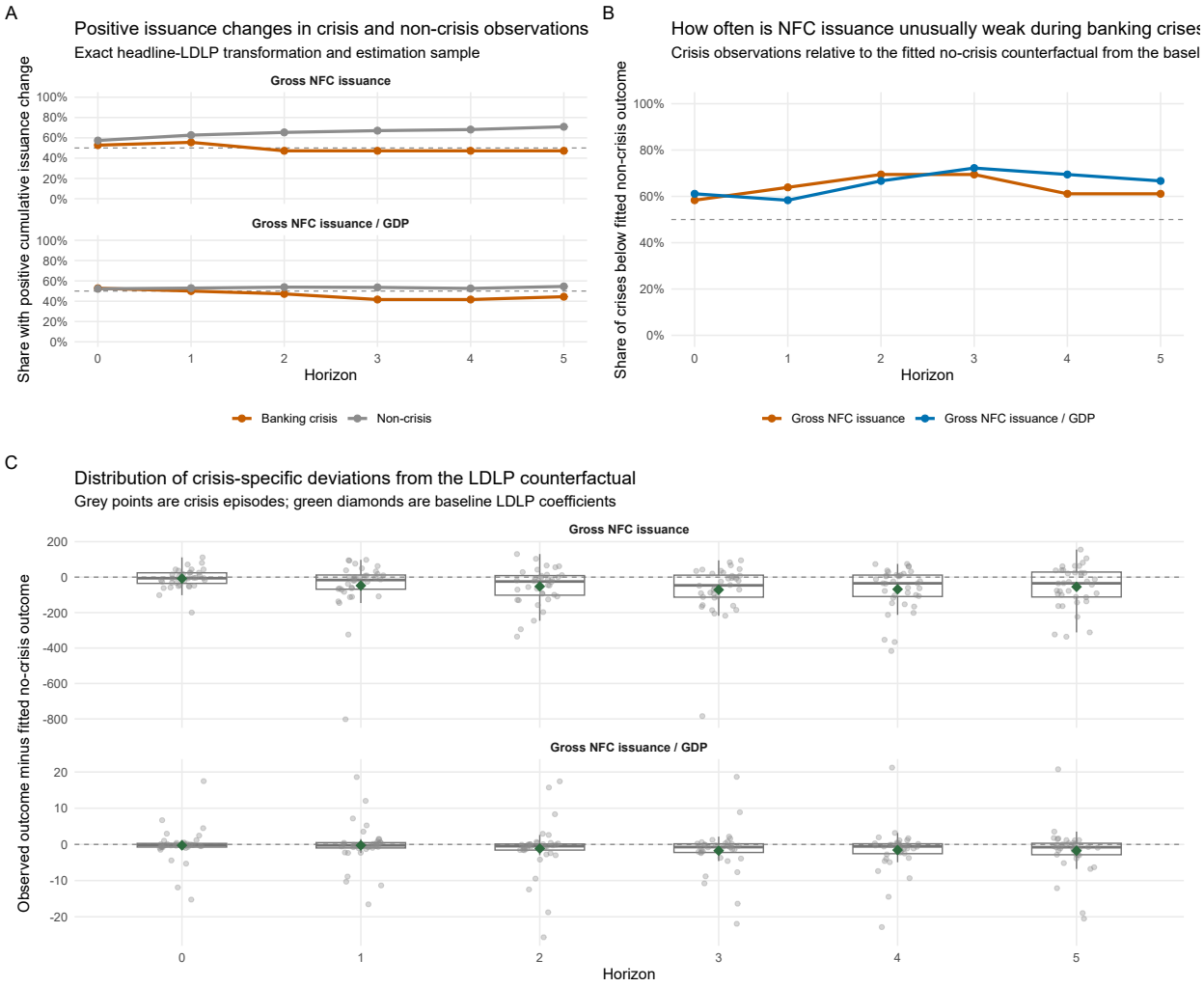


Figure 14: NFC bond issuance across banking-crisis episodes: raw and conditional patterns. Panel A reports the share of observations with positive cumulative changes in non-financial corporate bond issuance at each horizon, separately for banking-crisis and non-crisis observations. The outcome transformation and estimation sample are identical to those used in the baseline long-difference local projections. Panel B reports the share of banking-crisis observations for which the realized outcome lies below the fitted non-crisis outcome. The latter is obtained from the corresponding baseline local projection after setting the banking-crisis indicator to zero while holding country fixed effects, lagged issuance, and lagged macroeconomic controls at their observed values. Panel C plots the distribution of these crisis-specific deviations from the fitted non-crisis outcome. Grey points represent individual banking-crisis episodes, boxes summarize the cross-episode distribution, and diamonds report the corresponding baseline local-projection coefficient. The upper panels use gross NFC issuance and the lower panels issuance relative to GDP. The figure is descriptive: its purpose is to relate the average conditional responses estimated in the main text to heterogeneity in the underlying crisis observations.

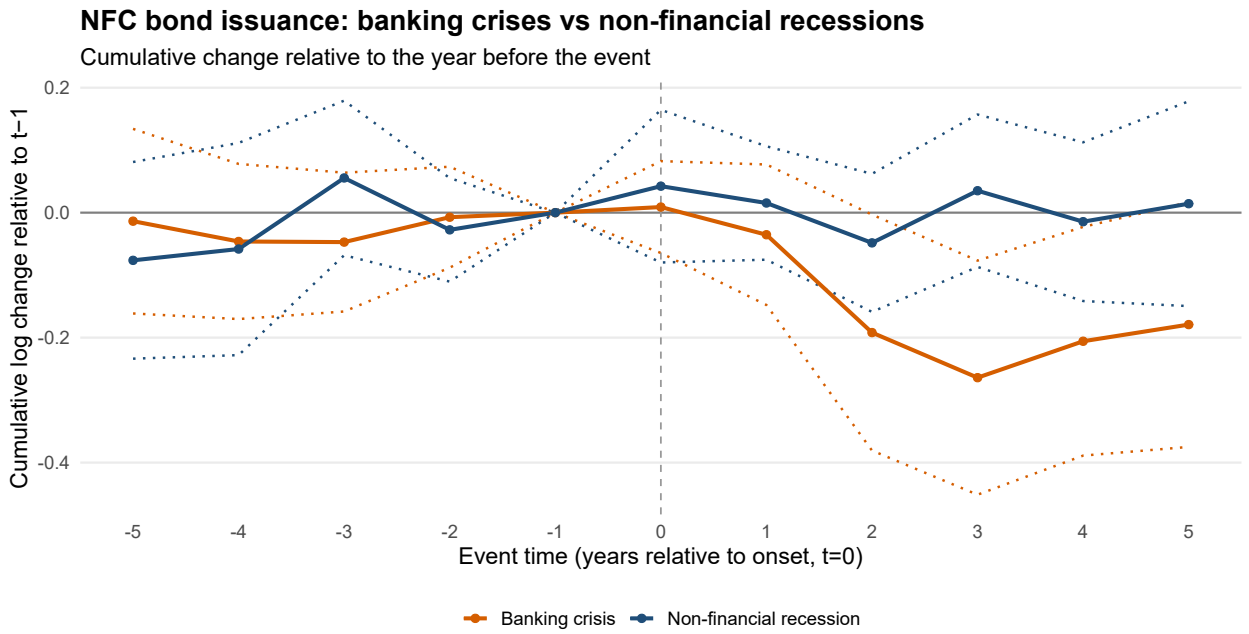


Figure 15: NFC bond issuance in banking crises and non-financial recessions. The figure compares event-study estimates of non-financial corporate bond issuance around systemic banking-crisis onsets and around non-financial recession onsets. The dependent variable is log NFC bond issuance relative to GDP, normalized to zero in the year prior to the event ($t-1$), so the plotted coefficients represent cumulative deviations from the pre-event baseline. Event time $t = 0$ denotes the onset year. Non-financial recession onsets are defined as the first year of negative real GDP growth following a non-negative year, excluding all cases that occur within a ±2-year window around a systemic banking-crisis onset. Estimates are obtained from horizon-specific country fixed-effects regressions. Dotted lines indicate 90% confidence intervals based on Driscoll–Kraay standard errors with horizon-dependent bandwidths.

Two observations emerge. First, the strongest and clearest composition shift is toward the public sector. The government share of total bond market issuance rises after banking crises, with the increase becoming visible after onset and remaining positive over the following years. This mirrors the decline in the total corporate share. Second, within the corporate segment, compositional shifts between non-financial and financial issuers are comparatively modest and noisier. The NFC and FC share paths do not display changes as large or as persistent as those observed for the government sector. Hence, the main compositional reallocation in banking crises appears to be between public and corporate issuers rather than within the corporate sector itself.

These figures are descriptive and should be interpreted as showing how primary bond market activity is redistributed across issuing sectors during crises. They complement the sector-level issuance dynamics in the main text by illustrating that the rise in government issuance is not only large in levels, but also sizable relative to total bond market activity.

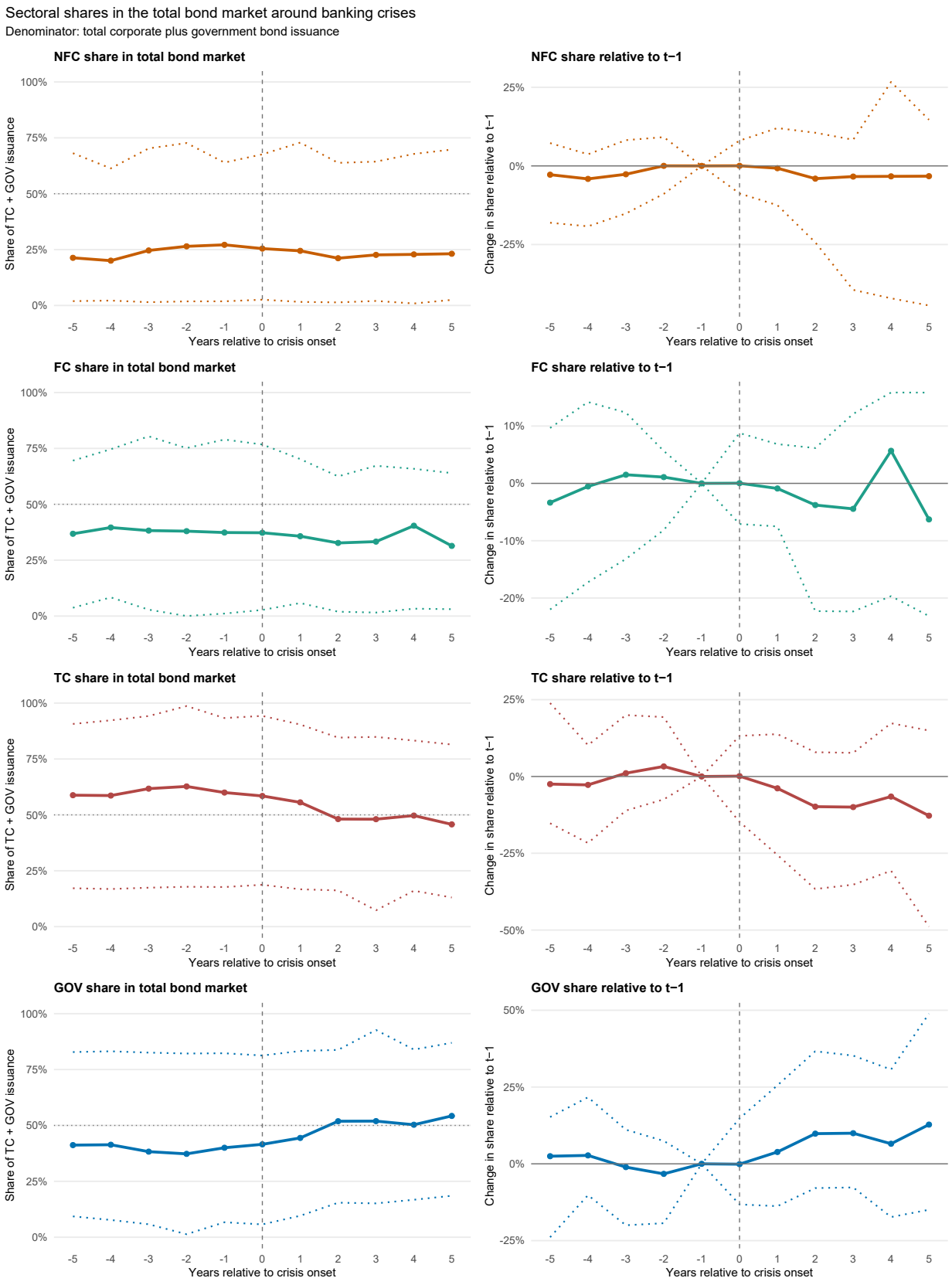


Figure 16: Sectoral shares in the total bond market around banking crises. The figure shows the evolution of issuance shares for non-financial corporations (NFC), financial corporations (FC), total corporations (TC), and government (GOV) around banking-crisis onsets. The denominator is total bond market issuance, defined as the sum of total corporate and government bond gross issuance. For each sector, the left column reports the average share level around crises and the right column reports the change in the share relative to its pre-crisis value in year $t-1$. Event time $t = 0$ denotes the crisis-onset year. Solid lines show cross-episode means and dotted lines indicate the 10th and 90th percentiles across crisis episodes.

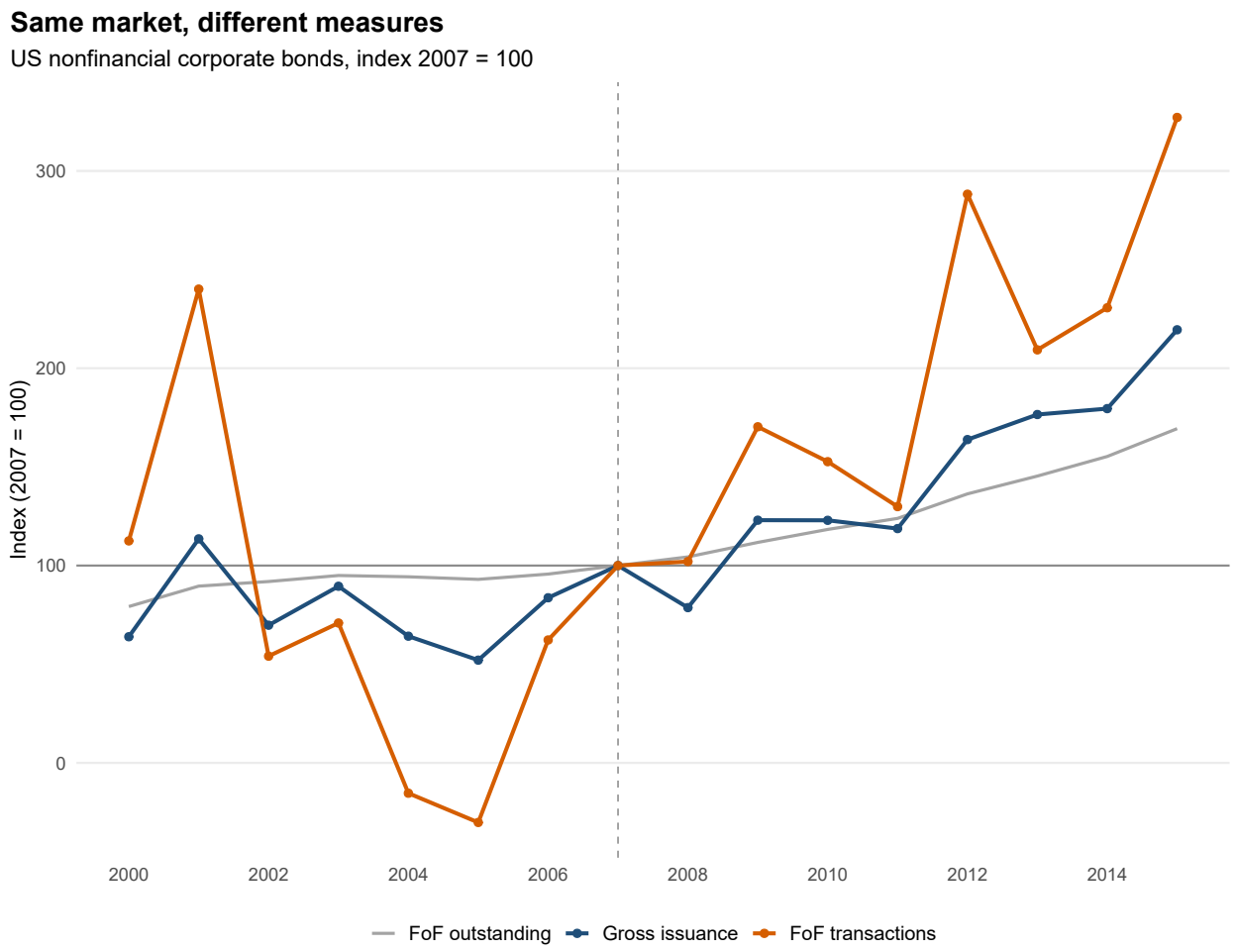


Figure 17: Same market, different measures. The figure compares three measures for U.S. nonfinancial corporate bonds: my hand-collected series on gross bond issuance, the Flow of Funds stock of outstanding corporate bond liabilities, and the Flow of Funds transactions series. The outstanding stock evolves smoothly by construction, while transactions are much more muted than gross issuance and can even turn negative. This reflects the fact that Flow of Funds transactions capture net changes in outstanding positions rather than actual primary-market issuance activity.

7.6 Comparison with Adrian et al. (2013): gross issuance versus Flow of Funds measures

A central benchmark for the spare-tire hypothesis is the U.S. Global Financial Crisis. Adrian et al. (2013) document substitution from bank lending toward bond financing using U.S. Flow of Funds evidence and micro-level issuance data. The comparison in this subsection clarifies how my primary-market gross issuance measure relates to the Flow of Funds objects used in that literature. This distinction matters because gross issuance, Flow of Funds transactions, and outstanding liabilities measure different margins of bond-market activity: new primary issuance, net changes in positions, and end-of-period stocks.

Using gross issuance rather than net transactions matters quantitatively. Flow of Funds transactions understate the scale of bond market activity and therefore compress the apparent response of bond finance during stress episodes. This distinction is particularly

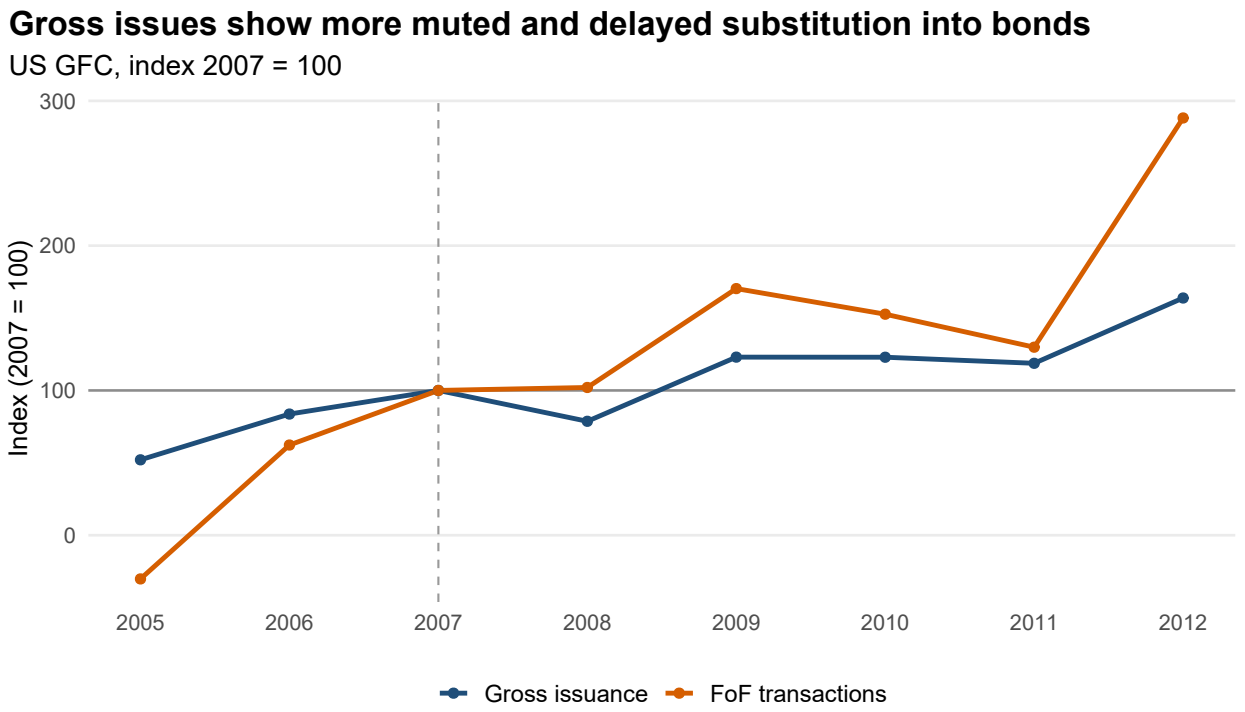


Figure 18: Gross issues show more muted and delayed substitution into bonds. The figure zooms in on the U.S. Global Financial Crisis and compares gross bond issuance with the Flow of Funds transactions series, both normalized to 2007 = 100. Both measures point to increased bond financing after the crisis onset, but gross issuance suggests a more muted and delayed adjustment than the standard Flow of Funds series. In particular, issuance declines in 2008 and rises only thereafter, indicating that substitution into bond markets was not immediate even in the benchmark U.S. case.

important in my setting, where the object of interest is gross primary issuance rather than the net evolution of outstanding credit. Figures 19, 18 show that these distinctions are quantitatively important even in the benchmark U.S. case.

The comparison does not contradict the evidence in Adrian et al. (2013). Rather, it shows that the timing and magnitude of the spare-tire response depend on the financing object being measured. Flow of Funds transactions capture net changes in outstanding positions, while gross issuance captures primary-market activity. In the U.S. GFC, both measures indicate a rebound in bond financing after the crisis onset, but gross issuance suggests that the adjustment was more delayed and less immediate than a net-transactions measure alone might imply.

7.7 Security-level evidence from Australia around the 1931 banking crisis

This section provides additional evidence underlying the Australian case discussed in Section 4.4. The Australian gross-issuance series used in the cross-country panel is reconstructed from individual bond issues compiled by Black et al. (2013). Appendix A provides the full source and sector-classification documentation. In particular, the NFC

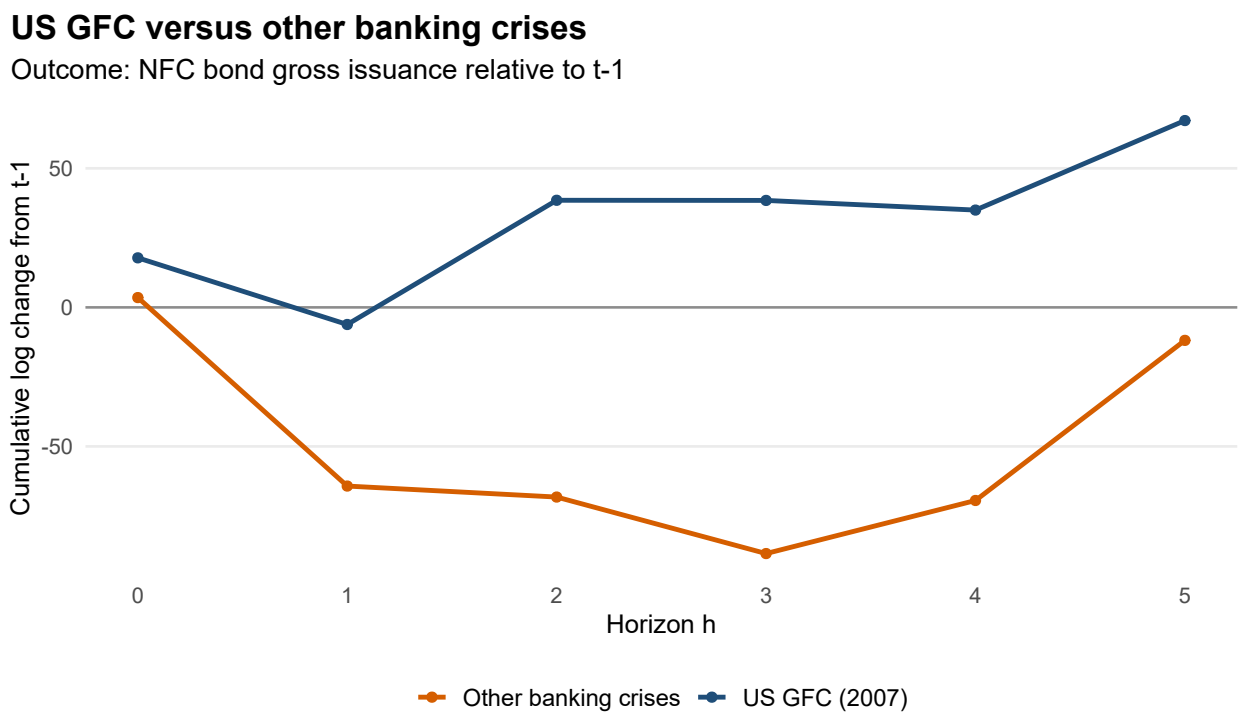


Figure 19: The U.S. GFC versus other banking crises. The figure compares the cumulative long-difference path of U.S. nonfinancial corporate bond issuance after 2007 with the average path observed in other banking crises in the cross-country historical sample. The U.S. episode displays a substantially more positive issuance response than the typical banking crisis, consistent with the interpretation of the GFC as an exceptional case rather than the norm.

aggregate combines private non-financial corporations (PNFCs) and commercially operated non-financial public trading enterprises (PTEs), consistent with the classification of the underlying source. Here I use the security-level records to examine timing, issuer participation, maturity, and robustness around the 1931 crisis.

Timing and composition of the 1931 aggregate The security-level data make clear why annual aggregates can be misleading around crisis dates. NFC issuance totals AUD 188.5 million in 1931, compared with AUD 60.0 million in 1930. Yet AUD 181.5 million, or 96.3 percent, is recorded in the first quarter of 1931. The bulk consists of two Melbourne Harbour Trust issues with a combined face value of AUD 179.5 million. These two PTE bonds alone represent 95.2 percent of total NFC issuance during the calendar year. After the April banking panic, only around AUD 7 million of NFC issuance is recorded during Q2–Q4.

The source codes issue dates at quarterly frequency: throughout 1925–1934, issue dates are recorded on March 1, June 1, September 1, or December 1. I therefore interpret Q1 as issuance recorded before the April panic and Q2–Q4 as subsequent issuance, rather than attempting to identify issuance at daily or monthly frequency.

This distinction also clarifies the sectoral composition of the aggregate. The large 1931 NFC observation is overwhelmingly generated by a public trading enterprise rather than

by broad issuance among private corporations. Because PTEs are commercially operated enterprises, the baseline cross-country series retains them within NFCs in order to preserve the institutional classification of the underlying source. Separating them here is useful precisely because it shows that a positive annual NFC observation need not imply widespread private corporate market access.

Extensive-margin adjustment and issuer selection I next restrict the analysis to PNFCs and compare the four years 1927–1930 with 1931–1933. Average annual gross issuance falls from AUD 16.6 million to AUD 5.2 million, or by 68.8 percent. The average number of active issuers falls almost one-for-one, from 8.5 to 2.7 per year, a decline of 68.6 percent. In contrast, issues per active issuer decline by only 6.2 percent and average issue size increases by 6.1 percent. Hence the contraction in private issuance is primarily an extensive-margin phenomenon.

Figure 20 provides a complementary issuer-level view. Sixteen distinct private issuers are observed during 1927–1930 after harmonizing a small number of obvious spelling variants. Only four of these firms issue again during 1931–1933: Broken Hill Proprietary, Australian Gas Light, Colonial Gas Association, and Mort's Dock. Importantly, these continuing issuers accounted for approximately 89 percent of all PNFC issuance already during 1927–1930. Broken Hill Proprietary alone represented 71 percent. Post-crisis access was therefore concentrated among firms that were already major market participants before the crisis.

At the same time, the primary market was not completely shut. Bond's Industries first appears as an issuer in 1931 and News in 1932. Together these first-observed issuers account for approximately 9 percent of PNFC issuance during 1931–1933. The Australian evidence is thus better described as strong selection than as a literal market closure: a small set of incumbent firms retained access and some new issuance remained possible, while the number of active private issuers contracted sharply.

Maturity structure The contraction also coincides with a marked shortening of the maturity structure. Among PNFC bonds with recorded original maturity of at least one year, the amount-weighted median maturity falls from 10.6 years in 1927–1930 to 7.9 years in 1931–1933. The share of issuance with maturity above ten years falls from 59.4 percent to 2.7 percent, while the share with maturity of five years or less rises from 20.6 percent to 33.5 percent.

The annualized series in Figure 21 shows that the disappearance of long-duration private issuance is concentrated in the crisis period. The share of PNFC issuance above ten years is 53 percent in 1927, 52 percent in 1928, 59 percent in 1929, and 69 percent in 1930. It falls to 6.6 percent in 1931 and zero in both 1932 and 1933. Long-maturity issuance reappears in 1934, when its share rises to 63 percent. While the small number of issues in individual years warrants caution, this pattern argues against interpreting the maturity change as a gradual pre-existing trend.

The supplied security-level file contains coupon rates but no issue-level yield-to-maturity field. I therefore do not interpret changes in coupons as changes in borrowing spreads. The price evidence used in the paper instead comes from the independent historical credit-spread data analyzed in Section 4.5 and Appendix 9.

Robustness Table 26 reports three robustness exercises. First, the raw Black et al. records contain exact duplicate observations. I retain the raw records in the baseline because they reproduce exactly the annual aggregates used in the cross-country dataset and because identical records may represent distinct placements or tranches. Around the crisis window, exact duplicate PNFC observations affect 1929 but none of the years 1930–1933. Mechanically retaining only the first record in each exact duplicate group changes the post/pre issuance index from 31.2 to 33.5, while leaving the active-issuer index unchanged at 31.4. The extensive-margin interpretation is therefore unaffected.

Second, because Broken Hill Proprietary is the dominant pre-crisis private issuer, I repeat the decomposition after excluding it. PNFC issuance during 1931–1933 remains only 44.5 percent of its 1927–1930 level, while the number of active issuers falls to 22.2 percent of its pre-crisis level. Conditional on issuing, average issue size rises sharply in this restricted sample, reinforcing rather than weakening the selection interpretation. The maturity result also survives: excluding Broken Hill Proprietary, the weighted median maturity declines from 6.0 to 3.9 years and the share above ten years from 29.3 percent to 6.5 percent.

Third, the result is not driven by the exact pre- and post-crisis windows. Using 1928–1930 versus 1931–1933, average PNFC issuance falls by 72.8 percent and active issuers by 65.2 percent. Using the wider windows 1926–1930 versus 1931–1935, issuance falls by 67.4 percent and active issuers by 60.0 percent. Across these alternatives, the extensive margin remains the dominant adjustment and long-maturity issuance remains substantially lower after 1930.

Overall, the Australian security-level evidence illustrates the distinction between continued bond-market access for selected issuers and stabilization of aggregate corporate financing. The market remained usable for some firms, including a small number of new issuers, but this was accompanied by a large decline in the number of active private issuers and an almost complete disappearance of long-maturity issuance. The case therefore provides a concrete microeconomic counterpart to the aggregation wedge identified in the cross-country estimates.

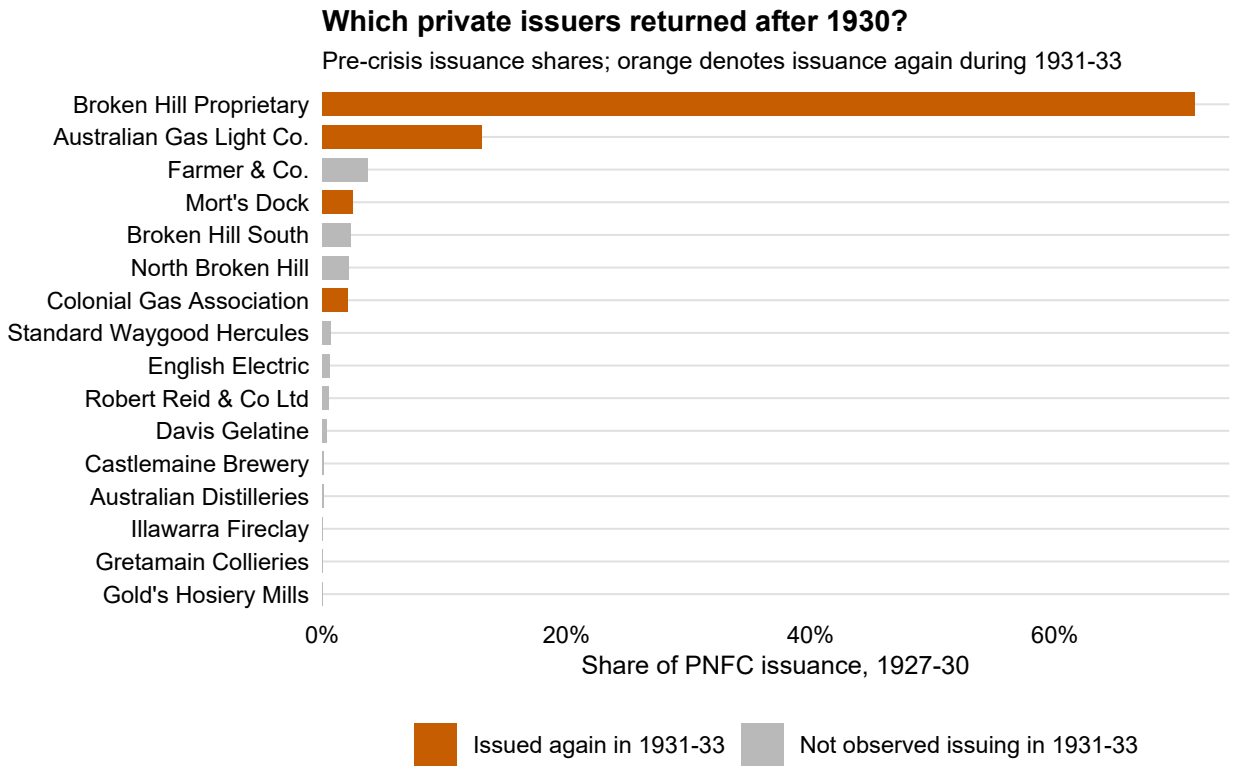


Figure 20: Issuer selection in the Australian private corporate bond market around 1931 The figure reports each private non-financial corporation's share of total PNFC gross bond issuance during 1927–1930. Issuers shown in orange are observed issuing again during 1931–1933; issuers shown in grey are not observed issuing during that period. Issuer names are harmonized using a small set of transparent spelling and naming corrections documented in the replication files. "Not observed issuing" refers only to issuance in these security-level data during 1931–1933 and should not be interpreted as firm failure or permanent exit from the bond market. Source: Black et al. (2013) underlying security-level bond issuance data and author's calculations.

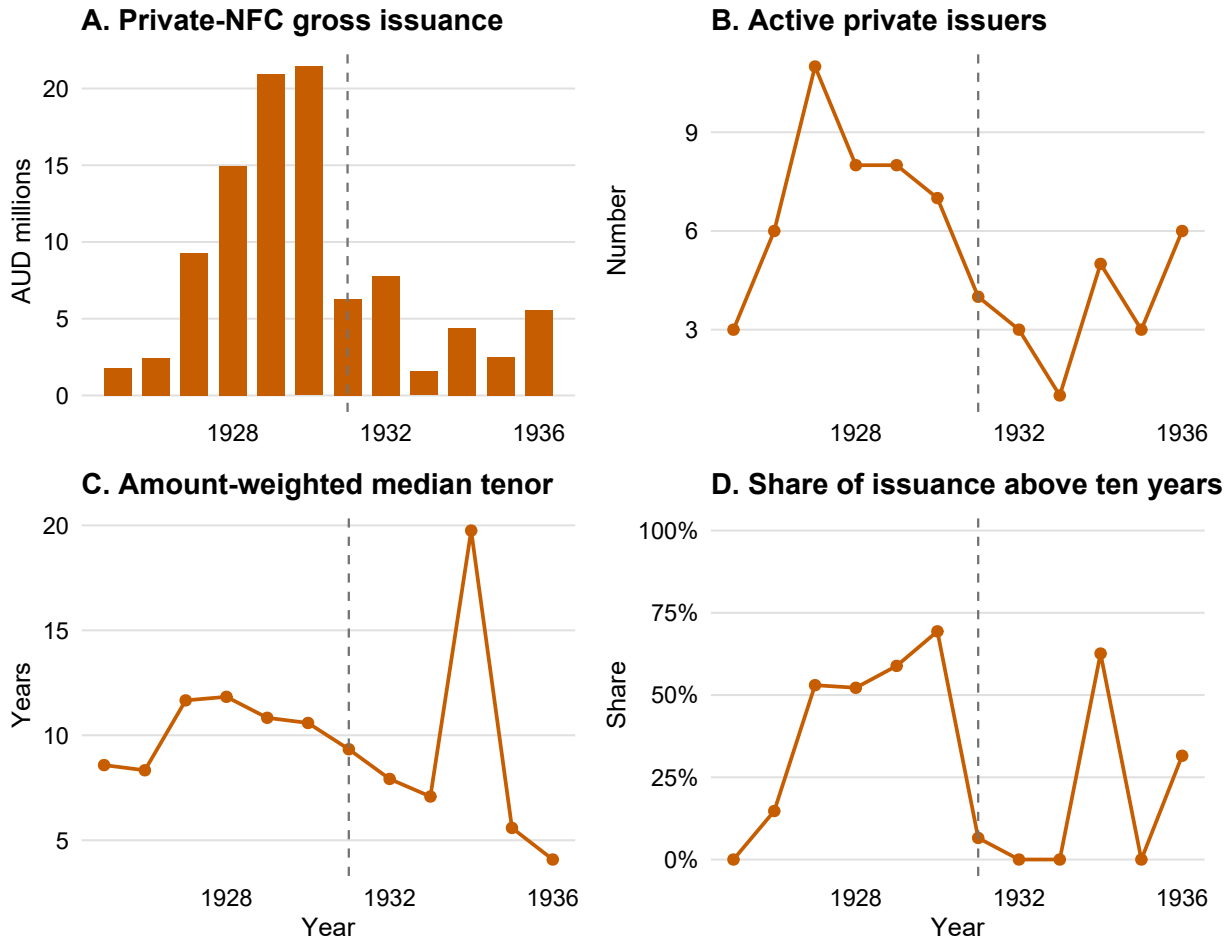


Figure 21: Private corporate bond issuance and maturity around the Australian crisis of 1931 The figure reports annual security-level diagnostics for Australian private non-financial corporations (PNFCs) over 1925–1936. Panel A shows gross bond issuance, Panel B the number of active issuers, Panel C the amount-weighted median original maturity, and Panel D the share of issuance with original maturity above ten years. The dashed vertical line denotes 1931. Maturity statistics are calculated using securities with recorded original maturity of at least one year. The annual series is shown to assess whether the pre/post differences reported in the main text reflect broader dynamics around the crisis rather than the particular choice of comparison windows. Source: Black et al. (2013) underlying security-level bond issuance data and author's calculations.

Table 26: Robustness of the Australian private-issuance decomposition

Specification	Post-period relative to pre-period = 100				Maturity	
	Gross issuance	Active issuers	Issues / active issuer	Average issue size	Median maturity	Share > 10 yrs (%)
Panel A. Sample and data robustness						
Baseline raw data	31.2	31.4	93.8	106.1	10.6 → 7.9	59.4 → 2.7
Exact duplicates: keep first only	33.5	31.4	96.3	111.0	10.6 → 7.9	57.9 → 2.7
Exclude Broken Hill Proprietary	44.5	22.2	79.2	252.9	6.0 → 3.9	29.3 → 6.5
Panel B. Alternative comparison windows						
1928–1930 vs. 1931–1933	27.2	34.8	88.9	88.0	10.6 → 7.9	60.6 → 2.7
1926–1930 vs. 1931–1935	32.6	40.0	90.4	90.2	10.6 → 7.9	57.6 → 14.0

Notes: The first four columns report the post-period value relative to the corresponding pre-period value, with the pre-period normalized to 100. The final two columns report the pre-period and post-period values, respectively, with the arrow indicating the change across periods. The baseline compares 1927–1930 with 1931–1933 using the raw security-level records. The exact-duplicate specification mechanically retains only the first observation among otherwise identical records; the raw records are retained in the baseline because they reproduce the aggregate Australian issuance series used in the paper. The third specification excludes Broken Hill Proprietary, the dominant private issuer before the crisis. Panel B varies the pre- and post-crisis comparison windows while retaining the raw records. Maturity statistics are amount weighted and restrict the sample to securities with recorded original maturity of at least one year. BHP denotes Broken Hill Proprietary. *Source:* Black et al. (2013) underlying security-level bond issuance data and author's calculations.

8 Appendix C: Robustness checks

This section presents robustness checks to the baseline results presented in the main text for the event study and the local projections.

8.1 Event study robustness

Alternative to 4 which uses log bond issuance in raw currency units as dependent variable. Otherwise the results shown follow the main specification as described in Equation 1.

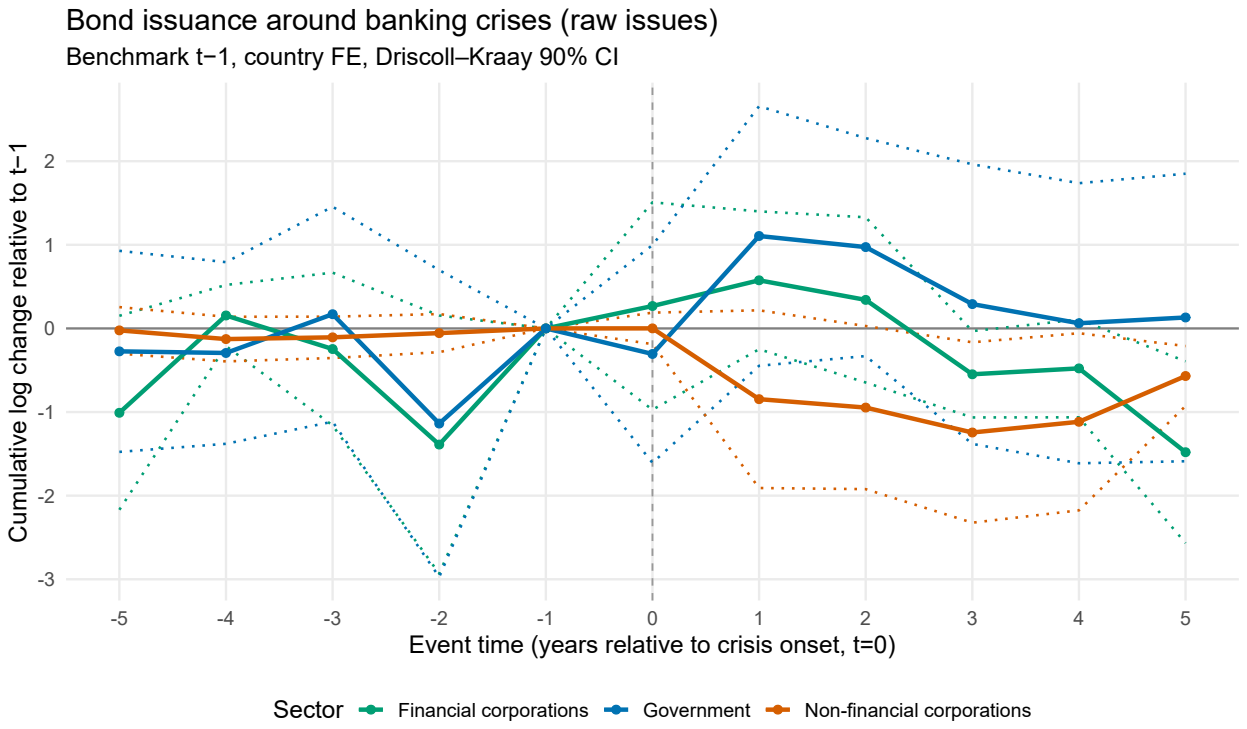


Figure 22: Event study of bond issuance around banking crises using raw issuance levels. The dependent variable is the cumulative change in log bond issuance relative to the year before crisis onset. All other details follow Figure 4 and the specification as explained in the main text.

8.2 Issuance local projections robustness

This appendix reports robustness checks for the long-difference local projection estimates in Section 4.2. I vary the treatment of outliers, the timing of winsorization, the dependent-variable definition, and the sample split used to assess historical coverage. Across these specifications, the qualitative results remain unchanged: non-financial corporate issuance contracts after banking crises, total corporate issuance declines, government issuance rises, and financial-sector issuance displays a more mixed response.

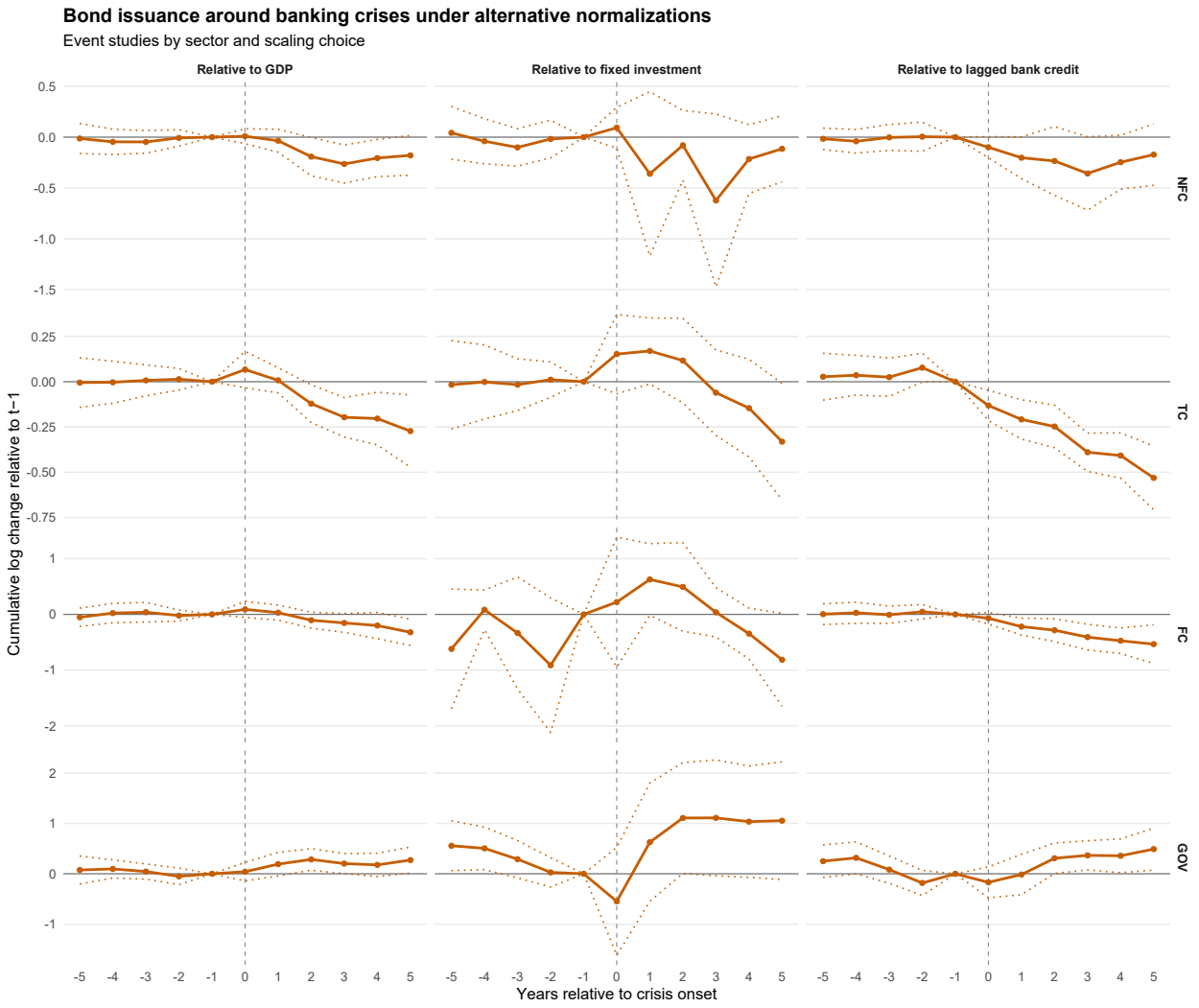


Figure 23: Bond issuance around banking crises under alternative normalizations. The figure reports event-study estimates of bond issuance around banking-crisis onsets for four issuer sectors: non-financial corporations (NFC), total corporations (TC), financial corporations (FC), and government (GOV). Within each sector, issuance is alternatively scaled by GDP, fixed investment, and lagged total corporate bank credit. All series are log-transformed and normalized relative to the year before crisis onset, so that the plotted coefficients represent cumulative log deviations from the pre-crisis benchmark year $t - 1$. Specification follows the main text. Event time $t = 0$ denotes the crisis-onset year. Each horizon is estimated separately using country fixed effects and Driscoll–Kraay standard errors with horizon-dependent bandwidths; dotted lines indicate 90% confidence intervals. Scaling issuance by fixed investment provides a demand-oriented normalization, i.e. showing whether bond issuance weakens relative to capital-spending. Normalization by lagged bank credit instead relates bond issuance to the pre-crisis size of the domestic bank-intermediated corporate credit base.

8.2.1 Sensitivity to outlier treatment

Bond issuance data contain extreme observations, especially around financial stress episodes and in periods of market development. The baseline specification therefore winsorizes the issuance series symmetrically at the 2% level before taking logs. To assess whether the main results are driven by this choice, Figure 24 compares the baseline specification to specifications without winsorization and with stronger 5% winsorization.

The estimated dynamics are stable across these choices. In particular, the post-crisis contraction in non-financial corporate issuance and the expansion in government issuance remain visible even without winsorization and under stronger trimming. The results are therefore not driven by a small number of extreme issuance observations.

8.2.2 Winsorizing the transformed outcome

The baseline specification winsorizes gross issuance before the log transformation. As an alternative, I leave gross issuance unchanged, construct the long-difference outcome $100 \times (y_{i,t+h} - y_{i,t-1})$, and winsorize this transformed outcome directly. This check asks whether the results are driven by the treatment of extreme cumulative log changes rather than extreme issuance levels.

Figure 25 shows that the results are essentially unchanged. The NFC response remains negative at all horizons, financial-sector issuance displays a temporary increase followed by a decline, and government issuance rises after crisis onset. This confirms that the baseline results are not an artifact of winsorizing the issuance level before taking logs.

8.2.3 Dependent-variable definition

The baseline local projections use the log of raw gross issuance. A natural concern is that issuance dynamics may partly reflect changes in the scale of the economy during crises. I therefore re-estimate the long-difference local projections using log issuance relative to GDP.

Figure 26 shows that the qualitative patterns are unchanged. Non-financial corporate and total corporate issuance decline relative to GDP after banking crises, while government issuance increases. The magnitudes differ because the dependent variable is now scaled by GDP before taking logs, but the sectoral ranking and dynamic pattern remain similar to the baseline raw-issuance specification.

8.2.4 Pre-crisis corporate credit growth

A potential concern is that the estimated decline in bond issuance after banking crises reflects the credit expansion that typically precedes financial crises rather than the crisis itself. I therefore augment the baseline specification with the cumulative change in

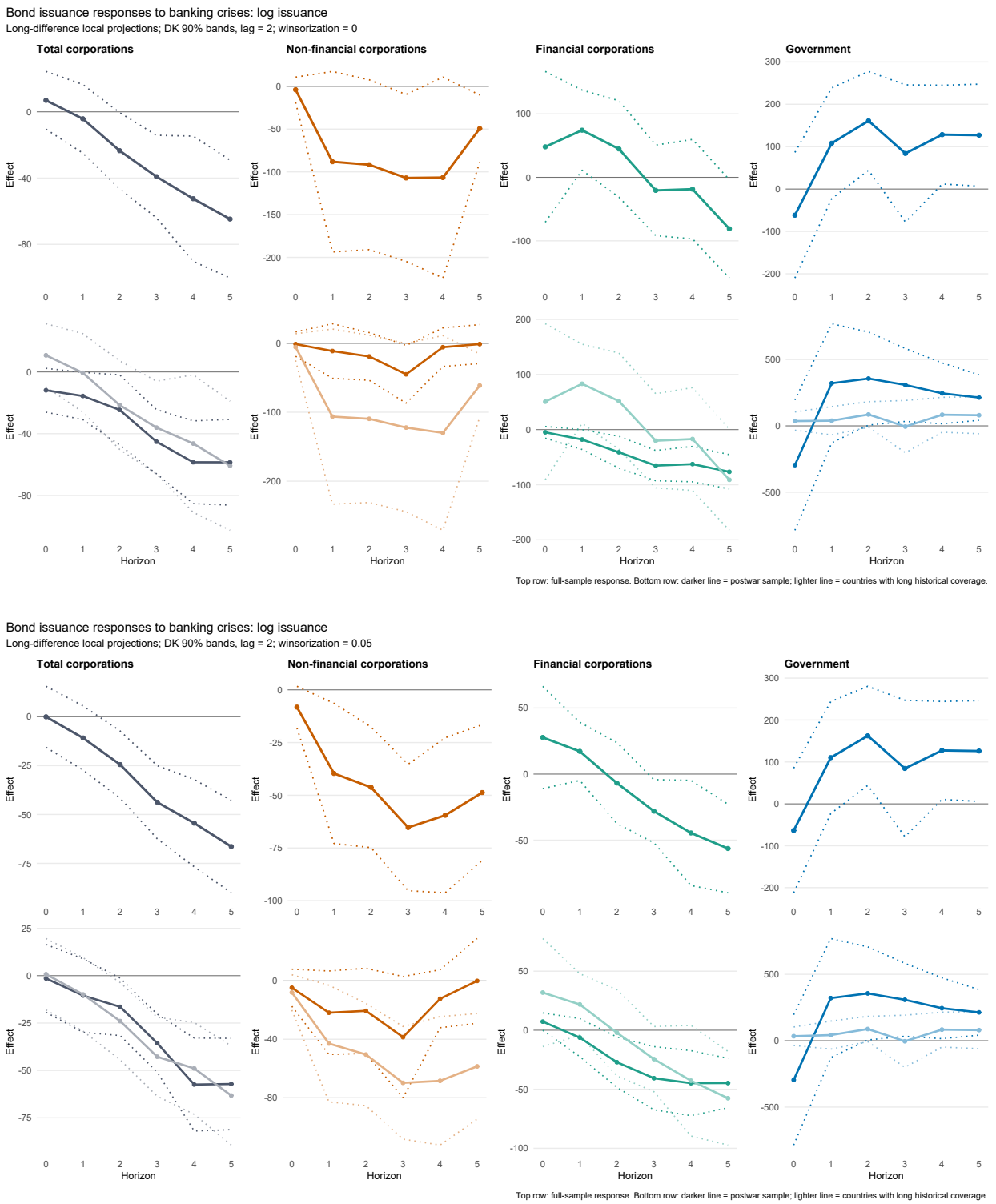


Figure 24: Sensitivity to outlier treatment in long-difference local projections. The top figure reports estimates without winsorization; the bottom figure reports estimates using 5% winsorization of the issuance series before log transformation. The dependent variable is $100 \times (y_{i,t+h} - y_{i,t-1})$, where $y_{i,t}$ denotes the log of gross bond issuance. Each figure reports total corporations, non-financial corporations, financial corporations, and government issuers. Within each figure, the top row shows the full-sample response and the bottom row allows responses to differ between countries with long historical coverage (darker) and countries observed only in the postwar sample (lighter). All specifications include country fixed effects, two lags of the dependent variable, and two lags of macroeconomic controls. Dotted lines denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors.

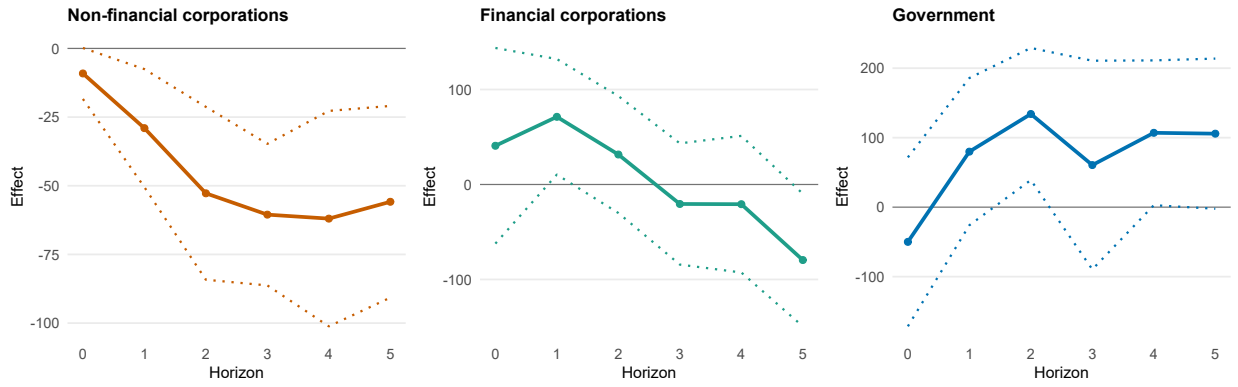


Figure 25: Robustness to winsorizing the transformed long-difference outcome. Gross issuance is not winsorized before logs. Instead, the dependent variable $100 \times (y_{i,t+h} - y_{i,t-1})$ is constructed first and then winsorized symmetrically at the 2% level. The figure reports estimates for non-financial corporations, financial corporations, and government issuers. All specifications include country fixed effects, two lags of the dependent variable, and two lags of macroeconomic controls. Dotted lines denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors.

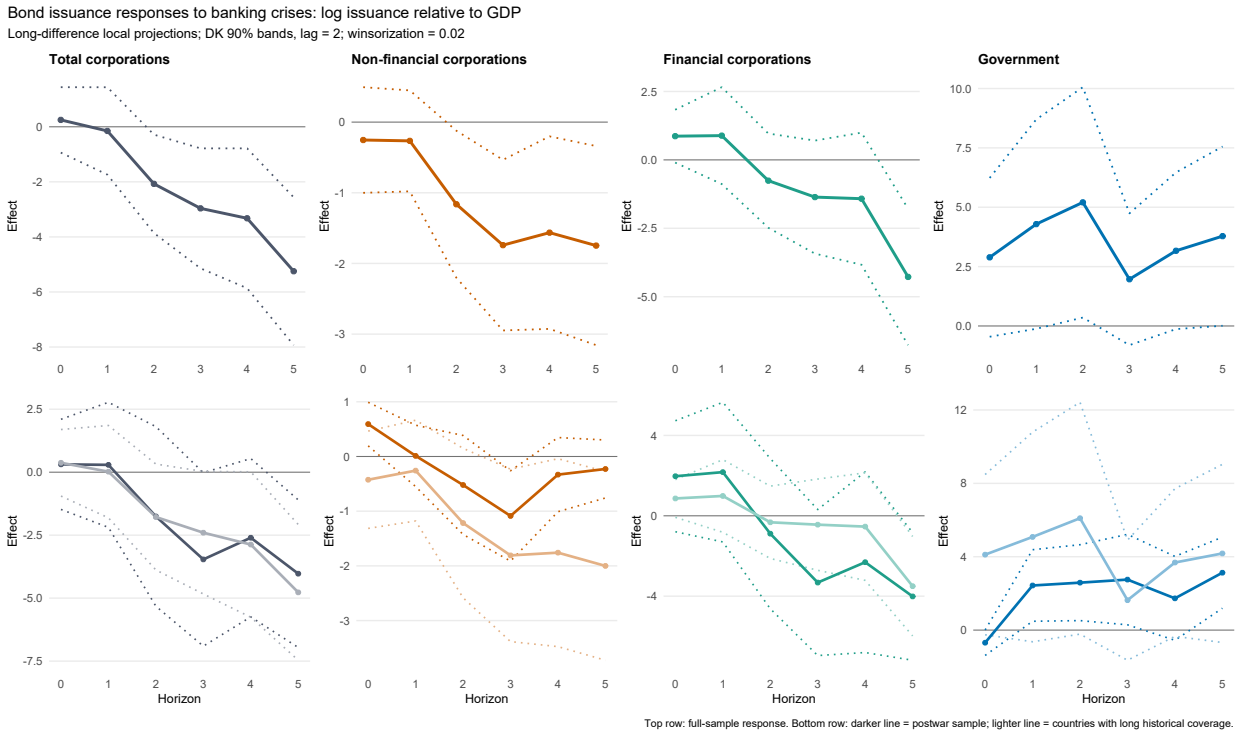


Figure 26: Outcome-definition robustness: issuance relative to GDP. The figure reports long-difference local projection estimates using $y_{i,t} = \log(1 + \text{Issues}_{i,t}/\text{GDP}_{i,t})$. The dependent variable is $100 \times (y_{i,t+h} - y_{i,t-1})$. The top row shows the full-sample response. The bottom row allows responses to differ between countries with long historical coverage and countries observed only in the postwar sample. All specifications include country fixed effects, two lags of the dependent variable, and two lags of macroeconomic controls. Issuance series are winsorized symmetrically at the 2% level before log transformation. Dotted lines denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors.

corporate bank credit relative to GDP between $t-4$ and $t-1$. Data comes from Müller and Verner (2023). This captures pre-crisis corporate credit conditions while remaining predetermined with respect to crisis onset.

Because historical corporate credit data are less widely available than bond issuance data, I compare the augmented specification to the baseline re-estimated on exactly the same credit-available sample. Figure 27 reports the results. Controlling for the pre-crisis corporate credit expansion leaves the main response patterns largely unchanged. For non-financial corporations, the decline in raw issuance is somewhat attenuated at medium horizons but remains pronounced. When issuance is scaled by GDP, the controlled and matched-sample baseline responses are virtually identical. The responses of financial corporations and government issuance are likewise qualitatively unchanged.

The post-crisis contraction in non-financial corporate bond issuance therefore does not appear to be driven by the corporate credit boom preceding the crisis.

8.2.5 Historical coverage heterogeneity

A separate concern is that the estimated crisis response may reflect differences in sample composition rather than stable issuance dynamics. Countries with long historical coverage are often those with more developed early bond markets, while countries entering the sample only in the postwar period may differ in financial structure, market depth, and data availability. To assess this, I allow the crisis response to differ between countries with pre-1945 issuance observations and countries observed only in the postwar sample.

This exercise is distinct from comparing pre- and post-1945 crises. The split used here is based on country-level data coverage, not on the date of the crisis episode. It therefore asks whether the baseline response is driven by the subset of countries for which long-run historical data are available.

The bottom rows of Figures 24 and 26 report these coverage-specific responses. The qualitative conclusions remain similar across samples. In particular, NFC issuance does not exhibit a systematic post-crisis expansion in either group, while government issuance rises after crises. This suggests that the main results are not mechanically driven by the historical coverage structure of the dataset.

8.2.6 Lag length

The baseline specification includes two lags of the dependent variable and macroeconomic controls. To assess sensitivity to the chosen lag structure, this subsection reports results using a richer and hence more conservative lag specification, as suggested in Montiel Olea and Plagborg-Møller (2021) to obtain lag-augmented local projections that are robust to highly persistent data for impulse responses at long horizons. Using four lags does not materially affect the estimated responses as shown in Figure 28.

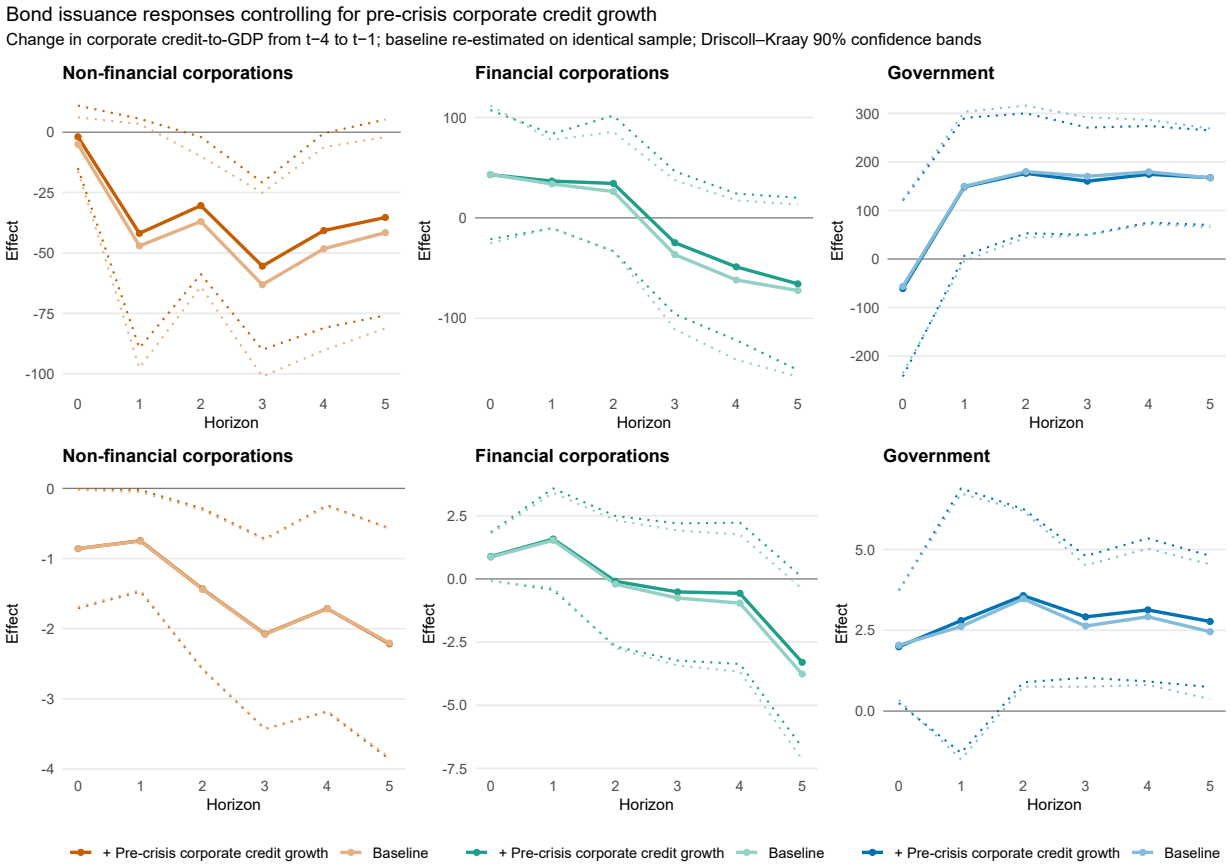


Figure 27: Bond issuance responses controlling for pre-crisis corporate credit growth. The figure reports long-difference local projection estimates with and without a control for the cumulative change in corporate bank credit relative to GDP between $t-4$ and $t-1$. Both specifications are estimated on the identical sample for which the pre-crisis credit measure is available. Panel A reports cumulative changes in log raw issuance and Panel B reports log issuance relative to GDP. Dark lines denote the specification including the pre-crisis corporate credit-growth control and lighter lines the matched-sample baseline. All remaining elements of the specification follow the baseline, including country fixed effects, two lags of the dependent variable and macroeconomic controls, 2% winsorization of issuance, and Driscoll–Kraay standard errors. Dotted lines denote 90% confidence intervals.

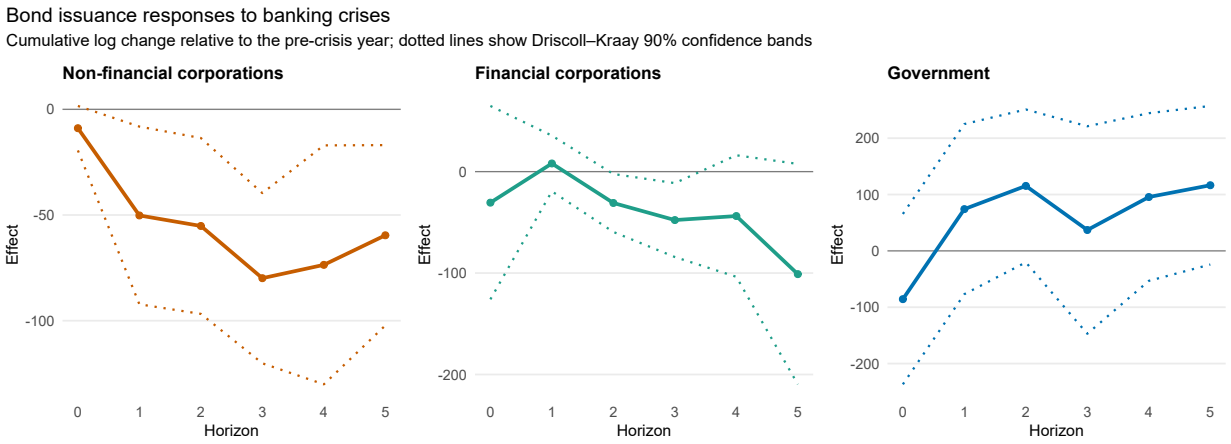


Figure 28: Lag-length robustness. The figure reports long-difference local projection estimates using four lags of the dependent variable and macroeconomic controls. The dependent variable is the log of raw bond issuance. All specifications include country fixed effects and dotted lines denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors.

8.2.7 Changing sample composition across horizons

Re-estimating each horizon on the common sample available through horizon five yields the same qualitative response profile, indicating that the impulse-response profile is not driven by changing sample composition across horizons.

8.2.8 Pre- versus post-1945 crisis episodes

This split differs from the historical-coverage split used above: here, crises are classified by the calendar date of the crisis onset, whereas the coverage split classifies countries by whether their issuance data extend before World War II.

8.2.9 Bond issuance responses before and after 1980

A potential concern with the long-run analysis is the pronounced increase in the size of bond markets since the 1980s. In particular, issuance relative to GDP exhibits a marked upward trend over the sample period, raising the possibility that the estimated crisis responses primarily reflect the substantially deeper bond markets of the recent period. I therefore allow the response to banking crises to differ between crisis episodes occurring before and after 1980.

Rather than estimating the two periods separately, I re-estimate the baseline long-difference local projections on the full sample and interact the banking-crisis indicator with an indicator for crisis years from 1980 onward. I also allow the coefficients on all lagged controls and lagged dependent variables to differ across the two periods. Specifi-

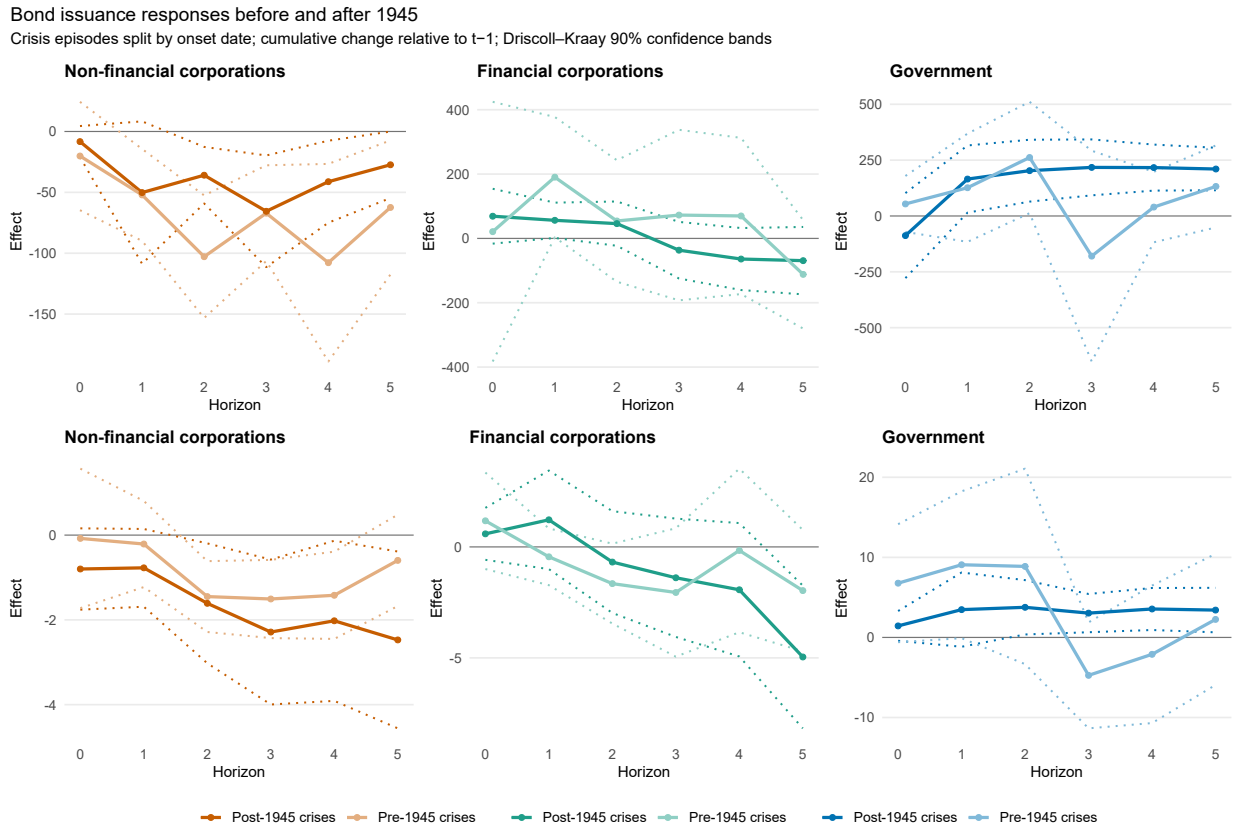


Figure 29: Pre- versus post-1945 crisis episodes. The figure reports long-difference local projection estimates of the cumulative response of bond issuance to banking crises, allowing the response to differ between crisis episodes with onset dates before 1945 and crisis episodes with onset dates in 1945 or later. Panel A uses the log of raw gross issuance as the dependent variable. Panel B uses log issuance relative to GDP. The dependent variable is $100 \times (y_{i,t+h} - y_{i,t-1})$. All specifications include country fixed effects, two lags of the dependent variable, and two lags of macroeconomic controls: real GDP growth, inflation, the short-term interest rate, and the current account balance. Issuance series are winsorized symmetrically at the 2% level before log transformation. Dotted lines denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors. Vertical scales differ across panels to highlight within-sector dynamics.

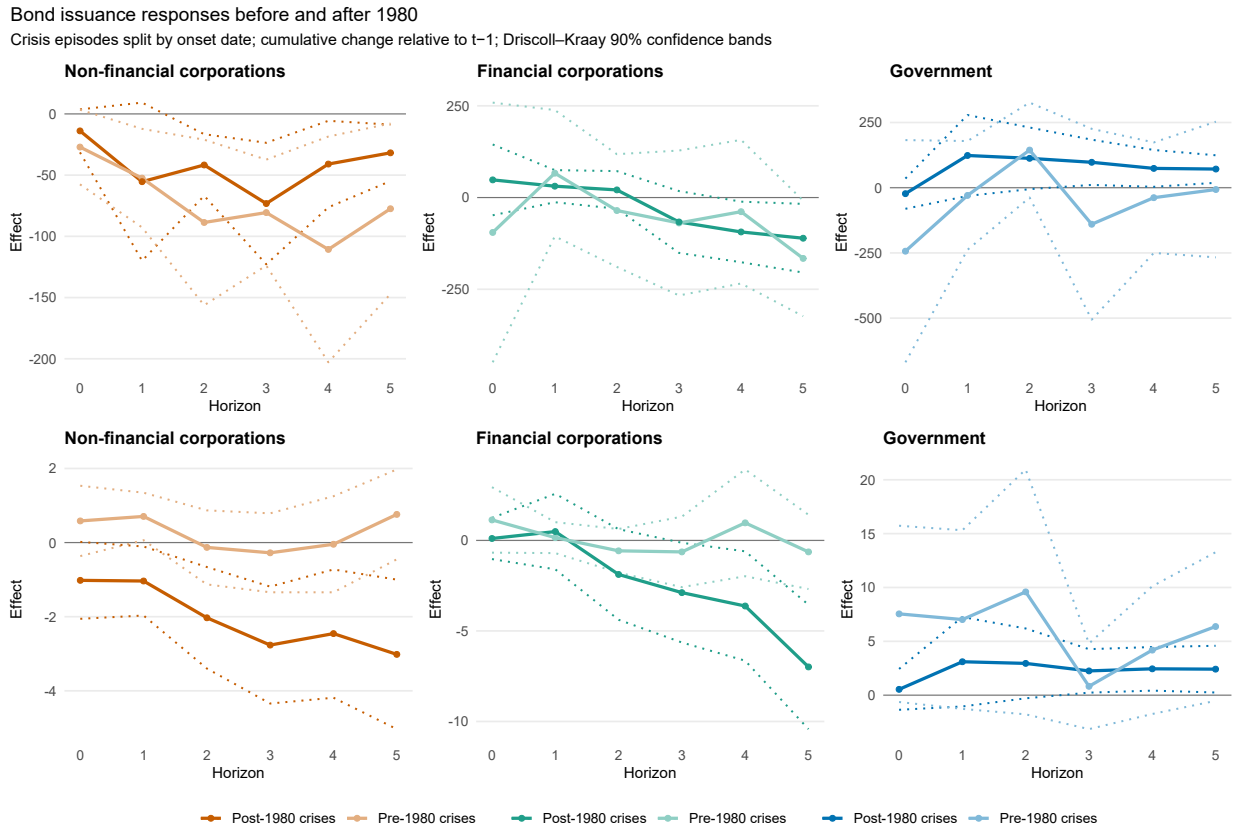


Figure 30: Bond issuance responses before and after 1980

Notes: The figure reports long-difference local-projection estimates of bond issuance responses to banking crises, allowing crisis responses and all baseline controls to differ between crisis episodes beginning before and after 1980. Panel A reports cumulative changes in log raw issuance and Panel B reports changes in log issuance relative to GDP. Dark lines denote post-1980 crisis episodes and light lines denote pre-1980 episodes. Dotted lines report 90% Driscoll–Kraay confidence intervals. The specification includes country fixed effects, two lags of the macroeconomic controls and dependent variable, and 2% winsorization.

cally, for each horizon h, I estimate

$$\begin{aligned}\Delta_h y_{i,t+h} = \alpha_i + \beta_h \text{Crisis}_{i,t} + \delta_h \left(\text{Crisis}_{i,t} \times \text{Post1980}_t\right) + \lambda_h \text{Post1980}_t \\ + \Gamma_h' X_{i,t} + \Theta_h' \left(X_{i,t} \times \text{Post1980}_t\right) + \varepsilon_{i,t+h},\end{aligned} \tag{5}$$

where the pre-1980 crisis response is given by β_h and the post-1980 response by $\beta_h + \delta_h$. The remaining specification follows the baseline, including country fixed effects, two lags of the macroeconomic controls and dependent variable, 2% winsorization, and Driscoll–Kraay standard errors.

Figure 30 presents the results. Panel A uses the baseline log issuance outcome. For non-financial corporations, the central result is present in both periods: bond issuance falls sharply following banking crises before as well as after 1980. The estimated trajectories are quantitatively similar, and formal tests of the interaction terms provide no evidence of systematic differences between the pre- and post-1980 responses. The responses of financial-corporate and government issuance are considerably noisier and likewise show no systematic evidence of a break around 1980.

The GDP-scaled specification in Panel B reveals more pronounced heterogeneity. For non-financial corporations, issuance relative to GDP changes little following pre-1980 crises, whereas post-1980 crises are followed by a persistent decline. The difference between the two periods becomes statistically significant at several horizons. A similar divergence emerges for financial corporations at longer horizons, while government issuance exhibits no systematic pre/post-1980 difference.

These results help distinguish two implications of the secular deepening of bond markets. First, the underlying contraction in non-financial corporate bond issuance following banking crises is not specific to the modern, high-issuance period: the response of raw issuance is already clearly visible before 1980. Second, the contraction becomes substantially more important relative to aggregate economic activity once bond markets deepen. The stronger post-1980 response of issuance relative to GDP is therefore consistent with the growing macroeconomic importance of bond-market financing rather than with the baseline issuance response being mechanically generated by the post-1980 expansion of bond markets.

8.2.10 Leave-One-Out analysis - countries

A potential concern with the baseline estimates is that the results may be driven by a small number of influential countries or crisis episodes. To assess the stability of the estimated effects, I conduct a systematic leave-one-out (LOO) robustness analysis. Starting from the baseline full-sample specification, I iteratively re-estimate the model while excluding one country at a time. For each exclusion, I recover the full impulse response function (IRF) for the issuance path following banking crises. The LOO analysis evaluates both the

dispersion of the estimated response across countries and the sensitivity of the baseline estimate to any single-country exclusion.

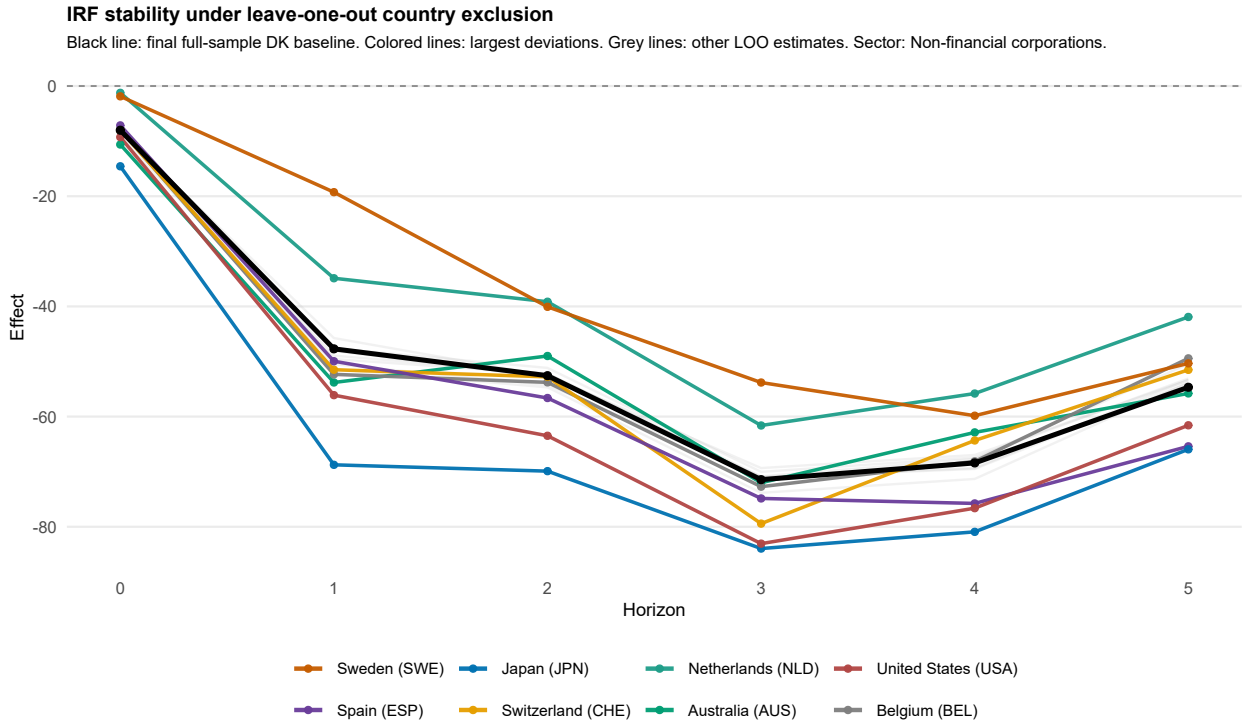


Figure 31: IRF stability under leave-one-out country exclusions. The figure overlays impulse response functions (IRFs) from the baseline LDLP specification estimated on the full sample (solid black line) and on samples excluding individual countries. Response variable is the log of non-financial corporations bonds issues. Grey dashed lines show IRFs from all leave-one-out estimations. Colored dashed lines highlight the eight most influential exclusions, defined as the countries whose exclusion yields the largest maximum absolute deviation from the full-sample IRF across horizons. Influence is measured as the maximum absolute difference between the leave-one-out and full-sample IRF coefficients across horizons, i.e. $\max_h |\hat{\beta}_h^{(-i)} - \hat{\beta}_h|$.

Figure 31 presents a leave-one-out (LOO) of crisis-period issuance coefficients across horizons. The solid black line reports the full-sample impulse response, while grey dashed lines show the IRFs obtained from all leave-one-out estimations. To enhance readability, the figure highlights the eight most influential exclusions, defined as the countries whose omission produces the largest maximum absolute deviation from the full-sample IRF across horizons. Even for these most influential cases, the IRFs closely track the full-sample response at all horizons. While the magnitude of the responses varies modestly, the qualitative pattern—an initial contraction followed by partial recovery—remains unchanged. The estimated responses are highly stable across LOO iterations: excluding any individual country does not materially affect the sign, timing, or overall shape of the impulse responses. This indicates that the baseline results are not driven by a small subset of influential countries.

8.2.11 Leave-One-Out analysis - crises

Similarly, Figure 32 repeats the same exercise for crisis rather than country exclusions, again confirming the stability of the baseline results. Specification exactly as for the country LOO.

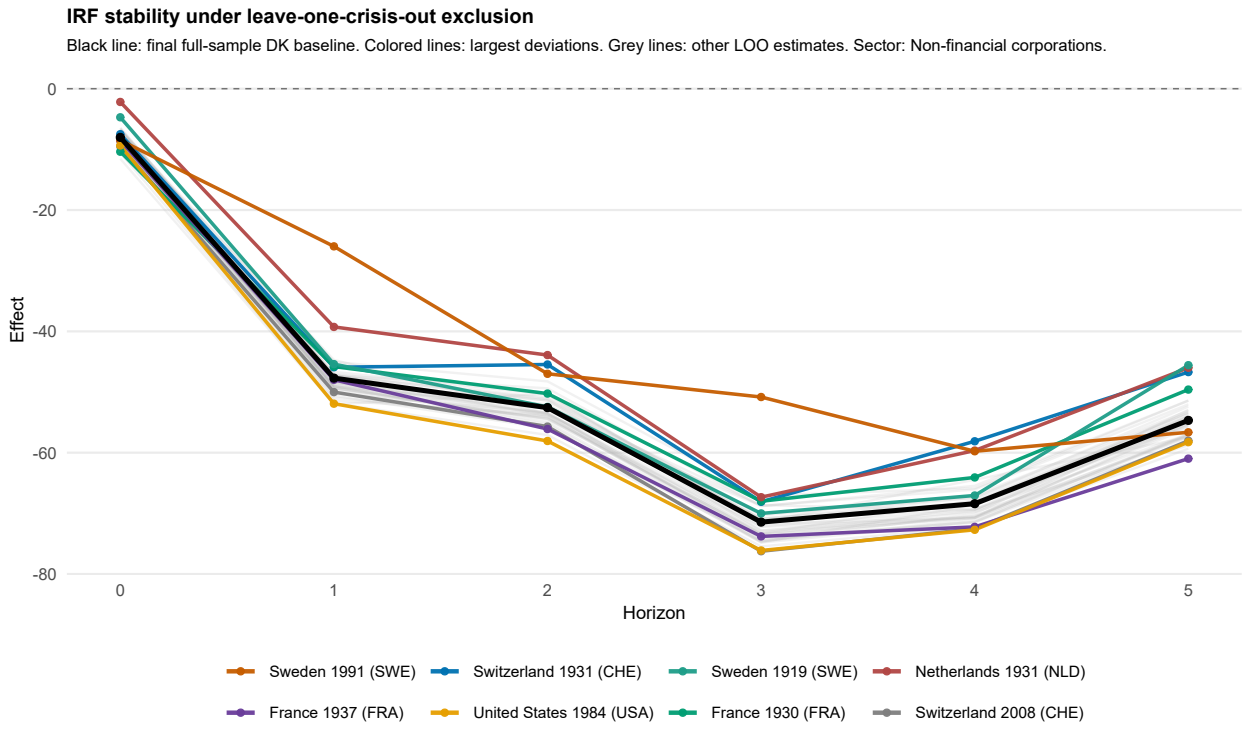


Figure 32: IRF stability under leave-one-out crisis exclusions. The figure overlays impulse response functions (IRFs) from the baseline LDLP specification estimated on the full sample (solid black line) and on samples excluding individual crisis. Response variable is the log of non-financial corporations bonds issues. Grey dashed lines show IRFs from all leave-one-out estimations. Colored dashed lines highlight the eight most influential exclusions, defined as the crisis whose exclusion yields the largest maximum absolute deviation from the full-sample IRF across horizons. Influence is measured as the maximum absolute difference between the leave-one-out and full-sample IRF coefficients across horizons, i.e. $\max_h |\hat{\beta}_h^{(-i)} - \hat{\beta}_h|$.

8.2.12 Year fixed effects

Banking crises are often clustered in periods of global financial and macroeconomic stress. A potential concern is therefore that the baseline estimates partly capture common shocks to bond issuance in crisis years rather than country-specific changes associated with banking crises. To address this concern, I augment the baseline long-difference local projection with calendar-year fixed effects:

$$\Delta_h y_{i,t} = \beta_h \text{Crisis}_{i,t} + \sum_{j=1}^{2} \Gamma'_{h,j} X_{i,t-j} + \sum_{j=1}^{2} \rho_{h,j} y_{i,t-j} + \alpha_i + \lambda_t + \varepsilon_{i,t+h}, \tag{6}$$

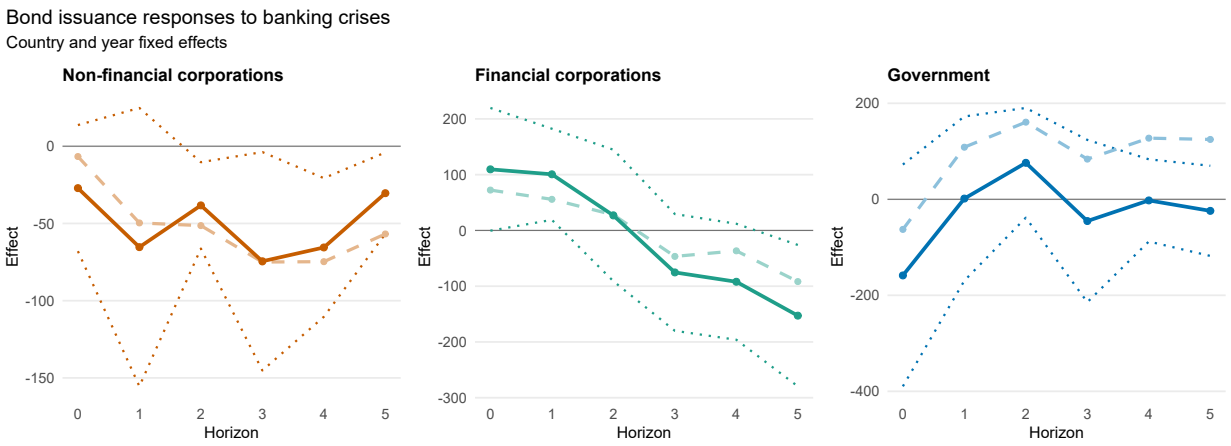


Figure 33: Bond issuance responses to banking crises: calendar-year fixed effects

Notes: The solid lines report long-difference local-projection estimates including country and calendar-year fixed effects; dotted lines denote 90% confidence intervals based on Driscoll–Kraay standard errors with five lags. The lighter dashed lines report the baseline country-fixed-effects specification re-estimated on the identical estimation sample. All other specification choices are unchanged from the baseline.

where λ_t denotes calendar-year fixed effects. All other elements of the baseline specification are unchanged, including the controls, lag structure, winsorization, and Driscoll–Kraay standard errors with five lags. The inclusion of year fixed effects absorbs shocks common to all countries in a given year, so that β_h is identified from cross-country differences within the same calendar year.

Figure 33 reports the resulting responses. Because introducing year fixed effects can alter the estimation sample, the figure also reports the baseline country-fixed-effects specification re-estimated on exactly the same observations retained by the country-and-year-fixed-effects model. This makes it possible to distinguish changes due to the additional fixed effects from changes in sample composition.

The main result for non-financial corporate issuance is robust. The medium-run contraction remains quantitatively close to the matched-sample baseline, particularly at horizons three and four, and remains statistically significant at the 90% level from horizons two through five. The response attenuates somewhat at the longest horizon, but does not reverse. The matched-sample country-fixed-effects estimates are themselves very close to the original baseline, indicating that changes in sample composition are not driving the result.

The responses of financial corporations and the government are more sensitive to common time shocks. Financial-corporation issuance continues to increase initially before contracting at longer horizons, while the positive medium-run government issuance response largely disappears. Overall, absorbing common calendar-year variation leaves the central finding—a persistent post-crisis contraction in non-financial corporate bond issuance—intact.

8.2.13 Alternative banking-crisis chronology

A potential concern is that the estimated issuance response depends on the classification or precise dating of banking crises. The baseline analysis uses the banking-crisis chronology of Baron et al. (2021). Historical crisis chronologies can differ both in whether an episode is classified as a systemic banking crisis and in the precise year assigned to crisis onset. I therefore re-estimate the headline long-difference local projection for non-financial corporate bond issuance using the crisis chronology of Jordà et al. (2017), specifically the JST Dataset version R6. Because the JST chronology covers a narrower set of countries than the baseline sample, I separate changes in sample composition from changes in crisis dating. I first re-estimate the baseline BVX specification using only country-year observations covered by JST and then replace the BVX crisis indicator with the JST indicator on exactly the same sample. All other features of the baseline specification are unchanged.

Figure 34 shows that restricting the BVX specification to the JST-covered sample has virtually no effect on the estimated issuance response. More importantly, replacing the BVX dates with the JST chronology yields the same pronounced and persistent contraction in NFC bond issuance. The JST estimates are somewhat more negative at intermediate horizons and less precisely estimated, but remain statistically below zero from horizons two through five. The estimated decline therefore does not depend on the particular banking-crisis chronology used in the baseline analysis.

8.3 Real effects robustness

This appendix reports robustness checks for the real-effects exercise in Section 4.6. The baseline specification relates post-crisis recovery to pre-crisis non-financial corporate bond-market depth, defined as the average NFC bond issuance-to-GDP ratio over the five years before crisis onset. Outcomes are measured from the crisis year t_0 to $t_0 + h$, so the estimates capture subsequent recovery conditional on the crisis having begun.

Figure 35 shows that the positive association between bond-market depth and investment recovery is not specific to the baseline depth measure. The pattern is similar when depth is measured using the median rather than the mean NFC issuance-to-GDP ratio, a longer ten-year pre-crisis window, a log transformation, issuance relative to lagged corporate bank credit, or total corporate rather than NFC issuance. The magnitude varies across definitions, but the medium-run investment slope remains positive.

Figure 36 shows that the investment result is not driven by the inclusion of the severity controls. The estimated slope remains positive when the crisis-year GDP shock is omitted, when the pre-crisis credit boom is omitted, and when both controls are excluded. This reduces the concern that the baseline result mechanically reflects either stronger credit expansions before crises or differences in the size of the initial output collapse.

Figure 37 and Table 27 report a full-panel interaction specification similar to Krishna-

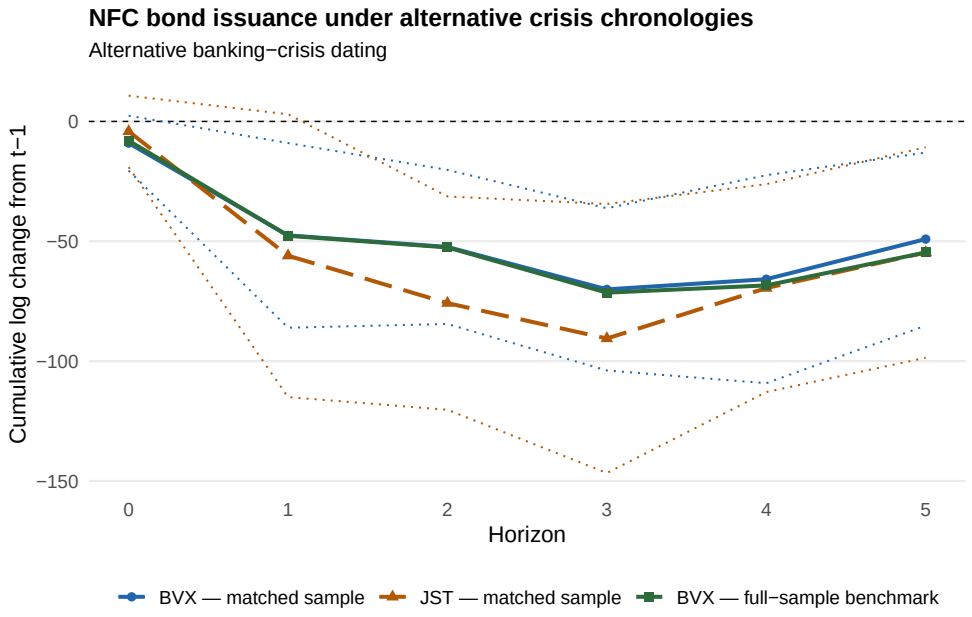


Figure 34: NFC bond issuance under alternative banking-crisis chronologies. The figure reports long-difference local projection estimates of the cumulative response of nonfinancial corporate bond issuance following banking-crisis onset. The solid line uses the Baron et al. (2021) crisis chronology and the dashed line uses the Jordà–Schularick–Taylor chronology, estimated on the common JST-covered sample. The dotted grey line reports the original full-sample BVX benchmark. The dependent variable is $100 \times (y_{i,t+h} - y_{i,t-1})$, where y denotes log gross NFC bond issuance. All specifications retain the baseline country fixed effects, lag structure, macroeconomic controls, and outlier treatment. Error bars denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors.

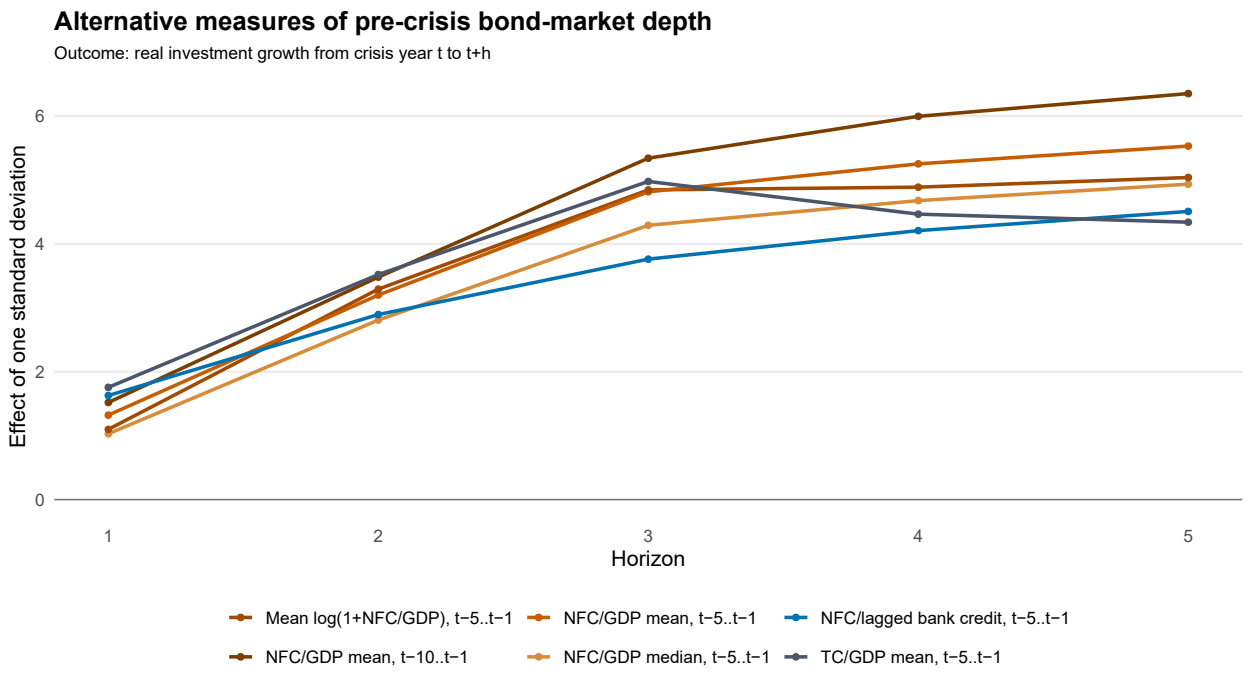


Figure 35: Alternative measures of pre-crisis bond-market depth. The figure reports crisis-event regressions of real investment recovery on alternative measures of pre-crisis bond-market depth. The dependent variable is $100 \times (\log I_{i,t_0+h} - \log I_{i,t_0})$, where I denotes real investment. Each depth measure is standardized across crisis episodes. All specifications control for the pre-crisis corporate credit boom and the crisis-year real GDP shock. Standard errors are clustered by country.

Table 27: Full-panel interaction: bond-market depth in crisis and non-crisis years

Outcome	Horizon	Observations	Countries	Crisis obs.	NFC depth × crisis onset	NFC depth × non-crisis years
Real GDP	1	741	16	29	0.14 (0.17)	0.07 (0.11)
Real GDP	2	740	16	29	0.22 (0.19)	0.17 (0.20)
Real GDP	3	739	16	29	0.31 (0.25)	0.20 (0.29)
Real GDP	4	738	16	29	0.44 (0.35)	0.19 (0.38)
Real GDP	5	737	16	29	0.63 (0.42)	-0.01 (0.40)
Real investment	1	677	16	27	0.29 (0.79)	0.14 (0.34)
Real investment	2	676	16	27	0.73 (0.90)	0.64 (0.63)
Real investment	3	675	16	27	1.05 (1.11)	1.01 (0.82)
Real investment	4	674	16	27	1.51 (1.46)	1.20 (0.92)
Real investment	5	673	16	27	1.81 (1.57)	1.24 (0.95)

Notes: The table reports a full-panel interaction specification in the spirit of Krishnamurthy and Muir (2025). The dependent variable is $100 \times (\log y_{i,t+h} - \log y_{i,t})$, where y is real GDP or real investment. The table reports the coefficient on pre-crisis NFC bond-market depth interacted with crisis-onset years and clean non-crisis years. NFC depth is the standardized five-year rolling average of NFC bond issuance relative to GDP. Regressions include country fixed effects, year fixed effects, two lags of real GDP growth, the crisis-onset indicator, and the pre-crisis credit boom interacted with the crisis-onset indicator. Standard errors are computed using Driscoll and Kraay (1998) standard errors with eight lags. Parentheses report standard errors. Sample is reduced as the depth measure requires data to be available well before crisis onsets which is often not the case historically.

murthy and Muir (2025). For each horizon $h = 1, \ldots, 5$, I estimate the full-panel interac-

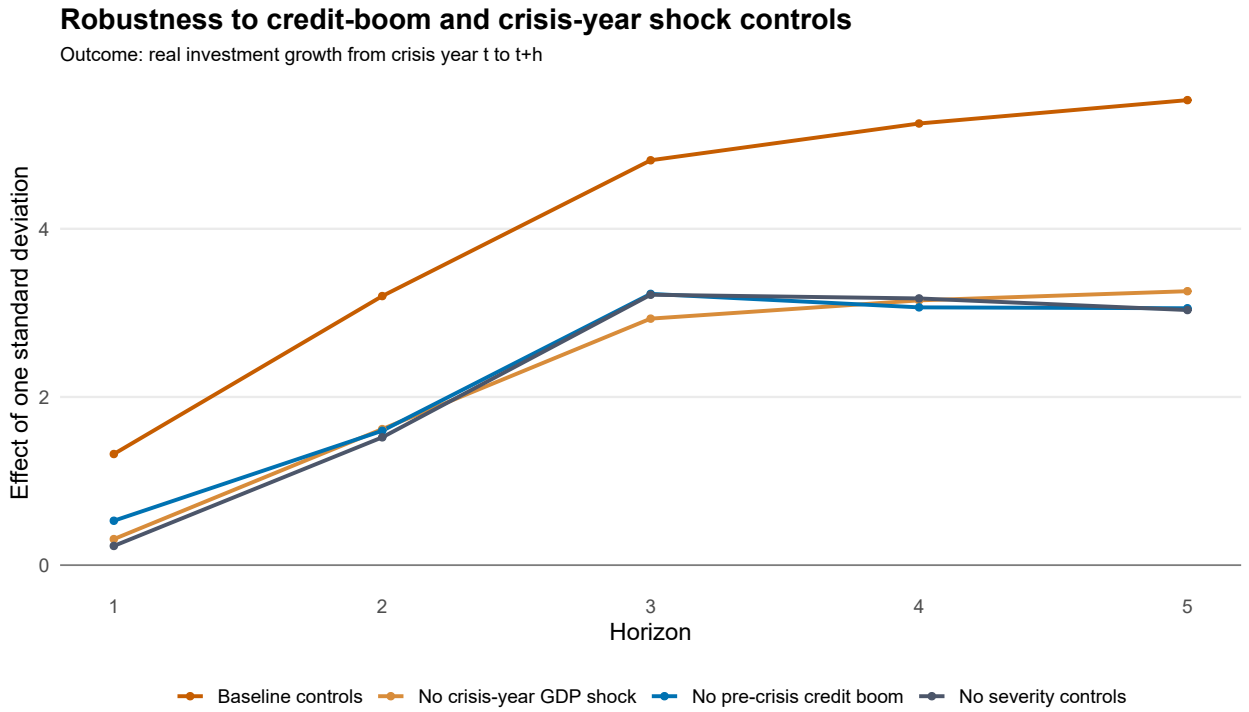


Figure 36: Robustness to credit-boom and crisis-year shock controls. The figure reports crisis-event regressions of real investment recovery on pre-crisis NFC bond-market depth under alternative control sets. The baseline controls for the pre-crisis corporate credit boom and the crisis-year GDP shock. The remaining specifications remove the crisis-year GDP shock, remove the pre-crisis credit boom, or remove both controls. Standard errors are clustered by country.

tion specification

$$\begin{aligned} 100 \times (\log y_{i,t+h} - \log y_{i,t}) = {} & \beta_h^C \left(\text{Depth}_{i,t} \times \text{Crisis}_{i,t}\right) + \beta_h^N \left(\text{Depth}_{i,t} \times \text{NonCrisis}_{i,t}\right) \\ & + \gamma_h \left(\text{CreditBoom}_{i,t} \times \text{Crisis}_{i,t}\right) + \delta_h \text{Crisis}_{i,t} + \rho_{1h} \Delta \log GDP_{i,t-1} \\ & + \rho_{2h} \Delta \log GDP_{i,t-2} + \alpha_i + \lambda_t + \varepsilon_{i,t+h}. \end{aligned} \tag{7}$$

Here $y_{i,t}$ denotes either real GDP or real investment. $\text{Depth}_{i,t}$ is the standardized five-year rolling average of NFC bond gross issuance relative to GDP, measured over $t-5$ to $t-1$. hence it is the pre-period bond-market depth in this full-panel specification, not always "pre-crisis depth," because for non-crisis years it is just lagged depth. $\text{Crisis}_{i,t}$ is an indicator for a systemic banking-crisis onset. $\text{NonCrisis}_{i,t}$ is an indicator for clean non-crisis years, defined as years with no crisis onset in the current year or over the subsequent five-year horizon. $\text{CreditBoom}_{i,t}$ is the change in corporate credit-to-GDP from $t-4$ to $t-1$. The specification includes country fixed effects α_i, year fixed effects λ_t, and two lags of real GDP growth. Standard errors are computed using Driscoll and Kraay (1998) standard errors with eight lags.

The coefficients of interest are β_h^C and β_h^N. β_h^C measures the association between pre-crisis NFC bond-market depth and subsequent real outcomes in banking-crisis onset years,

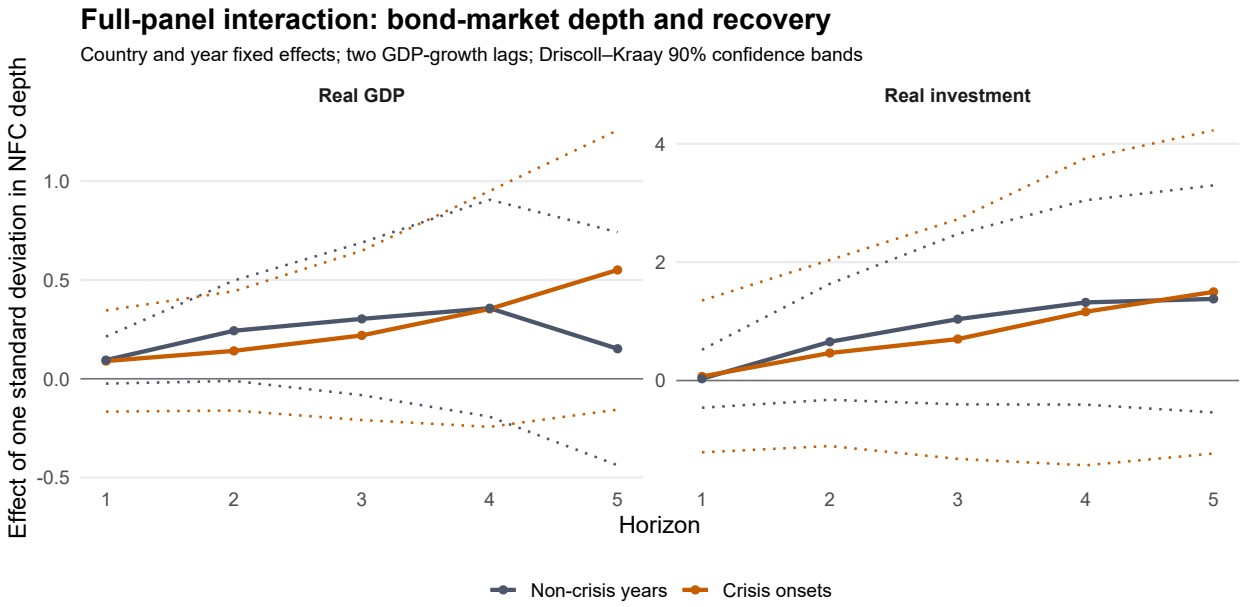


Figure 37: Full-panel interaction: bond-market depth in crisis and non-crisis years. The figure reports coefficient paths from a full-panel interaction specification. The dependent variable is $100 \times (\log y_{i,t+h} - \log y_{i,t})$, where y is real GDP or real investment. The figure plots the implied slope of real outcomes with respect to five-year lagged NFC bond-market depth separately for crisis-onset years and non-crisis years. Regressions include country fixed effects, year fixed effects, two lags of real GDP growth, the crisis-onset indicator, and the pre-crisis credit boom interacted with the crisis-onset indicator. Dotted lines denote 90% confidence intervals based on Driscoll and Kraay (1998) standard errors with eight lags following Krishnamurthy and Muir (2025).

while β_h^N measures the corresponding association in clean non-crisis years. A crisis-specific spare-tire mechanism would imply $\beta_h^C > \beta_h^N$, especially for investment. The coefficients on NFC bond-market depth are positive for investment in both crisis-onset and non-crisis years, especially at medium horizons. However, the crisis and non-crisis slopes are not sharply separated, and the difference between them is imprecise. The full-panel evidence therefore supports a cautious interpretation: pre-crisis bond-market depth is associated with stronger investment growth, including after banking crises, but the data do not cleanly establish that this association is unique to crisis periods.

This full-panel exercise is useful because it maps the real-effects result back to the theory. If bond markets acted as a crisis-specific spare tire, the coefficient on NFC depth should be substantially larger in crisis-onset years than in clean non-crisis years. The estimates do not show such a sharp separation. Investment slopes are positive in both states and their difference imprecise. Thus, the real-effects evidence is consistent with bond-market depth being associated with stronger investment performance, but it does not provide strong support for a distinct crisis-state substitution mechanism.

Table 28: Sample coverage for the real-effects crisis-event regressions

Statistic	Count
Crisis episodes	59
Real GDP outcome available	59
Real investment outcome available	57
Pre-crisis NFC depth available	38
Credit boom available	39
Crisis-year GDP shock available	58
Baseline GDP sample	28
Baseline investment sample	26

Notes: The table reports sample coverage for the main crisis-event real-effects specification at horizon $h = 5$. The baseline sample requires the outcome, pre-crisis NFC bond-market depth, the pre-crisis corporate credit boom, and the crisis-year real GDP shock to be observed.

9 Appendix D: Credit-market repricing and NFC issuance

This appendix describes the credit-spread data, their treatment and overlap with the bond-issuance panel, and the full set of price–quantity results underlying Section 4.5. The exercise is designed as interpretation evidence. It asks whether the severity of risky debt-market repricing is systematically related to the severity of the subsequent NFC primary-issuance contraction. It does not claim to recover a structural bond-supply shock.

9.1 Credit-spread data and construction

The credit-spread series come from the replication data of Krishnamurthy and Muir (2025). Their historical measure, available for 1869–1929, is constructed from bond prices and yields in the *Investor's Monthly Manual* as a within-country risky-minus-safe spread, averaged over the final quarter of each year. The underlying bond cross-section is broad and includes railroads, banks, sovereign or quasi-sovereign issuers, and other corporations; it should therefore be interpreted as a price measure for the risky bond market rather than as an NFC-specific borrowing rate. Modern observations combine the U.S. Moody's Baa–Aaa spread with corporate spread series from central banks and commercial providers. Table 29 summarizes the main source regimes and their treatment.

I use the authors' mean-normalized country spread, `spreadnorm`, and their supplied annual change, `chgspreadnorm`. I do not reconstruct the change mechanically from adjacent observations because the supplied variable deliberately suppresses comparisons across source breaks and gaps. The spread panel is merged to the issuance data by ISO3 code and calendar year, while crisis onsets continue to follow **?**, as in the main analysis. No spread observations are interpolated. The strict price sample requires the supplied normalized spread change at crisis onset and excludes Sweden from 1987 onward because that segment measures bank-loan rather than bond-market pricing. Hence, the Swedish crises of 1991 and 2008 do not enter the strict sample; the earlier Swedish bond observations remain eligible. The baseline uses the same-year change from $t-1$ to t, with maximum repricing over $t, t+1$ and $t, t+1, t+2$ reported as timing robustness checks.

9.2 Overlap and exact crisis sample

The full NFC baseline contains 36 crisis origins with the controls and outcomes required by the exact LDLP at the reported horizons. Restricting the panel to countries that ever have a strict KM bond spread leaves the baseline response essentially unchanged, but requiring a spread change at the crisis origin selects a much milder set of episodes. The constant-origin price sample used for the main heterogeneity analysis contains 18 crisis onsets from nine countries. The full fixed-effects estimation panel contains 12 countries because non-crisis observations from additional spread-covered countries enter the control and fixed-effects

Table 29: Summary of KM spread sources and treatment in the issuance merge

Period and source	Spread concept	Treatment
1869–1929: *Investor's Monthly Manual*	Within-country high-yield minus low-yield bond spread; broad risky bond market rather than NFC-only	Included when the author-supplied normalized change is observed
United States, 1928 onward: Moody's	Baa minus Aaa corporate bond spread	Included
Japan and South Korea: central banks	Corporate credit-spread series; Korea uses three-year AA- corporate bonds	Included
Modern Europe and other countries	Corporate-minus-government or comparable bond-market spread	Included where classified as a bond spread
Sweden, 1987 onward	Bank-loan spread to non-financial firms, with short average maturity	Excluded from strict bond sample
World-war years	Government benchmarks distorted by wartime borrowing	Retained as missing

The table summarizes the data appendix and replication variables of **?**. Their final replication file does not provide a source flag for every country-year. I therefore retain the author-supplied normalized variables as the source of record and impose only restrictions that can be identified unambiguously.

structure.

The exact 18-event sample is shown in Table 30. The same-year measure places the U.S. 2007 onset in the low-repricing group because the large U.S. spread increase materialized mainly in 2008. This is why the onset-window timing checks are important and are reported below.

Table 30: Exact constant-origin crisis sample by same-year spread-repricing tercile

Low repricing	Middle repricing	High repricing
Japan 1990 (−0.85)	United States 1990 (0.33)	Japan 1997 (0.98)
Sweden 1878 (−0.31)	Japan 1927 (0.46)	Spain 2008 (1.31)
Japan 1923 (−0.18)	Japan 2001 (0.52)	Germany 2008 (1.77)
Sweden 1907 (−0.16)	Japan 1922 (0.76)	France 2008 (2.18)
United States 1984 (0.05)	Switzerland 2008 (0.84)	United Kingdom 2008 (2.98)
United States 2007 (0.07)	Spain 2010 (0.85)	South Korea 1997 (3.77)

Parentheses report the author-supplied normalized spread change from $t-1$ to the BVX crisis-onset year. The group cutoffs are defined on this exact 18-event constant-origin sample. The modern Swedish bank-loan segment is excluded; only the historical Swedish bond observations enter.

9.3 Empirical specifications

Nested-sample baseline. The first exercise re-estimates the exact baseline NFC LDLP on nested samples. It retains the two-percent winsorization of issuance levels, the log transformation, two lags of the dependent variable, two lags of real GDP growth, inflation, the short-term interest rate, and the current account balance, country fixed effects, and Driscoll–Kraay standard errors. The samples are: (i) the full NFC baseline; (ii) countries ever covered by strict KM spreads; (iii) crisis origins with an observed strict spread change; and (iv) crisis origins without one. The crisis coefficient in the third and fourth samples is restricted to the relevant origins while ordinary country-year observations remain in the estimation panel.

Continuous spread interaction. Equation (3) is estimated separately at horizons $h = 0, \ldots, 5$. The spread interaction is centered at the mean spread change among observed crisis origins. The reported "effect of one standard deviation" rescales θ_h by the cross-crisis standard deviation of the normalized spread change. Conditional effects at selected spread percentiles use the full covariance between the crisis coefficient and the interaction coefficient.

Tercile specification. As a transparent non-parametric presentation, the exact 18-event sample is divided into low, middle, and high same-year spread-repricing terciles. The three crisis indicators are entered jointly in the baseline LDLP. Formal Wald tests use the

full Driscoll–Kraay covariance matrix, with the high-minus-low contrast pre-specified as the central comparison.

Demand-oriented outcomes. To assess whether the spread gradient merely reflects lower financing demand, I replace raw issuance with

$$100\,\Delta_h \log\left(1 + \frac{\text{NFC issuance}}{D}\right),$$

where D is nominal GDP, fixed investment, broader investment, or lagged corporate bank credit. Investment variables from the Global Macro Database are converted into the same legacy local-currency units as the issuance series using the country-year conversion factor implied by nominal GDP. Lagged corporate credit is measured prior to the outcome horizon to avoid a contemporaneous mechanical relationship.

9.4 Sample composition and external validity

Figure 38 separates country composition from crisis-origin composition. Panel A shows that restricting the baseline to countries that ever appear in the strict KM sample leaves the NFC response nearly unchanged. Panel B shows that the subset of crisis origins with an observed strict spread change has a much milder issuance response than the complementary no-spread set. Table 31 reports the corresponding coefficients, and Table 32 formally tests their difference.

At $h = 3$, the spread-observed crisis coefficient is approximately −20, compared with about −123 among crisis origins without an observed spread change. The jointly estimated difference is roughly 104 log points ($p = 0.056$), and the gap remains large at $h = 5$ ($p = 0.025$). Historical spread coverage therefore does not span the crisis block generating the strongest full-sample contraction. This limits external validity: the price data can characterize heterogeneity within observed crises, but cannot establish that spread repricing explains the full-sample average issuance decline.

9.5 Spread repricing and NFC issuance

Figure 39 reports the continuous conditional crisis effect over the observed spread distribution. The spread gradient is weak at short horizons but becomes economically and statistically important at medium horizons. At $h = 3$, the raw gradient is −32.0 normalized log points per unit of spread change; a one-standard-deviation larger spread increase implies an additional 38.1 log-point contraction. The corresponding $h = 5$ estimate is an additional 32.3 log-point contraction. Table 33 reports the coefficients.

Conditional effects clarify the economic content. At the 25th percentile of same-year repricing, the estimated $h = 3$ response is slightly positive and imprecise. At the median

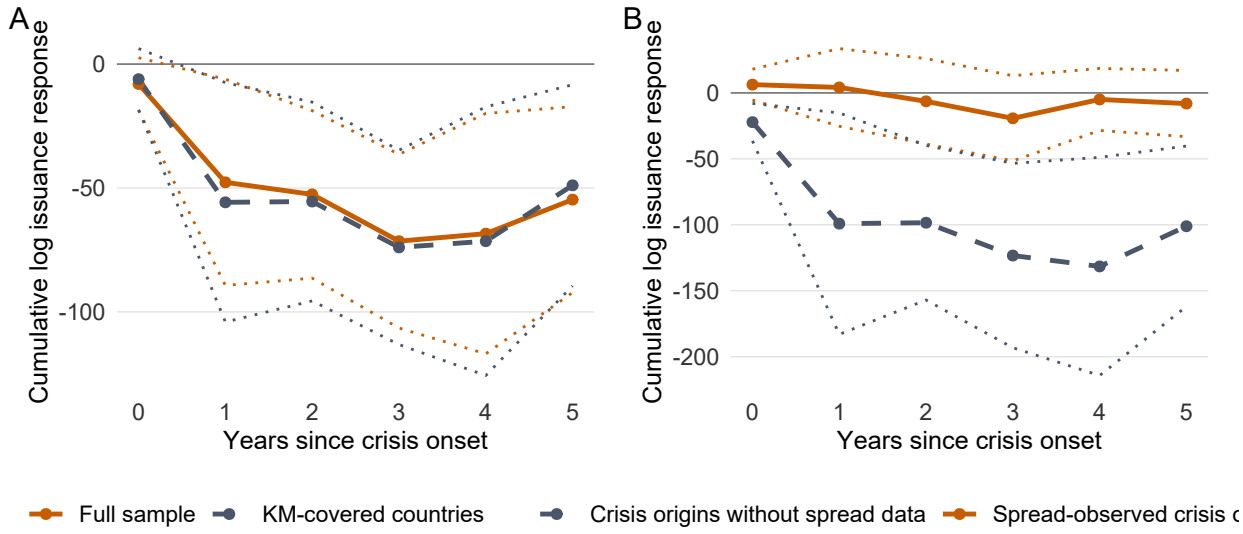


Figure 38: NFC issuance responses across spread-data samples. Panel A compares the full-sample baseline with the same specification estimated in countries ever covered by the strict KM bond-spread series. Panel B jointly estimates separate crisis effects for onsets with and without an observed strict spread change. All specifications use the baseline LDLP controls, country fixed effects, two-percent winsorization, and Driscoll–Kraay 90% confidence intervals.

Table 31: NFC issuance responses across nested spread-data samples

Sample	$h = 1$	$h = 3$	$h = 5$	Crisis onsets
Full NFC baseline	-47.7*	-71.4***	-54.7**	36
	(25.3)	(21.3)	(22.8)	
Countries ever covered by strict KM spreads	-55.8*	-74.0***	-48.9**	31
	(29.4)	(23.8)	(24.7)	
Only crisis onsets with strict spread changes	3.9	-20.0	-5.9	18
	(17.8)	(19.5)	(15.2)	
Only crisis onsets without strict spread changes	-99.7*	-123.4***	-101.0***	18
	(51.0)	(42.3)	(37.2)	

Entries report LDLP coefficients with Driscoll–Kraay standard errors in parentheses. The dependent variable is the cumulative log change in NFC gross bond issuance relative to $t - 1$. All specifications use the exact baseline controls and country fixed effects. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

Table 32: Formal tests of the difference between spread-observed and no-spread crisis effects

Horizon	Difference	Std. error	90% CI	p-value
$h = 1$	103.3	60.5	[3.7, 202.9]	0.088
$h = 3$	104.0	54.4	[14.6, 193.5]	0.056
$h = 5$	93.0	41.5	[24.7, 161.3]	0.025

The difference is the jointly estimated spread-observed crisis coefficient minus the coefficient for crisis origins without an observed strict spread change. Positive values indicate a less negative issuance response in the spread-observed sample. Inference uses the full Driscoll–Kraay covariance matrix.

it is modestly negative. At the 75th percentile it reaches about −30.5 log points, and at the 90th percentile it reaches about −68.8 log points. The latter corresponds to an issuance decline of roughly one half relative to the pre-crisis year. These estimates, reported in Table 34, explain why raw crisis plots contain several large issuance increases: low-repricing crises do not display the same quantity contraction as high-repricing crises.

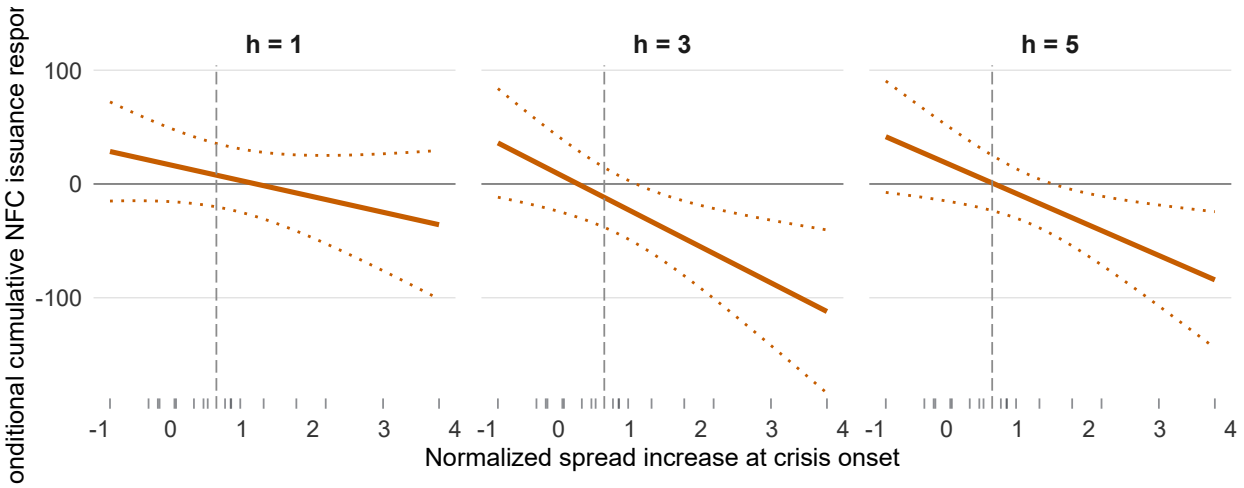


Figure 39: Conditional NFC crisis effects across the spread-repricing distribution. Each horizon is estimated using the exact baseline LDLP augmented with a centered interaction between the crisis indicator and the author-supplied KM spread change. Solid lines are point estimates; dotted lines are Driscoll–Kraay 90% confidence intervals. Rug marks show observed crisis spread changes, and the vertical dashed line marks their median.

Figure 40 provides the same evidence in grouped form. The low-repricing path remains positive through $h = 5$, the middle group remains close to zero, and the high-repricing group falls sharply. At $h = 3$, the high-minus-low difference is −115.6 log points ($p = 0.006$); at $h = 5$ it is −90.2 log points ($p = 0.008$). The middle group is not individually distinguishable from either tail at every horizon, so the grouped evidence should be interpreted primarily as a sharp high-versus-low contrast rather than as a precisely estimated linear ordering across all terciles.

Table 33: NFC issuance responses and the size of crisis spread repricing

Horizon	Crisis effect at mean spread	Spread gradient	Effect of one SD larger spread	Crisis onse
	4.8	-13.9	-16.5	
$h = 1$	(16.8)	(12.2)	(14.5)	18
	-18.6	-32.0**	-38.1**	
$h = 3$	(15.6)	(14.1)	(16.8)	18
	-4.9	-27.2**	-32.3**	
$h = 5$	(13.6)	(13.2)	(15.6)	18

The spread interaction is centered at the mean spread change among observed crisis onsets. The third column reports the a one-unit increase in the normalized KM spread; the fourth rescales this coefficient by the crisis-sample standard deviatio standard errors are in parentheses.

Table 34: Conditional NFC issuance effects at selected spread-repricing percentiles

Horizon	Spread percentile	Spread change	Conditional effect	90% CI
			6.9	
$h = 3$	25th percentile	0.06	(19.5)	[-25.1, 38.9]
			-11.8	
$h = 3$	Median	0.64	(16.0)	[-38.1, 14.5]
			-30.5*	
$h = 3$	75th percentile	1.22	(16.4)	[-57.4, -3.6]
			-68.8**	
$h = 3$	90th percentile	2.42	(26.7)	[-112.7, -24.9]
			16.7	
$h = 5$	25th percentile	0.06	(19.7)	[-15.6, 49.1]
			0.9	
$h = 5$	Median	0.64	(14.8)	[-23.4, 25.2]
			-15.0	
$h = 5$	75th percentile	1.22	(12.9)	[-36.3, 6.3]
			-47.5**	
$h = 5$	90th percentile	2.42	(20.8)	[-81.7, -13.3]

Conditional effects are linear combinations of the crisis coefficient and the centered crisis-spread interaction. Standard errors use the full Driscoll–Kraay covariance between the two coefficients.

Table 35: NFC issuance responses by crisis spread-repricing tercile

Spread-repricing group	$h = 1$	$h = 3$	$h = 5$
	29.4	31.3	45.4
Low spread repricing	(26.2)	(23.7)	(27.7)
	-7.8	-1.7	-15.1
Middle spread repricing	(29.8)	(33.7)	(29.0)
	-7.5	-84.3***	-44.8***
High spread repricing	(24.8)	(31.2)	(15.6)
	-36.9	-115.6***	-90.2***
High minus low	(36.2)	(42.2)	(34.3)

The low, middle, and high groups contain six crisis onsets each and are estimated jointly. The final row reports the high-minus-low Wald contrast using the full Driscoll–Kraay covariance matrix. Standard errors are in parentheses.

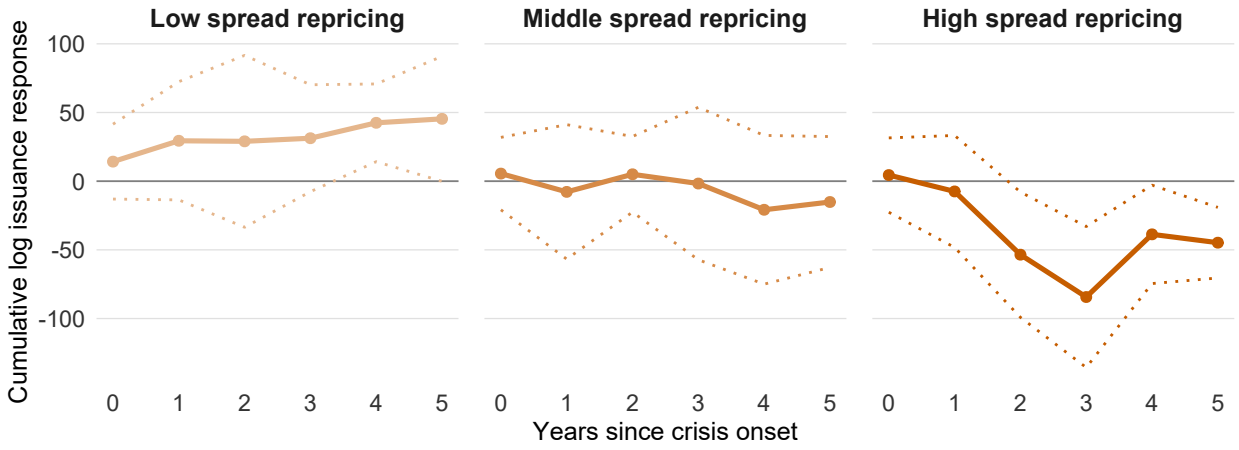


Figure 40: NFC issuance responses by crisis spread-repricing tercile. Crisis origins are divided into terciles using the normalized same-year KM spread change among the exact 18-event constant-origin sample. Each panel contains six crisis onsets. The three coefficients are estimated jointly at every horizon using the baseline controls and country fixed effects. Solid lines are point estimates; dotted lines are Driscoll–Kraay 90% confidence intervals.

9.6 Issuance relative to GDP, investment, and lagged credit

Figure 41 shows the conditional $h = 3$ effects for three scaled outcomes. The negative spread gradient survives when issuance is divided by GDP, fixed investment, or lagged corporate credit. Table 36 reports the full horizon results and also includes broader investment as an alternative denominator.

At $h = 3$, a one-standard-deviation larger spread increase predicts an additional decline of 1.62 units in the transformed NFC issuance-to-GDP outcome and an additional decline of 6.06 units relative to fixed investment. The broader investment definition produces a similar coefficient of −5.61. Relative to lagged corporate credit, the corresponding estimate is −11.15, although this specification has only 13 observed crisis origins. These estimates do not make GDP, investment, or credit exogenous measures of financing demand. They show instead that the issuance contraction is disproportionate to several relevant scales of corporate financing activity.

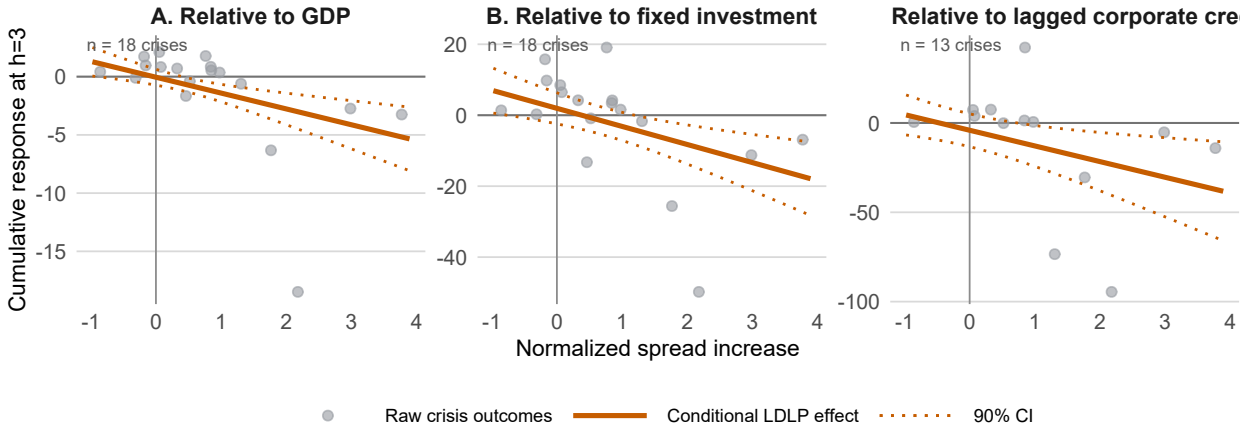


Figure 41: Crisis repricing and NFC issuance relative to GDP, fixed investment, and lagged corporate credit at $h = 3$. Grey dots show the raw crisis-level cumulative changes for the observed price sample. Solid orange lines show conditional effects from the exact baseline LDLP augmented with the centered crisis-spread interaction; dotted lines denote Driscoll–Kraay 90% confidence intervals. Outcomes are $100 \times \Delta \log(1 +$ NFC issuance/denominator) relative to $t - 1$. Fixed investment uses the legacy-currency-adjusted GMD fixed-investment series. The raw dots document the underlying observations but are not adjusted for fixed effects or baseline controls.

Table 36: Spread repricing and NFC issuance relative to GDP, investment, and lagged credit

Denominator	Horizon	Crisis effect at mean spread	Effect of one SD larger spread	Crisis onsets	Countries
		-0.42	-0.77		
GDP	$h = 1$	(0.56)	(0.59)	18	12
		-1.21***	-1.62***		
GDP	$h = 3$	(0.42)	(0.55)	18	12
		-0.75*	-1.37***		
GDP	$h = 5$	(0.43)	(0.48)	18	12
		1.39	-2.40		
Fixed investment	$h = 1$	(2.48)	(1.67)	18	12
		-2.41	-6.06***		
Fixed investment	$h = 3$	(2.37)	(2.17)	18	12
		0.53	-5.79**		
Fixed investment	$h = 5$	(2.99)	(2.51)	18	12
		0.76	-0.82		
Broad investment	$h = 1$	(1.88)	(1.44)	18	12
		-2.70	-5.61***		
Broad investment	$h = 3$	(1.71)	(2.05)	18	12
		-0.80	-4.41***		
Broad investment	$h = 5$	(1.75)	(1.67)	18	12
		-15.00**	-4.56		
Lagged corporate credit	$h = 1$	(6.92)	(4.83)	13	12
		-14.06*	-11.15**		
Lagged corporate credit	$h = 3$	(7.31)	(5.16)	13	12
		0.61	-8.33*		
Lagged corporate credit	$h = 5$	(5.60)	(4.43)	13	12

Outcomes are cumulative changes in $100 \log(1+\text{NFC issuance/denominator})$ relative to $t-1$. The reported gradient is the additional response associated with a one-standard-deviation larger crisis spread increase. Driscoll–Kraay standard errors are in parentheses.

9.7 Timing, influence, and cross-crisis robustness

Annual timing. The same-year baseline is transparent and uses the author-supplied change without researcher-selected timing. Annual crisis chronologies are nevertheless coarse. The U.S. 2007 onset is the clearest example: the supplied $t-1$ to t change is small, while the large repricing occurs in 2008. Replacing the same-year measure with the maximum spread over $t, t+1$ relative to $t-1$ strengthens rather than weakens the relationship. In simple crisis-level regressions, the $h=3$ slope is -43.4 for the supplied same-year change, -46.2 for the $t/t+1$ maximum, and -54.1 for the $t/t+1/t+2$ maximum. The corresponding rank correlations are also strongly negative. I retain the supplied same-year change as the formal baseline because it minimizes discretion, while treating the onset-window measures as important robustness checks.

Table 37: Robustness to alternative timing of crisis spread repricing at $h=3$

Spread measure	Slope	Std. error	Spearman ρ	Crisis origins
Supplied $t-1$ to t change	-43.4^{***}	12.1	-0.67	18
Maximum over $t, t+1$ relative to $t-1$	-46.2^{***}	10.7	-0.68	17
Maximum over $t, t+1, t+2$ relative to $t-1$	-54.1^{***}	9.6	-0.76	17

Entries are crisis-level associations between spread repricing and the cumulative NFC issuance response at $h=3$. Standard errors are clustered by country. The timing exercise is a robustness check for annual crisis-date uncertainty; the controlled LDLP uses the supplied same-year change as its baseline.

Influence and synchronized episodes. Individual country, crisis, and event-family leave-outs preserve a negative spread gradient. The result also remains negative when crises are collapsed to one observation per country or one observation per country-event family. Excluding all 2007–2011 crises or the European GFC/euro cluster reduces the magnitude but does not reverse the sign. This indicates that the relationship is not generated by one observation, although the large European GFC repricing materially strengthens the full-sample estimate. Because some block exclusions leave only four or five countries, conventional clustered inference should be interpreted cautiously; wild-cluster or CR2 inference should be added before final publication.

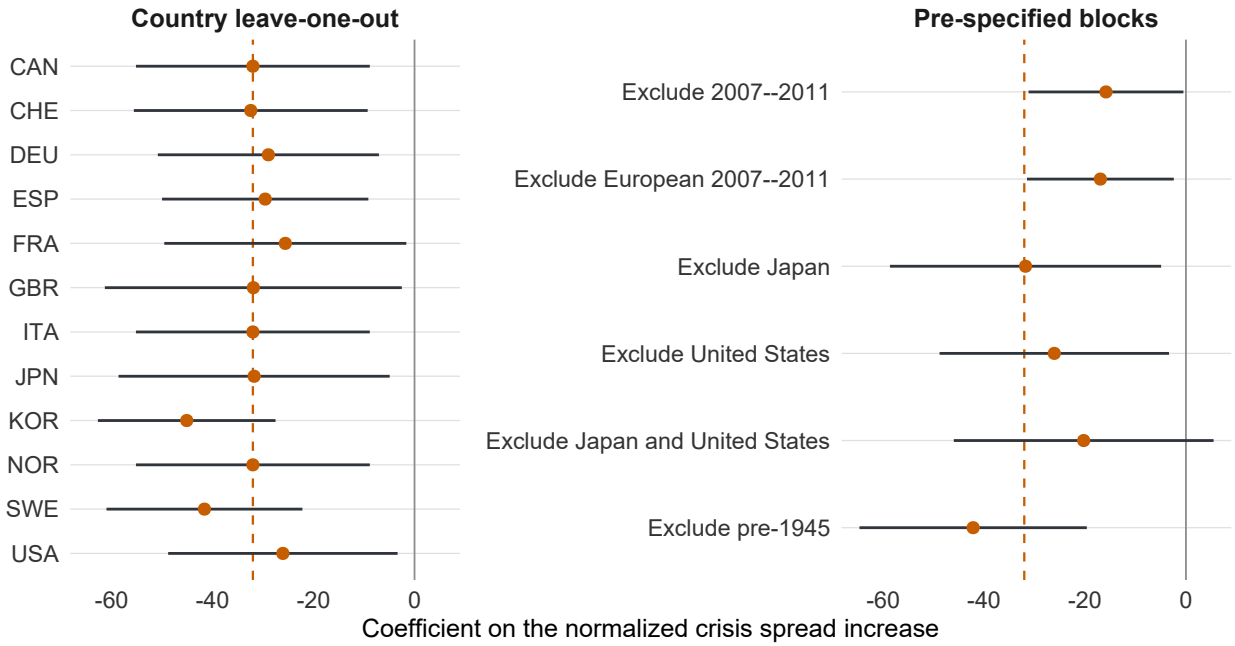


Figure 42: Influence of pre-specified block and country exclusions on the controlled $h = 3$ spread gradient. The dashed vertical line marks the full strict-sample estimate. Intervals use the same LDLP and Driscoll–Kraay inference as the baseline price specification.

Table 38: Cross-crisis robustness of the spread–NFC issuance relationship at $h = 3$

Specification	OLS slope	Spearman ρ	Rank p	Huber slope	Theil–Sen slope	Units	Countries
Full strict sample	-43.4*** (12.1)	-0.67	0.002	-43.3	-48.0	18	9
Exclude all 2007–2011 crises	-30.0** (8.6)	-0.52	0.102	-29.6	-27.5	11	4
Exclude European 2007–2011 crises	-29.8** (8.3)	-0.54	0.071	-29.3	-28.2	12	4
Exclude Japan and United States	-28.4** (10.3)	-0.57	0.112	-28.5	-28.9	9	7
Exclude pre-1945 crises	-48.3** (13.8)	-0.80	< 0.001	-48.0	-54.4	13	8
Earliest onset per event family only	-41.1** (12.3)	-0.65	0.007	-41.0	-44.9	16	9
Country means	-35.5** (13.7)					9	9
Country-event-family means	-40.5*** (11.5)					16	9

The baseline event-level OLS slopes use country-clustered standard errors. Country means are estimated with one observation per country; event-family means average overlapping crisis onsets within country. Robust and rank estimates are shown where meaningful. The structural LDLP evidence is reported separately in Tables 33 and 35.

Observable crisis-sample composition. Spread-observed and no-spread crisis origins have similar prewar shares and bank-equity losses, but the observed spread sample has stronger initial GDP growth and substantially lower pre-crisis NFC issuance growth. The latter difference is particularly visible in ranks. These comparisons help explain why the spread-covered crises display a milder unconditional issuance response; they are descriptive diagnostics of historical data selection rather than balance tests for a treatment design.

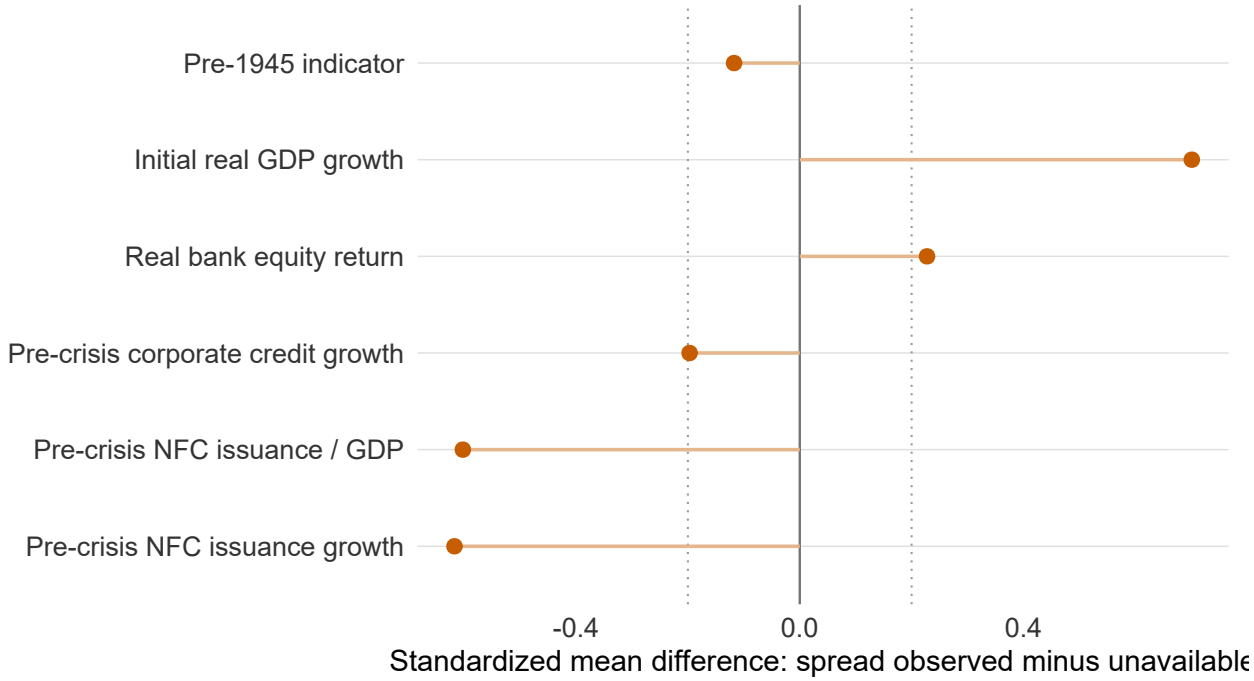


Figure 43: Observable composition of crisis origins with and without strict spread data. Points show standardized mean differences; the underlying group means and rank tests are reported in Table 39.

Table 39: Observable differences between crisis origins with and without spread data

Observable	Spread observed	Spread unavailable	Difference	Std. difference	Welch p	Rank p
Pre-1945 indicator	0.3 [18]	0.3 [18]	-0.1	-0.12	0.727	0.736
Initial real GDP growth	1.9 [18]	-0.2 [18]	2.1	0.70	0.043	0.189
Real bank equity return	-35.3 [17]	-40.2 [18]	4.9	0.23	0.503	0.564
Pre-crisis corporate credit growth	24.3 [13]	29.6 [14]	-5.3	-0.20	0.610	0.790
Pre-crisis NFC issuance / GDP	4.3 [17]	17.7 [18]	-13.4	-0.60	0.083	0.830
Pre-crisis NFC issuance growth	14.4 [16]	56.0 [18]	-41.6	-0.62	0.088	0.022

Samples are restricted to crisis origins entering the exact baseline NFC LDLP at $h = 3$. Brackets report the number of non-missing observations. Standardized differences use the pooled within-group standard deviation. These comparisons diagnose historical spread-data selection; they are not causal tests.

9.8 Sectoral comparison on a common crisis sample

A final exercise asks whether the price–quantity gradient differs across issuing sectors. To avoid comparing coefficients estimated on different event sets, the sample is restricted to 11 crises in five countries for which NFC, total-corporate, financial-corporate, and government issuance are all observed. The resulting pattern is suggestive: larger corporate spread increases predict a much larger contraction in NFC issuance, a smaller decline in total-corporate issuance, little clear change in financial-corporate issuance, and an increase in government issuance. This mirrors the sectoral reallocation documented in the full issuance sample. The comparison is nevertheless based on very few country clusters and should be treated as descriptive supporting evidence rather than a definitive sectoral test.

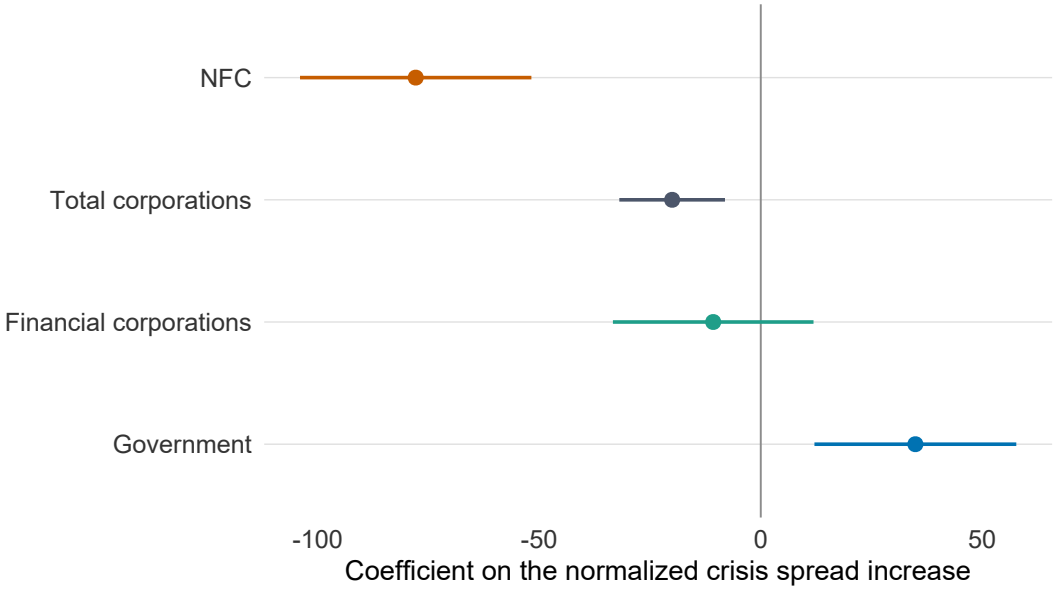


Figure 44: Sectoral price–quantity gradients on an identical crisis sample. The figure reports crisis-level $h = 3$ slopes for NFC, total-corporate, financial-corporate, and government issuance using the same 11 crisis origins.

9.9 Interpretation and limitations

The price evidence sharpens but does not overturn the paper's quantity-first interpretation. The full issuance sample shows a persistent average NFC contraction after banking crises. Historical spread coverage selects a smaller and more issuance-resilient subset, so the average response in that subset is not itself representative of the full crisis sample. Within observed price episodes, however, the degree of credit-market repricing is highly informative: large spread increases are followed by substantially weaker medium-run NFC issuance, including relative to GDP, investment, and lagged corporate credit.

Table 40: Sectoral price–quantity gradients on an identical crisis sample

Sector	Spread gradient	90% CI	Crisis onsets	Countries
NFC	-77.9*** (12.2)	[-104.0, -51.8]	11	5
Total corporations	-20.0** (5.6)	[-31.9, -8.1]	11	5
Financial corporations	-10.7 (10.6)	[-33.4, 11.9]	11	5
Government	34.9** (10.7)	[12.1, 57.7]	11	5

All outcomes and the strict KM spread are observed for the same 11 crisis onsets. Coefficients come from crisis-level regressions with country-clustered standard errors. The small common sample means the sectoral comparison is suggestive rather than definitive.

Four limitations bound the conclusion. First, the spread is a country-level risky debt-market price, not the average primary-market yield paid by the NFC issuers in the issuance dataset. Second, the historical spread is broad and the modern source varies across countries. Third, the spread and issuance responses are jointly endogenous to crisis severity, expected defaults, liquidity, risk premia, investor risk-bearing capacity, and issuer composition. Fourth, price coverage is missing for many of the crises that generate the strongest full-sample issuance decline, particularly several Great Depression and modern European episodes. The estimates therefore do not identify an exogenous shift in bond supply and do not show that spread repricing explains the full-sample average contraction.

Subject to these limitations, the evidence is difficult to reconcile with a simple financing-demand account in which issuance falls only because firms proportionally reduce investment. The contraction is largest when risky debt prices deteriorate most and remains visible relative to several financing scales. The most defensible conclusion is therefore that weaker financing demand, deteriorating borrower quality, and tighter market conditions interact during systemic banking crises. Bond markets can remain usable for selected firms and may even support higher issuance in low-repricing crises, but they do not provide a broad aggregate spare tire when the crisis is accompanied by substantial credit-market repricing.